

Markov Chains

4

4.1 Introduction

Consider a process that has a value in each time period. Let X_n denote its value in time period n , and suppose we want to make a probability model for the sequence of successive values $X_0, X_1, X_2, \dots$. The simplest model would probably be to assume that the X_n are independent random variables, but often such an assumption is clearly unjustified. For instance, starting at some time suppose that X_n represents the price of one share of some security, such as Google, at the end of n additional trading days. Then it certainly seems unreasonable to suppose that the price at the end of day $n + 1$ is independent of the prices on days $n, n - 1, n - 2$ and so on down to day 0. However, it might be reasonable to suppose that the price at the end of trading day $n + 1$ depends on the previous end-of-day prices only through the price at the end of day n . That is, it might be reasonable to assume that the conditional distribution of X_{n+1} given all the past end-of-day prices $X_n, X_{n-1}, \dots, X_0$ depends on these past prices only through the price at the end of day n . Such an assumption defines a Markov chain, a type of stochastic process that will be studied in this chapter, and which we now formally define.

Let $\{X_n, n = 0, 1, 2, \dots\}$ be a stochastic process that takes on a finite or countable number of possible values. Unless otherwise mentioned, this set of possible values of the process will be denoted by the set of nonnegative integers $\{0, 1, 2, \dots\}$. If $X_n = i$, then the process is said to be in state i at time n . We suppose that whenever the process is in state i , there is a fixed probability P_{ij} that it will next be in state j . That is, we suppose that

$$P\{X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0\} = P_{ij} \quad (4.1)$$

for all states $i_0, i_1, \dots, i_{n-1}, i, j$ and all $n \geq 0$. Such a stochastic process is known as a *Markov chain*. Eq. (4.1) may be interpreted as stating that, for a Markov chain, the conditional distribution of any future state X_{n+1} , given the past states $X_0, X_1, \dots, X_{n-1}$ and the present state X_n , is independent of the past states and depends only on the present state.

The value P_{ij} represents the probability that the process will, when in state i , next make a transition into state j . Since probabilities are nonnegative and since the process must make a transition into some state, we have

$$P_{ij} \geq 0, \quad i, j \geq 0; \quad \sum_{j=0}^{\infty} P_{ij} = 1, \quad i = 0, 1, \dots$$

Let $\mathbf{P}$ denote the matrix of one-step transition probabilities P_{ij} , so that

$$\mathbf{P} = \begin{bmatrix} P_{00} & P_{01} & P_{02} & \cdots \\ P_{10} & P_{11} & P_{12} & \cdots \\ \vdots & \vdots & \vdots & \\ P_{i0} & P_{i1} & P_{i2} & \cdots \\ \vdots & \vdots & \vdots & \end{bmatrix}$$

Example 4.1 (Forecasting the Weather). Suppose that the chance of rain tomorrow depends on previous weather conditions only through whether or not it is raining today and not on past weather conditions. Suppose also that if it rains today, then it will rain tomorrow with probability α ; and if it does not rain today, then it will rain tomorrow with probability β .

If we say that the process is in state 0 when it rains and state 1 when it does not rain, then the preceding is a two-state Markov chain whose transition probabilities are given by

$$\mathbf{P} = \begin{bmatrix} \alpha & 1 - \alpha \\ \beta & 1 - \beta \end{bmatrix} \quad \blacksquare$$

Example 4.2 (A Communications System). Consider a communications system that transmits the digits 0 and 1. Each digit transmitted must pass through several stages, at each of which there is a probability p that the digit entered will be unchanged when it leaves. Letting X_n denote the digit entering the n th stage, then $\{X_n, n = 0, 1, \dots\}$ is a two-state Markov chain having a transition probability matrix

$$\mathbf{P} = \begin{bmatrix} p & 1 - p \\ 1 - p & p \end{bmatrix} \quad \blacksquare$$

Example 4.3. On any given day Gary is either cheerful (C), so-so (S), or glum (G). If he is cheerful today, then he will be C , S , or G tomorrow with respective probabilities 0.5, 0.4, 0.1. If he is feeling so-so today, then he will be C , S , or G tomorrow with probabilities 0.3, 0.4, 0.3. If he is glum today, then he will be C , S , or G tomorrow with probabilities 0.2, 0.3, 0.5.

Letting X_n denote Gary's mood on the n th day, then $\{X_n, n \geq 0\}$ is a three-state Markov chain (state 0 = C , state 1 = S , state 2 = G) with transition probability matrix

$$\mathbf{P} = \begin{bmatrix} 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix} \quad \blacksquare$$

Example 4.4 (Transforming a Process into a Markov Chain). Suppose that whether or not it rains today depends on previous weather conditions through the last two days. Specifically, suppose that if it has rained for the past two days, then it will rain tomorrow with probability 0.7; if it rained today but not yesterday, then it will rain

tomorrow with probability 0.5; if it rained yesterday but not today, then it will rain tomorrow with probability 0.4; if it has not rained in the past two days, then it will rain tomorrow with probability 0.2.

If we let the state at time n depend only on whether or not it is raining at time n , then the preceding model is not a Markov chain (why not?). However, we can transform this model into a Markov chain by saying that the state at any time is determined by the weather conditions during both that day and the previous day. In other words, we can say that the process is in

- state 0 if it rained both today and yesterday,
- state 1 if it rained today but not yesterday,
- state 2 if it rained yesterday but not today,
- state 3 if it did not rain either yesterday or today.

The preceding would then represent a four-state Markov chain having a transition probability matrix

$$\mathbf{P} = \begin{bmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{bmatrix}$$

You should carefully check the matrix $\mathbf{P}$, and make sure you understand how it was obtained. ■

Example 4.5 (A Random Walk Model). A Markov chain whose state space is given by the integers $i = 0, \pm 1, \pm 2, \dots$ is said to be a *random walk* if, for some number $0 < p < 1$,

$$P_{i,i+1} = p = 1 - P_{i,i-1}, \quad i = 0, \pm 1, \dots$$

The preceding Markov chain is called a random walk for we may think of it as being a model for an individual walking on a straight line who at each point of time either takes one step to the right with probability p or one step to the left with probability $1 - p$. ■

Example 4.6 (A Gambling Model). Consider a gambler who, at each play of the game, either wins \$1 with probability p or loses \$1 with probability $1 - p$. If we suppose that our gambler quits playing either when he goes broke or he attains a fortune of $\$N$, then the gambler's fortune is a Markov chain having transition probabilities

$$P_{i,i+1} = p = 1 - P_{i,i-1}, \quad i = 1, 2, \dots, N-1, \\ P_{00} = P_{NN} = 1$$

States 0 and N are called *absorbing* states since once entered they are never left. Note that the preceding is a finite state random walk with absorbing barriers (states 0 and N). ■

Example 4.7. In most of Europe and Asia annual automobile insurance premiums are determined by use of a Bonus Malus (Latin for Good-Bad) system. Each policyholder is given a positive integer valued state and the annual premium is a function of this state (along, of course, with the type of car being insured and the level of insurance). A policyholder's state changes from year to year in response to the number of claims made by that policyholder. Because lower numbered states correspond to lower annual premiums, a policyholder's state will usually decrease if he or she had no claims in the preceding year, and will generally increase if he or she had at least one claim. (Thus, no claims is good and typically results in a decreased premium, while claims are bad and typically result in a higher premium.)

For a given Bonus Malus system, let $s_i(k)$ denote the next state of a policyholder who was in state i in the previous year and who made a total of k claims in that year. If we suppose that the number of yearly claims made by a particular policyholder is a Poisson random variable with parameter λ , then the successive states of this policyholder will constitute a Markov chain with transition probabilities

$$P_{i,j} = \sum_{k: s_i(k)=j} e^{-\lambda} \frac{\lambda^k}{k!}, \quad j \geq 0$$

Whereas there are usually many states (20 or so is not atypical), the following table specifies a hypothetical Bonus Malus system having four states.

State	Annual Premium	Next state if			
		0 claims	1 claim	2 claims	≥ 3 claims
1	200	1	2	3	4
2	250	1	3	4	4
3	400	2	4	4	4
4	600	3	4	4	4

Thus, for instance, the table indicates that $s_2(0) = 1$; $s_2(1) = 3$; $s_2(k) = 4$, $k \geq 2$. Consider a policyholder whose annual number of claims is a Poisson random variable with parameter λ . If a_k is the probability that such a policyholder makes k claims in a year, then

$$a_k = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k \geq 0$$

For the Bonus Malus system specified in the preceding table, the transition probability matrix of the successive states of this policyholder is

$$\mathbf{P} = \begin{pmatrix} a_0 & a_1 & a_2 & 1 - a_0 - a_1 - a_2 \\ a_0 & 0 & a_1 & 1 - a_0 - a_1 \\ 0 & a_0 & 0 & 1 - a_0 \\ 0 & 0 & a_0 & 1 - a_0 \end{pmatrix}$$

■

4.2 Chapman–Kolmogorov Equations

We have already defined the one-step transition probabilities P_{ij} . We now define the n -step transition probabilities P_{ij}^n to be the probability that a process in state i will be in state j after n additional transitions. That is,

$$P_{ij}^n = P\{X_{n+k} = j | X_k = i\}, \quad n \geq 0, i, j \geq 0$$

Of course $P_{ij}^1 = P_{ij}$. The *Chapman–Kolmogorov equations* provide a method for computing these n -step transition probabilities. These equations are

$$P_{ij}^{n+m} = \sum_{k=0}^{\infty} P_{ik}^n P_{kj}^m \quad \text{for all } n, m \geq 0, \text{ all } i, j \quad (4.2)$$

and are most easily understood by noting that $P_{ik}^n P_{kj}^m$ represents the probability that starting in i the process will go to state j in $n + m$ transitions through a path which takes it into state k at the n th transition. Hence, summing over all intermediate states k yields the probability that the process will be in state j after $n + m$ transitions. Formally, we have

$$\begin{aligned} P_{ij}^{n+m} &= P\{X_{n+m} = j | X_0 = i\} \\ &= \sum_{k=0}^{\infty} P\{X_{n+m} = j, X_n = k | X_0 = i\} \\ &= \sum_{k=0}^{\infty} P\{X_{n+m} = j | X_n = k, X_0 = i\} P\{X_n = k | X_0 = i\} \\ &= \sum_{k=0}^{\infty} P_{kj}^m P_{ik}^n \end{aligned}$$

If we let $\mathbf{P}^{(n)}$ denote the matrix of n -step transition probabilities P_{ij}^n , then Eq. (4.2) asserts that

$$\mathbf{P}^{(n+m)} = \mathbf{P}^{(n)} \cdot \mathbf{P}^{(m)}$$

where the dot represents matrix multiplication.¹ Hence, in particular,

$$\mathbf{P}^{(2)} = \mathbf{P}^{(1+1)} = \mathbf{P} \cdot \mathbf{P} = \mathbf{P}^2$$

and by induction

$$\mathbf{P}^{(n)} = \mathbf{P}^{(n-1+1)} = \mathbf{P}^{n-1} \cdot \mathbf{P} = \mathbf{P}^n$$

¹ If $\mathbf{A}$ is an $N \times M$ matrix whose element in the i th row and j th column is a_{ij} and $\mathbf{B}$ is an $M \times K$ matrix whose element in the i th row and j th column is b_{ij} , then $\mathbf{A} \cdot \mathbf{B}$ is defined to be the $N \times K$ matrix whose element in the i th row and j th column is $\sum_{k=1}^M a_{ik} b_{kj}$.

That is, the n -step transition matrix may be obtained by multiplying the matrix $\mathbf{P}$ by itself n times.

Example 4.8. Consider Example 4.1 in which the weather is considered as a two-state Markov chain. If $\alpha = 0.7$ and $\beta = 0.4$, then calculate the probability that it will rain four days from today given that it is raining today.

Solution: The one-step transition probability matrix is given by

$$\mathbf{P} = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

Hence,

$$\begin{aligned} \mathbf{P}^{(2)} = \mathbf{P}^2 &= \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} \cdot \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} \\ &= \begin{bmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{bmatrix}, \\ \mathbf{P}^{(4)} = (\mathbf{P}^2)^2 &= \begin{bmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{bmatrix} \cdot \begin{bmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{bmatrix} \\ &= \begin{bmatrix} 0.5749 & 0.4251 \\ 0.5668 & 0.4332 \end{bmatrix} \end{aligned}$$

and the desired probability P_{00}^4 equals 0.5749. ■

Example 4.9. Consider Example 4.4. Given that it rained on Monday and Tuesday, what is the probability that it will rain on Thursday?

Solution: The two-step transition matrix is given by

$$\begin{aligned} \mathbf{P}^{(2)} = \mathbf{P}^2 &= \begin{bmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{bmatrix} \cdot \begin{bmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{bmatrix} \\ &= \begin{bmatrix} 0.49 & 0.12 & 0.21 & 0.18 \\ 0.35 & 0.20 & 0.15 & 0.30 \\ 0.20 & 0.12 & 0.20 & 0.48 \\ 0.10 & 0.16 & 0.10 & 0.64 \end{bmatrix} \end{aligned}$$

Since rain on Thursday is equivalent to the process being in either state 0 or state 1 on Thursday, the desired probability is given by $P_{00}^2 + P_{01}^2 = 0.49 + 0.12 = 0.61$. ■

Example 4.10. An urn always contains 2 balls. Ball colors are red and blue. At each stage a ball is randomly chosen and then replaced by a new ball, which with probability 0.8 is the same color, and with probability 0.2 is the opposite color, as the ball it replaces. If initially both balls are red, find the probability that the fifth ball selected is red.

Solution: To find the desired probability we first define an appropriate Markov chain. This can be accomplished by noting that the probability that a selection is red is determined by the composition of the urn at the time of the selection. So, let us define X_n to be the number of red balls in the urn after the n th selection and subsequent replacement. Then $X_n, n \geq 0$, is a Markov chain with states 0, 1, 2 and with transition probability matrix $\mathbf{P}$ given by

$$\begin{pmatrix} 0.8 & 0.2 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.2 & 0.8 \end{pmatrix}$$

To understand the preceding, consider for instance $P_{1,0}$. Now, to go from 1 red ball in the urn to 0 red balls, the ball chosen must be red (which occurs with probability 0.5) and it must then be replaced by a ball of opposite color (which occurs with probability 0.2), showing that

$$P_{1,0} = (0.5)(0.2) = 0.1$$

To determine the probability that the fifth selection is red, condition on the number of red balls in the urn after the fourth selection. This yields

$$\begin{aligned} &P(\text{fifth selection is red}) \\ &= \sum_{i=0}^2 P(\text{fifth selection is red} | X_4 = i) P(X_4 = i | X_0 = 2) \\ &= (0)P_{2,0}^4 + (0.5)P_{2,1}^4 + (1)P_{2,2}^4 \\ &= 0.5P_{2,1}^4 + P_{2,2}^4 \end{aligned}$$

To calculate the preceding we compute $\mathbf{P}^4$. Doing so yields

$$P_{2,1}^4 = 0.4352, \quad P_{2,2}^4 = 0.4872$$

giving the answer $P(\text{fifth selection is red}) = 0.7048$. ■

Example 4.11. Suppose that balls are successively distributed among 8 urns, with each ball being equally likely to be put in any of these urns. What is the probability that there will be exactly 3 nonempty urns after 9 balls have been distributed?

Solution: If we let X_n be the number of nonempty urns after n balls have been distributed, then $X_n, n \geq 0$ is a Markov chain with states 0, 1, ..., 8 and transition probabilities

$$P_{i,i} = i/8 = 1 - P_{i,i+1}, \quad i = 0, 1, \dots, 8$$

The desired probability is $P_{0,3}^9 = P_{1,3}^8$, where the equality follows because $P_{0,1} = 1$. Now, starting with 1 occupied urn, if we had wanted to determine the entire probability distribution of the number of occupied urns after 8 additional balls

had been distributed we would need to consider the transition probability matrix with states $1, 2, \dots, 8$. However, because we only require the probability, starting with a single occupied urn, that there are 3 occupied urns after an additional 8 balls have been distributed we can make use of the fact that the state of the Markov chain cannot decrease to collapse all states $4, 5, \dots, 8$ into a single state 4 with the interpretation that the state is 4 whenever four or more of the urns are occupied. Consequently, we need only determine the eight-step transition probability $P_{1,3}^8$ of the Markov chain with states $1, 2, 3, 4$ having transition probability matrix $\mathbf{P}$ given by

$$\begin{pmatrix} 1/8 & 7/8 & 0 & 0 \\ 0 & 2/8 & 6/8 & 0 \\ 0 & 0 & 3/8 & 5/8 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Raising the preceding matrix to the power 4 yields the matrix $\mathbf{P}^4$ given by

$$\begin{pmatrix} 0.0002 & 0.0256 & 0.2563 & 0.7178 \\ 0 & 0.0039 & 0.0952 & 0.9009 \\ 0 & 0 & 0.0198 & 0.9802 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Hence,

$$\begin{aligned} P_{1,3}^8 &= 0.0002 \times 0.2563 + 0.0256 \times 0.0952 + 0.2563 \times 0.0198 \\ &\quad + 0.7178 \times 0 = 0.00756 \end{aligned} \quad \blacksquare$$

Consider a Markov chain with transition probabilities P_{ij} . Let $\mathcal{A}$ be a set of states, and suppose we are interested in the probability that the Markov chain ever enters any of the states in $\mathcal{A}$ by time m . That is, for a given state $i \notin \mathcal{A}$, we are interested in determining

$$\beta = P(X_k \in \mathcal{A} \text{ for some } k = 1, \dots, m | X_0 = i)$$

To determine the preceding probability we will define a Markov chain $\{W_n, n \geq 0\}$ whose states are the states that are not in $\mathcal{A}$ plus an additional state, which we will call A in our general discussion (though in specific examples we will usually give it a different name). Once the $\{W_n\}$ Markov chain enters state A it remains there forever.

The new Markov chain is defined as follows. Letting X_n denote the state at time n of the Markov chain with transition probabilities P_{ij} , define

$$N = \min\{n : X_n \in \mathcal{A}\}$$

and let $N = \infty$ if $X_n \notin \mathcal{A}$ for all n . In words, N is the first time the Markov chain enters the set of states $\mathcal{A}$. Now, define

$$W_n = \begin{cases} X_n, & \text{if } n < N \\ A, & \text{if } n \geq N \end{cases}$$

So the state of the $\{W_n\}$ process is equal to the state of the original Markov chain up to the point when the original Markov chain enters a state in $\mathcal{A}$. At that time the new process goes to state A and remains there forever. From this description it follows that $W_n, n \geq 0$ is a Markov chain with states $i, i \notin \mathcal{A}, A$ and with transition probabilities $Q_{i,j}$, given by

$$\begin{aligned} Q_{i,j} &= P_{i,j}, \quad \text{if } i \notin \mathcal{A}, j \notin \mathcal{A} \\ Q_{i,A} &= \sum_{j \in \mathcal{A}} P_{i,j}, \quad \text{if } i \notin \mathcal{A} \\ Q_{A,A} &= 1 \end{aligned}$$

Because the original Markov chain will have entered a state in $\mathcal{A}$ by time m if and only if the state at time m of the new Markov chain is A , we see that

$$\begin{aligned} P(X_k \in \mathcal{A} \text{ for some } k = 1, \dots, m | X_0 = i) \\ = P(W_m = A | X_0 = i) = P(W_m = A | W_0 = i) = Q_{i,A}^m \end{aligned}$$

That is, the desired probability is equal to an m -step transition probability of the new chain.

Example 4.12. In a sequence of independent flips of a fair coin, let N denote the number of flips until there is a run of three consecutive heads. Find

- (a) $P(N \leq 8)$ and
- (b) $P(N = 8)$.

Solution: To determine $P(N \leq 8)$, define a Markov chain with states 0, 1, 2, 3 where for $i < 3$ state i means that we currently are on a run of i consecutive heads, and where state 3 means that a run of three consecutive heads has already occurred. Thus, the transition probability matrix is

$$\mathbf{P} = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 1/2 & 0 & 0 & 1/2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where, for instance, the values for row 2 are obtained by noting that if we currently are on a run of size 1 then the next state will be 0 if the next flip is a tail, or 2 if it is a head. Hence, $P_{1,0} = P_{1,2} = 1/2$. Because there would be a run of three consecutive heads within the first eight flips if and only if $X_8 = 3$, the desired probability is $P_{0,3}^8$. Squaring $\mathbf{P}$ to obtain $\mathbf{P}^2$, then squaring the result to obtain $\mathbf{P}^4$, and then squaring that matrix gives the result

$$\mathbf{P}^8 = \begin{pmatrix} 81/256 & 44/256 & 24/256 & 107/256 \\ 68/256 & 37/256 & 20/256 & 131/256 \\ 44/256 & 24/256 & 13/256 & 175/256 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Hence, the probability that there will be a run of three consecutive heads within the first eight flips is $107/256 \approx .4180$.

(b) Noting that $N = 8$ if the pattern hasn't yet occurred in the first 7 transitions, the state after 7 transitions is 2, and the next flip lands heads, shows that

$$P(N = 8) = \frac{1}{2} P_{0,2}^7. \quad \blacksquare$$

We can also use Markov chains to determine probabilities concerning the time until a pattern appears when the data itself comes from a Markov chain. We illustrate this by an example.

Example 4.13. Let $\{X_n, n \geq 0\}$ be a Markov chain with states 0, 1, 2, 3 and transition probabilities $P_{i,j}, i, j = 0, 1, 2, 3$, and let N denote the number of transitions, starting in state 0, until the pattern 1, 2, 1, 2 appears. That is,

$$N = \min\{n \geq 4 : X_{n-3} = 1, X_{n-2} = 2, X_{n-1} = 1, X_n = 2\}.$$

Suppose we are interested in evaluating $P(N \leq k)$ for some specified value k . To do so, we define a new Markov chain $\{Y_n, n \geq 0\}$, that tracks the progress towards the pattern. The Y_n are defined as follows:

- If the pattern has appeared by the n th transition—that is, if $X_0, \dots, X_n$ includes 1, 2, 1, 2—then $Y_n = 4$.
- If the pattern has not yet appeared by the n th transition

$$Y_n = 1 \text{ if } X_n = 1 \text{ and } (X_{n-2}, X_{n-1}) \neq (1, 2).$$

$$Y_n = 2 \text{ if } X_{n-1} = 1, X_n = 2.$$

$$Y_n = 3 \text{ if } X_{n-2} = 1, X_{n-1} = 2, X_n = 1.$$

$$Y_n = 5 \text{ if } X_n = 2, X_{n-1} \neq 1.$$

$$Y_n = 6 \text{ if } X_n = 0.$$

$$Y_n = 7 \text{ if } X_n = 3$$

Thus, for $i = 1, 2, 3, 4$, $Y_n = i$ signifies that we are i steps into the pattern (or in the case $i = 4$ that the pattern has appeared). $Y_n = 5$ (or 6 or 7) if there is no current progress with regards to the pattern and the current state is 2 (or 0 or 3). The desired probability $P(N \leq k)$ is equal to the probability that the number of transitions of the Markov chain $\{Y_n\}$ to go from state 6 to state 4 is less than or equal to k . Because state 4 is an absorbing state of this chain, this probability is $Q_{6,4}^k$ where $Q_{i,j}$ are the transition probabilities of the Markov chain $\{Y_n\}$. $\blacksquare$

Suppose now that we want to compute the probability that the $\{X_n, n \geq 0\}$ chain, starting in state i , enters state j at time m without ever entering any of the states in $\mathcal{A}$, where neither i nor j is in $\mathcal{A}$. That is, for $i, j \notin \mathcal{A}$, we are interested in

$$\alpha = P(X_m = j, X_k \notin \mathcal{A}, k = 1, \dots, m-1 | X_0 = i)$$

Noting that the event that $X_m = j, X_k \notin \mathcal{A}, k = 1, \dots, m-1$ is equivalent to the event that $W_m = j$, it follows that for $i, j \notin \mathcal{A}$,

$$\begin{aligned} P(X_m = j, X_k \notin \mathcal{A}, k = 1, \dots, m-1 | X_0 = i) &= P(W_m = j | X_0 = i) \\ &= P(W_m = j | W_0 = i) = Q_{i,j}^m. \end{aligned}$$

Example 4.14. Consider a Markov chain with states 1, 2, 3, 4, 5, and suppose that we want to compute

$$P(X_4 = 2, X_3 \leq 2, X_2 \leq 2, X_1 \leq 2 | X_0 = 1)$$

That is, we want the probability that, starting in state 1, the chain is in state 2 at time 4 and has never entered any of the states in the set $\mathcal{A} = \{3, 4, 5\}$.

To compute this probability all we need to know are the transition probabilities $P_{11}, P_{12}, P_{21}, P_{22}$. So, suppose that

$$\begin{aligned} P_{11} &= 0.3 & P_{12} &= 0.3 \\ P_{21} &= 0.1 & P_{22} &= 0.2 \end{aligned}$$

Then we consider the Markov chain having states 1, 2, 3 (we are giving state 4 the name 3), and having the transition probability matrix $\mathbf{Q}$ as follows:

$$\begin{pmatrix} 0.3 & 0.3 & 0.4 \\ 0.1 & 0.2 & 0.7 \\ 0 & 0 & 1 \end{pmatrix}$$

The desired probability is Q_{12}^4 . Raising $\mathbf{Q}$ to the power 4 yields the matrix

$$\begin{pmatrix} 0.0219 & 0.0285 & 0.9496 \\ 0.0095 & 0.0124 & 0.9781 \\ 0 & 0 & 1 \end{pmatrix}$$

Hence, the desired probability is $\alpha = 0.0285$. ■

When $i \notin \mathcal{A}$ but $j \in \mathcal{A}$ we can determine the probability

$$\alpha = P(X_m = j, X_k \notin \mathcal{A}, k = 1, \dots, m-1 | X_0 = i)$$

as follows.

$$\begin{aligned} \alpha &= \sum_{r \notin \mathcal{A}} P(X_m = j, X_{m-1} = r, X_k \notin \mathcal{A}, k = 1, \dots, m-2 | X_0 = i) \\ &= \sum_{r \notin \mathcal{A}} P(X_m = j | X_{m-1} = r, X_k \notin \mathcal{A}, k = 1, \dots, m-2, X_0 = i) \\ &\quad \times P(X_{m-1} = r, X_k \notin \mathcal{A}, k = 1, \dots, m-2 | X_0 = i) \\ &= \sum_{r \notin \mathcal{A}} P_{r,j} P(X_{m-1} = r, X_k \notin \mathcal{A}, k = 1, \dots, m-2 | X_0 = i) \\ &= \sum_{r \notin \mathcal{A}} P_{r,j} Q_{i,r}^{m-1} \end{aligned}$$

Also, when $i \in \mathcal{A}$ we could determine

$$\alpha = P(X_m = j, X_k \notin \mathcal{A}, k = 1, \dots, m-1 | X_0 = i)$$

by conditioning on the first transition to obtain

$$\begin{aligned} \alpha &= \sum_{r \notin \mathcal{A}} P(X_m = j, X_k \notin \mathcal{A}, \\ &\quad k = 1, \dots, m-1 | X_0 = i, X_1 = r) P(X_1 = r | X_0 = i) \\ &= \sum_{r \notin \mathcal{A}} P(X_{m-1} = j, X_k \notin \mathcal{A}, k = 1, \dots, m-2 | X_0 = r) P_{i,r} \end{aligned}$$

For instance, if $i \in \mathcal{A}$, $j \notin \mathcal{A}$ then the preceding equation yields

$$P(X_m = j, X_k \notin \mathcal{A}, k = 1, \dots, m-1 | X_0 = i) = \sum_{r \notin \mathcal{A}} Q_{r,j}^{m-1} P_{i,r}$$

We can also compute the conditional probability of X_n given that the chain starts in state i and has not entered any state in $\mathcal{A}$ by time n , as follows. For $i, j \notin \mathcal{A}$,

$$\begin{aligned} &P\{X_n = j | X_0 = i, X_k \notin \mathcal{A}, k = 1, \dots, n\} \\ &= \frac{P\{X_n = j, X_k \notin \mathcal{A}, k = 1, \dots, n | X_0 = i\}}{P\{X_k \notin \mathcal{A}, k = 1, \dots, n | X_0 = i\}} = \frac{Q_{i,j}^n}{\sum_{r \notin \mathcal{A}} Q_{i,r}^n} \end{aligned}$$

Remark. So far, all of the probabilities we have considered are conditional probabilities. For instance, P_{ij}^n is the probability that the state at time n is j given that the initial state at time 0 is i . If the unconditional distribution of the state at time n is desired, it is necessary to specify the probability distribution of the initial state. Let us denote this by

$$\alpha_i \equiv P\{X_0 = i\}, \quad i \geq 0 \quad \left(\sum_{i=0}^{\infty} \alpha_i = 1 \right)$$

All unconditional probabilities may be computed by conditioning on the initial state. That is,

$$\begin{aligned} P\{X_n = j\} &= \sum_{i=0}^{\infty} P\{X_n = j | X_0 = i\} P\{X_0 = i\} \\ &= \sum_{i=0}^{\infty} P_{ij}^n \alpha_i \end{aligned}$$

For instance, if $\alpha_0 = 0.4$, $\alpha_1 = 0.6$, in Example 4.8, then the (unconditional) probability that it will rain four days after we begin keeping weather records is

$$P\{X_4 = 0\} = 0.4P_{00}^4 + 0.6P_{10}^4$$

$$\begin{aligned}
&= (0.4)(0.5749) + (0.6)(0.5668) \\
&= 0.5700
\end{aligned}$$

4.3 Classification of States

State j is said to be *accessible* from state i if $P_{ij}^n > 0$ for some $n \geq 0$. Note that this implies that state j is accessible from state i if and only if, starting in i , it is possible that the process will ever enter state j . This is true since if j is not accessible from i , then

$$\begin{aligned}
P\{\text{ever be in } j | \text{start in } i\} &= P\left\{\bigcup_{n=0}^{\infty} \{X_n = j\} \mid X_0 = i\right\} \\
&\leq \sum_{n=0}^{\infty} P\{X_n = j | X_0 = i\} \\
&= \sum_{n=0}^{\infty} P_{ij}^n \\
&= 0
\end{aligned}$$

Two states i and j that are accessible to each other are said to *communicate*, and we write $i \leftrightarrow j$.

Note that any state communicates with itself since, by definition,

$$P_{ii}^0 = P\{X_0 = i | X_0 = i\} = 1$$

The relation of communication satisfies the following three properties:

- (i) State i communicates with state i , all $i \geq 0$.
- (ii) If state i communicates with state j , then state j communicates with state i .
- (iii) If state i communicates with state j , and state j communicates with state k , then state i communicates with state k .

Properties (i) and (ii) follow immediately from the definition of communication. To prove (iii) suppose that i communicates with j , and j communicates with k . Thus, there exist integers n and m such that $P_{ij}^n > 0$, $P_{jk}^m > 0$. Now by the Chapman-Kolmogorov equations, we have

$$P_{ik}^{n+m} = \sum_{r=0}^{\infty} P_{ir}^n P_{rk}^m \geq P_{ij}^n P_{jk}^m > 0$$

Hence, state k is accessible from state i . Similarly, we can show that state i is accessible from state k . Hence, states i and k communicate.

Two states that communicate are said to be in the same *class*. It is an easy consequence of (i), (ii), and (iii) that any two classes of states are either identical or disjoint.

In other words, the concept of communication divides the state space up into a number of separate classes. The Markov chain is said to be *irreducible* if there is only one class, that is, if all states communicate with each other.

Example 4.15. Consider the Markov chain consisting of the three states 0, 1, 2 and having transition probability matrix

$$\mathbf{P} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

It is easy to verify that this Markov chain is irreducible. For example, it is possible to go from state 0 to state 2 since

$$0 \rightarrow 1 \rightarrow 2$$

That is, one way of getting from state 0 to state 2 is to go from state 0 to state 1 (with probability $\frac{1}{2}$) and then go from state 1 to state 2 (with probability $\frac{1}{4}$). ■

Example 4.16. Consider a Markov chain consisting of the four states 0, 1, 2, 3 and having transition probability matrix

$$\mathbf{P} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The classes of this Markov chain are $\{0, 1\}$, $\{2\}$, and $\{3\}$. Note that while state 0 (or 1) is accessible from state 2, the reverse is not true. Since state 3 is an absorbing state, that is, $P_{33} = 1$, no other state is accessible from it. ■

For any state i we let f_i denote the probability that, starting in state i , the process will ever reenter state i . State i is said to be *recurrent* if $f_i = 1$ and *transient* if $f_i < 1$.

Suppose that the process starts in state i and i is recurrent. Hence, with probability 1, the process will eventually reenter state i . However, by the definition of a Markov chain, it follows that the process will be starting over again when it reenters state i and, therefore, state i will eventually be visited again. Continual repetition of this argument leads to the conclusion that *if state i is recurrent then, starting in state i , the process will reenter state i again and again and again—in fact, infinitely often.*

On the other hand, suppose that state i is transient. Hence, each time the process enters state i there will be a positive probability, namely, $1 - f_i$, that it will never again enter that state. Therefore, starting in state i , the probability that the process will be in state i for exactly n time periods equals $f_i^{n-1}(1 - f_i)$, $n \geq 1$. In other words, *if state i is transient then, starting in state i , the number of time periods that the process will be in state i has a geometric distribution with finite mean $1/(1 - f_i)$.*

From the preceding two paragraphs, it follows that *state i is recurrent if and only if, starting in state i , the expected number of time periods that the process is in state i is infinite*. But, letting

$$I_n = \begin{cases} 1, & \text{if } X_n = i \\ 0, & \text{if } X_n \neq i \end{cases}$$

we have that $\sum_{n=0}^{\infty} I_n$ represents the number of periods that the process is in state i . Also,

$$\begin{aligned} E \left[\sum_{n=0}^{\infty} I_n \mid X_0 = i \right] &= \sum_{n=0}^{\infty} E[I_n \mid X_0 = i] \\ &= \sum_{n=0}^{\infty} P\{X_n = i \mid X_0 = i\} \\ &= \sum_{n=0}^{\infty} P_{ii}^n \end{aligned}$$

We have thus proven the following.

Proposition 4.1. *State i is*

$$\begin{aligned} &\text{recurrent if } \sum_{n=1}^{\infty} P_{ii}^n = \infty, \\ &\text{transient if } \sum_{n=1}^{\infty} P_{ii}^n < \infty \end{aligned}$$

The argument leading to the preceding proposition is doubly important because it also shows that a transient state will only be visited a finite number of times (hence the name transient). This leads to the conclusion that in a finite-state Markov chain not all states can be transient. To see this, suppose the states are $0, 1, \dots, M$ and suppose that they are all transient. Then after a finite amount of time (say, after time T_0) state 0 will never be visited, and after a time (say, T_1) state 1 will never be visited, and after a time (say, T_2) state 2 will never be visited, and so on. Thus, after a finite time $T = \max\{T_0, T_1, \dots, T_M\}$ no states will be visited. But as the process must be in some state after time T we arrive at a contradiction, which shows that at least one of the states must be recurrent.

Another use of Proposition 4.1 is that it enables us to show that recurrence is a class property.

Corollary 4.2. *If state i is recurrent, and state i communicates with state j , then state j is recurrent.*

Proof. To prove this we first note that, since state i communicates with state j , there exist integers k and m such that $P_{ij}^k > 0$, $P_{ji}^m > 0$. Now, for any integer n

$$P_{jj}^{m+n+k} \geq P_{ji}^m P_{ii}^n P_{ij}^k$$

This follows since the left side of the preceding is the probability of going from j to j in $m+n+k$ steps, while the right side is the probability of going from j to j in $m+n+k$ steps via a path that goes from j to i in m steps, then from i to i in an additional n steps, then from i to j in an additional k steps.

From the preceding we obtain, by summing over n , that

$$\sum_{n=1}^{\infty} P_{jj}^{m+n+k} \geq P_{ji}^m P_{ij}^k \sum_{n=1}^{\infty} P_{ii}^n = \infty$$

since $P_{ji}^m P_{ij}^k > 0$ and $\sum_{n=1}^{\infty} P_{ii}^n$ is infinite since state i is recurrent. Thus, by Proposition 4.1 it follows that state j is also recurrent. ■

- Remarks.** (i) Corollary 4.2 also implies that transience is a class property. For if state i is transient and communicates with state j , then state j must also be transient. For if j were recurrent then, by Corollary 4.2, i would also be recurrent and hence could not be transient.
- (ii) Corollary 4.2 along with our previous result that not all states in a finite Markov chain can be transient leads to the conclusion that all states of a finite irreducible Markov chain are recurrent.

Example 4.17. Let the Markov chain consisting of the states 0, 1, 2, 3 have the transition probability matrix

$$\mathbf{P} = \begin{bmatrix} 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Determine which states are transient and which are recurrent.

Solution: It is a simple matter to check that all states communicate and, hence, since this is a finite chain, all states must be recurrent. ■

Example 4.18. Consider the Markov chain having states 0, 1, 2, 3, 4 and

$$\mathbf{P} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{4} & 0 & 0 & \frac{1}{2} \end{bmatrix}$$

Determine the recurrent state.

Solution: This chain consists of the three classes $\{0, 1\}$, $\{2, 3\}$, and $\{4\}$. The first two classes are recurrent and the third transient. ■

Example 4.19 (A Random Walk). Consider a Markov chain whose state space consists of the integers $i = 0, \pm 1, \pm 2, \dots$, and has transition probabilities given by

$$P_{i,i+1} = p = 1 - P_{i,i-1}, \quad i = 0, \pm 1, \pm 2, \dots$$

where $0 < p < 1$. In other words, on each transition the process either moves one step to the right (with probability p) or one step to the left (with probability $1 - p$). One colorful interpretation of this process is that it represents the wanderings of a drunken man as he walks along a straight line. Another is that it represents the winnings of a gambler who on each play of the game either wins or loses one dollar.

Since all states clearly communicate, it follows from Corollary 4.2 that they are either all transient or all recurrent. So let us consider state 0 and attempt to determine if $\sum_{n=1}^{\infty} P_{00}^n$ is finite or infinite.

Since it is impossible to be even (using the gambling model interpretation) after an odd number of plays we must, of course, have that

$$P_{00}^{2n-1} = 0, \quad n = 1, 2, \dots$$

On the other hand, we would be even after $2n$ trials if and only if we won n of these and lost n of these. Because each play of the game results in a win with probability p and a loss with probability $1 - p$, the desired probability is thus the binomial probability

$$P_{00}^{2n} = \binom{2n}{n} p^n (1 - p)^n = \frac{(2n)!}{n!n!} (p(1 - p))^n, \quad n = 1, 2, 3, \dots$$

By using an approximation, due to Stirling, which asserts that

$$n! \sim n^{n+1/2} e^{-n} \sqrt{2\pi} \quad (4.3)$$

where we say that $a_n \sim b_n$ when $\lim_{n \rightarrow \infty} a_n/b_n = 1$, we obtain

$$\binom{2n}{n} \sim \frac{(2n)^{2n+1/2} e^{-2n} \sqrt{2\pi}}{n^{2n+1} e^{-2n} (2\pi)} = \frac{2^{2n}}{\sqrt{n\pi}}$$

Hence,

$$P_{00}^{2n} \sim \frac{(4p(1 - p))^n}{\sqrt{\pi n}}$$

Now it is easy to verify, for positive a_n, b_n , that if $a_n \sim b_n$, then $\sum_n a_n < \infty$ if and only if $\sum_n b_n < \infty$. Hence, $\sum_{n=1}^{\infty} P_{00}^n$ will converge if and only if

$$\sum_{n=1}^{\infty} \frac{(4p(1 - p))^n}{\sqrt{\pi n}}$$

does. However, $4p(1-p) \leq 1$ with equality holding if and only if $p = \frac{1}{2}$. Hence, $\sum_{n=1}^{\infty} P_{00}^n = \infty$ if and only if $p = \frac{1}{2}$. Thus, the chain is recurrent when $p = \frac{1}{2}$ and transient if $p \neq \frac{1}{2}$.

When $p = \frac{1}{2}$, the preceding process is called a *symmetric random walk*. We could also look at symmetric random walks in more than one dimension. For instance, in the two-dimensional symmetric random walk the process would, at each transition, either take one step to the left, right, up, or down, each having probability $\frac{1}{4}$. That is, the state is the pair of integers (i, j) and the transition probabilities are given by

$$P_{(i,j),(i+1,j)} = P_{(i,j),(i-1,j)} = P_{(i,j),(i,j+1)} = P_{(i,j),(i,j-1)} = \frac{1}{4}$$

By using the same method as in the one-dimensional case, we now show that this Markov chain is also recurrent.

Since the preceding chain is irreducible, it follows that all states will be recurrent if state $\mathbf{0} = (0, 0)$ is recurrent. So consider $P_{\mathbf{00}}^{2n}$. Now after $2n$ steps, the chain will be back in its original location if for some i , $0 \leq i \leq n$, the $2n$ steps consist of i steps to the left, i to the right, $n-i$ up, and $n-i$ down. Since each step will be either of these four types with probability $\frac{1}{4}$, it follows that the desired probability is a multinomial probability. That is,

$$\begin{aligned} P_{\mathbf{00}}^{2n} &= \sum_{i=0}^n \frac{(2n)!}{i!i!(n-i)!(n-i)!} \left(\frac{1}{4}\right)^{2n} \\ &= \sum_{i=0}^n \frac{(2n)!}{n!n!} \frac{n!}{(n-i)!i!} \frac{n!}{(n-i)!i!} \left(\frac{1}{4}\right)^{2n} \\ &= \left(\frac{1}{4}\right)^{2n} \binom{2n}{n} \sum_{i=0}^n \binom{n}{i} \binom{n}{n-i} \\ &= \left(\frac{1}{4}\right)^{2n} \binom{2n}{n} \binom{2n}{n} \end{aligned} \tag{4.4}$$

where the last equality uses the combinatorial identity

$$\binom{2n}{n} = \sum_{i=0}^n \binom{n}{i} \binom{n}{n-i}$$

which follows upon noting that both sides represent the number of subgroups of size n one can select from a set of n white and n black objects. Now,

$$\begin{aligned} \binom{2n}{n} &= \frac{(2n)!}{n!n!} \\ &\sim \frac{(2n)^{2n+1/2} e^{-2n} \sqrt{2\pi}}{n^{2n+1} e^{-2n} (2\pi)} \quad \text{by Stirling's approximation} \end{aligned}$$

$$= \frac{4^n}{\sqrt{\pi n}}$$

Hence, from Eq. (4.4) we see that

$$P_{00}^{2n} \sim \frac{1}{\pi n}$$

which shows that $\sum_n P_{00}^{2n} = \infty$, and thus all states are recurrent.

Interestingly enough, whereas the symmetric random walks in one and two dimensions are both recurrent, all higher-dimensional symmetric random walks turn out to be transient. (For instance, the three-dimensional symmetric random walk is at each transition equally likely to move in any of six ways—either to the left, right, up, down, in, or out.) ■

Remark. For the one-dimensional random walk of Example 4.19 here is a direct argument for establishing recurrence in the symmetric case, and for determining the probability that it ever returns to state 0 in the nonsymmetric case. Let

$$\beta = P\{\text{ever return to } 0\}$$

To determine β , start by conditioning on the initial transition to obtain

$$\beta = P\{\text{ever return to } 0 | X_1 = 1\}p + P\{\text{ever return to } 0 | X_1 = -1\}(1 - p) \quad (4.5)$$

Now, let α denote the probability that the Markov chain will ever return to state 0 given that it is currently in state 1. Because the Markov chain will always increase by 1 with probability p or decrease by 1 with probability $1 - p$ no matter what its current state, note that α is also the probability that the Markov chain currently in state i will ever enter state $i - 1$, for any i . To obtain an equation for α , condition on the next transition to obtain

$$\begin{aligned} \alpha &= P\{\text{ever return} | X_1 = 1, X_2 = 0\}(1 - p) + P\{\text{ever return} | X_1 = 1, X_2 = 2\}p \\ &= 1 - p + P\{\text{ever return} | X_1 = 1, X_2 = 2\}p \\ &= 1 - p + p\alpha^2 \end{aligned}$$

where the final equation follows by noting that in order for the chain to ever go from state 2 to state 0 it must first go to state 1—and the probability of that ever happening is α —and if it does eventually go to state 1 then it must still go to state 0—and the conditional probability of that ever happening is also α . Therefore,

$$\alpha = 1 - p + p\alpha^2$$

The two roots of this equation are $\alpha = 1$ and $\alpha = (1 - p)/p$. Consequently, in the case of the symmetric random walk where $p = 1/2$ we can conclude that $\alpha = 1$. By symmetry, the probability that the symmetric random walk will ever enter state 0 given

that it is currently in state -1 is also 1 , proving that the symmetric random walk is recurrent.

Suppose now that $p > 1/2$. In this case, it can be shown (see Exercise 17 at the end of this chapter) that $P\{\text{ever return to } 0 | X_1 = -1\} = 1$. Consequently, Eq. (4.5) reduces to

$$\beta = \alpha p + 1 - p$$

Because the random walk is transient in this case we know that $\beta < 1$, showing that $\alpha \neq 1$. Therefore, $\alpha = (1 - p)/p$, yielding that

$$\beta = 2(1 - p), \quad p > 1/2$$

Similarly, when $p < 1/2$ we can show that $\beta = 2p$. Thus, in general

$$P\{\text{ever return to } 0\} = 2 \min(p, 1 - p) \quad \blacksquare$$

In our next example we use the recurrence of the symmetric random walk to construct an example where $E[\sum_{n=1}^{\infty} X_n] \neq \sum_{n=1}^{\infty} E[X_n]$.

Example 4.20. Whereas it is true that $E[\sum_{n=1}^{\infty} X_n] = \sum_{n=1}^{\infty} E[X_n]$ when the random variables $X_n, n \geq 1$ are all nonnegative, it is not true in general. For an example where it does not hold, suppose that $Y_1, Y_2, \dots$ are independent and identically distributed with $P(Y_n = 1) = P(Y_n = -1) = 1/2, n \geq 1$. Note that $E[Y_n] = 0$. Let

$$N = \min(k : Y_1 + \dots + Y_k = 1)$$

and note that it follows from the fact that the symmetric random walk is recurrent that N will be finite with probability 1. Now, let

$$I_n = \begin{cases} 1, & \text{if } n \leq N \\ 0, & \text{if } n > N \end{cases}$$

Because $I_n = 1$ if $N > n - 1$, and it is equal to 0 otherwise, it follows that the value of I_n is determined by $Y_1, \dots, Y_{n-1}$. Indeed, because N is defined to be the first time the sum of the Y 's is equal to 1, it follows that

$$\{I_n = 1\} = \{N > n - 1\} = \{Y_1 \neq 1, Y_1 + Y_2 \neq 1, \dots, Y_1 + \dots + Y_{n-1} \neq 1\}$$

which shows that I_n and Y_n are independent. Now, define $X_n, n \geq 1$, by

$$X_n = Y_n I_n = \begin{cases} Y_n, & \text{if } n \leq N \\ 0, & \text{if } n > N \end{cases}$$

By the independence of Y_n and I_n ,

$$E[X_n] = E[Y_n] E[I_n] = 0$$

showing that

$$\sum_{n=1}^{\infty} E[X_n] = 0$$

However,

$$\sum_{n=1}^{\infty} X_n = \sum_{n=1}^{\infty} Y_n I_n = \sum_{n=1}^N Y_n = 1$$

and so

$$E\left[\sum_{n=1}^{\infty} X_n\right] = 1$$

Thus,

$$E\left[\sum_{n=1}^{\infty} X_n\right] = 1 \quad \text{and} \quad \sum_{n=1}^{\infty} E[X_n] = 0 \quad \blacksquare$$

Example 4.21 (On the Ultimate Instability of the Aloha Protocol). Consider a communications facility in which the numbers of messages arriving during each of the time periods $n = 1, 2, \dots$ are independent and identically distributed random variables. Let $a_i = P\{i \text{ arrivals}\}$, and suppose that $a_0 + a_1 < 1$. Each arriving message will transmit at the end of the period in which it arrives. If exactly one message is transmitted, then the transmission is successful and the message leaves the system. However, if at any time two or more messages simultaneously transmit, then a collision is deemed to occur and these messages remain in the system. Once a message is involved in a collision it will, independently of all else, transmit at the end of each additional period with probability p —the so-called Aloha protocol (because it was first instituted at the University of Hawaii). We will show that such a system is asymptotically unstable in the sense that the number of successful transmissions will, with probability 1, be finite.

To begin let X_n denote the number of messages in the facility at the beginning of the n th period, and note that $\{X_n, n \geq 0\}$ is a Markov chain. Now for $k \geq 0$ define the indicator variables I_k by

$$I_k = \begin{cases} 1, & \text{if the first time that the chain departs state } k \text{ it} \\ & \text{directly goes to state } k-1 \\ 0, & \text{otherwise} \end{cases}$$

and let it be 0 if the system is never in state $k, k \geq 0$. (For instance, if the successive states are $0, 1, 3, 4, \dots$, then $I_3 = 0$ since when the chain first departs state 3 it goes to state 4; whereas, if they are $0, 3, 3, 2, \dots$, then $I_3 = 1$ since this time it goes to state 2.)

Now,

$$\begin{aligned}
 E\left[\sum_{k=0}^{\infty} I_k\right] &= \sum_{k=0}^{\infty} E[I_k] \\
 &= \sum_{k=0}^{\infty} P\{I_k = 1\} \\
 &\leq \sum_{k=0}^{\infty} P\{I_k = 1 | k \text{ is ever visited}\} \tag{4.6}
 \end{aligned}$$

Now, $P\{I_k = 1 | k \text{ is ever visited}\}$ is the probability that when state k is departed the next state is $k - 1$. That is, it is the conditional probability that a transition from k is to $k - 1$ given that it is not back into k , and so

$$P\{I_k = 1 | k \text{ is ever visited}\} = \frac{P_{k,k-1}}{1 - P_{k,k}}$$

Because

$$\begin{aligned}
 P_{k,k-1} &= a_0 k p (1 - p)^{k-1}, \\
 P_{k,k} &= a_0 [1 - k p (1 - p)^{k-1}] + a_1 (1 - p)^k
 \end{aligned}$$

which is seen by noting that if there are k messages present on the beginning of a day, then (a) there will be $k - 1$ at the beginning of the next day if there are no new messages that day and exactly one of the k messages transmits; and (b) there will be k at the beginning of the next day if either

- (i) there are no new messages and it is not the case that exactly one of the existing k messages transmits, or
- (ii) there is exactly one new message (which automatically transmits) and none of the other k messages transmits.

Substitution of the preceding into Eq. (4.6) yields

$$\begin{aligned}
 E\left[\sum_{k=0}^{\infty} I_k\right] &\leq \sum_{k=0}^{\infty} \frac{a_0 k p (1 - p)^{k-1}}{1 - a_0 [1 - k p (1 - p)^{k-1}] - a_1 (1 - p)^k} \\
 &< \infty
 \end{aligned}$$

where the convergence follows by noting that when k is large the denominator of the expression in the preceding sum converges to $1 - a_0$ and so the convergence or divergence of the sum is determined by whether or not the sum of the terms in the numerator converge and $\sum_{k=0}^{\infty} k (1 - p)^{k-1} < \infty$.

Hence, $E[\sum_{k=0}^{\infty} I_k] < \infty$, which implies that $\sum_{k=0}^{\infty} I_k < \infty$ with probability 1 (for if there was a positive probability that $\sum_{k=0}^{\infty} I_k$ could be ∞ , then its mean would be ∞). Hence, with probability 1, there will be only a finite number of states that are

initially departed via a successful transmission; or equivalently, there will be some finite integer N such that whenever there are N or more messages in the system, there will never again be a successful transmission. From this (and the fact that such higher states will eventually be reached—why?) it follows that, with probability 1, there will only be a finite number of successful transmissions. ■

Remark. For a (slightly less than rigorous) probabilistic proof of Stirling's approximation, let $X_1, X_2, \dots$ be independent Poisson random variables each having mean 1. Let $S_n = \sum_{i=1}^n X_i$, and note that both the mean and variance of S_n are equal to n . Now,

$$\begin{aligned}
 P\{S_n = n\} &= P\{n-1 < S_n \leq n\} \\
 &= P\{-1/\sqrt{n} < (S_n - n)/\sqrt{n} \leq 0\} \\
 &\approx \int_{-1/\sqrt{n}}^0 (2\pi)^{-1/2} e^{-x^2/2} dx && \text{when } n \text{ is large, by the} \\
 &\approx (2\pi)^{-1/2} (1/\sqrt{n}) && \text{central limit theorem} \\
 &= (2\pi n)^{-1/2}
 \end{aligned}$$

But S_n is Poisson with mean n , and so

$$P\{S_n = n\} = \frac{e^{-n} n^n}{n!}$$

Hence, for n large

$$\frac{e^{-n} n^n}{n!} \approx (2\pi n)^{-1/2}$$

or, equivalently

$$n! \approx n^{n+1/2} e^{-n} \sqrt{2\pi}$$

which is Stirling's approximation.

4.4 Long-Run Proportions and Limiting Probabilities

For pairs of states $i \neq j$, let $f_{i,j}$ denote the probability that the Markov chain, starting in state i , will ever make a transition into state j . That is,

$$f_{i,j} = P(X_n = j \text{ for some } n > 0 | X_0 = i)$$

We then have the following result.

Proposition 4.3. *If i is recurrent and i communicates with j , then $f_{i,j} = 1$.*

Proof. Because i and j communicate there is a value n such that $P_{i,j}^n > 0$. Let $X_0 = i$ and say that the first opportunity is a success if $X_n = j$, and note that the first opportunity is a success with probability $P_{i,j}^n > 0$. If the first opportunity is not a success then consider the next time (after time n) that the chain enters state i . (Because state i is recurrent we can be certain that it will eventually reenter state i after time n .) Say that the second opportunity is a success if n time periods later the Markov chain is in state j . If the second opportunity is not a success then wait until the next time the chain enters state i and say that the third opportunity is a success if n time periods later the Markov chain is in state j . Continuing in this manner, we can define an unlimited number of opportunities, each of which is a success with the same positive probability $P_{i,j}^n$. Because the number of opportunities until the first success occurs is geometric with parameter $P_{i,j}^n$, it follows that with probability 1 a success will eventually occur and so, with probability 1, state j will eventually be entered. ■

If state j is recurrent, let m_j denote the expected number of transitions that it takes the Markov chain when starting in state j to return to that state. That is, with

$$N_j = \min\{n > 0 : X_n = j\}$$

equal to the number of transitions until the Markov chain makes a transition into state j ,

$$m_j = E[N_j | X_0 = j]$$

Definition. Say that the recurrent state j is *positive recurrent* if $m_j < \infty$ and say that it is *null recurrent* if $m_j = \infty$.

Now suppose that the Markov chain is irreducible and recurrent. In this case we now show that the long-run proportion of time that the chain spends in state j is equal to $1/m_j$. That is, letting π_j denote the long-run proportion of time that the Markov chain is in state j , we have the following proposition.

Proposition 4.4. *If the Markov chain is irreducible and recurrent, then for any initial state*

$$\pi_j = 1/m_j$$

Proof. Suppose that the Markov chain starts in state i , and let T_1 denote the number of transitions until the chain enters state j ; then let T_2 denote the additional number of transitions from time T_1 until the Markov chain next enters state j ; then let T_3 denote the additional number of transitions from time $T_1 + T_2$ until the Markov chain next enters state j , and so on. Note that T_1 is finite because Proposition 4.3 tells us that with probability 1 a transition into j will eventually occur. Also, for $n \geq 2$, because T_n is the number of transitions between the $(n - 1)$ th and the n th transition into state j , it follows from the Markovian property that $T_2, T_3, \dots$ are independent and identically distributed with mean m_j . Because the n th transition into state j occurs at time $T_1 +$

$\dots + T_n$ we obtain that π_j , the long-run proportion of time that the chain is in state j , is

$$\begin{aligned}\pi_j &= \lim_{n \rightarrow \infty} \frac{n}{\sum_{i=1}^n T_i} \\ &= \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{n} \sum_{i=1}^n T_i} \\ &= \lim_{n \rightarrow \infty} \frac{1}{\frac{T_1}{n} + \frac{T_2 + \dots + T_n}{n}} \\ &= \frac{1}{m_j}\end{aligned}$$

where the last equality follows because $\lim_{n \rightarrow \infty} T_1/n = 0$ and, from the strong law of large numbers, $\lim_{n \rightarrow \infty} \frac{T_2 + \dots + T_n}{n} = \lim_{n \rightarrow \infty} \frac{T_2 + \dots + T_n}{n-1} \frac{n-1}{n} = m_j$. ■

Because $m_j < \infty$ is equivalent to $1/m_j > 0$, it follows from the preceding that state j is positive recurrent if and only if $\pi_j > 0$. We now exploit this to show that positive recurrence is a class property.

Proposition 4.5. *If i is positive recurrent and $i \leftrightarrow j$ then j is positive recurrent.*

Proof. Suppose that i is positive recurrent and that $i \leftrightarrow j$. Now, let n be such that $P_{i,j}^n > 0$. Because π_i is the long-run proportion of time that the chain is in state i , and $P_{i,j}^n$ is the long-run proportion of time when the Markov chain is in state i that it will be in state j after n transitions

$$\begin{aligned}\pi_i P_{i,j}^n &= \text{long-run proportion of time the chain is in } i \\ &\quad \text{and will be in } j \text{ after } n \text{ transitions} \\ &= \text{long-run proportion of time the chain is in } j \\ &\quad \text{and was in } i \text{ } n \text{ transitions ago} \\ &\leq \text{long-run proportion of time the chain is in } j\end{aligned}$$

Hence, $\pi_j \geq \pi_i P_{i,j}^n > 0$, showing that j is positive recurrent. ■

Remarks. (i) It follows from the preceding result that null recurrence is also a class property. For suppose that i is null recurrent and $i \leftrightarrow j$. Because i is recurrent and $i \leftrightarrow j$ we can conclude that j is recurrent. But if j were positive recurrent then by the preceding proposition i would also be positive recurrent. Because i is not positive recurrent, neither is j .

(ii) An irreducible finite state Markov chain must be positive recurrent. For we know that such a chain must be recurrent; hence, all its states are either positive recurrent or null recurrent. If they were null recurrent then all the long run proportions would equal 0, which is impossible when there are only a finite number of states. Consequently, we can conclude that the chain is positive recurrent.

- (iii) The classical example of a null recurrent Markov chain is the one dimensional symmetric random walk of Example 4.18. One way to show it is null recurrent is to argue that the mean time to return to a state is infinite. (For another argument, see Exercise 39.) To show this, let $m_{i,j}$ denote the mean number of transitions, starting in state i , until a transition into state j occurs. Now, in Example 3.16, it was shown that the mean number of transitions to go from state 1 to either 0 or n is $n - 1$, implying that

$$m_{1,0} \geq n - 1.$$

As the preceding is true for all n , we obtain upon letting $n \rightarrow \infty$ that

$$m_{1,0} = \infty.$$

Conditioning on the result of the first bet gives

$$m_{0,0} = 1 + m_{1,0} \frac{1}{2} + m_{-1,0} \frac{1}{2}.$$

Hence, $m_{0,0} = \infty$, establishing that the symmetric random walk is null recurrent. ■

To determine the long-run proportions $\{\pi_j, j \geq 1\}$, note, because π_i is the long-run proportion of transitions that come from state i , that

$$\pi_i P_{i,j} = \text{long-run proportion of transitions that go from state } i \text{ to state } j$$

Summing the preceding over all i now yields that

$$\pi_j = \sum_i \pi_i P_{i,j}$$

Indeed, the following important theorem can be proven.

Theorem 4.1. *Consider an irreducible Markov chain. If the chain is positive recurrent then the long-run proportions are the unique solution of the equations*

$$\begin{aligned} \pi_j &= \sum_i \pi_i P_{i,j}, \quad j \geq 1 \\ \sum_j \pi_j &= 1 \end{aligned} \tag{4.7}$$

Moreover, if there is no solution of the preceding linear equations, then the Markov chain is either transient or null recurrent and all $\pi_j = 0$.

Example 4.22. Consider Example 4.1, in which we assume that if it rains today, then it will rain tomorrow with probability α ; and if it does not rain today, then it will rain

tomorrow with probability β . If we say that the state is 0 when it rains and 1 when it does not rain, then by Theorem 4.1 the long-run proportions π_0 and π_1 are given by

$$\begin{aligned}\pi_0 &= \alpha\pi_0 + \beta\pi_1, \\ \pi_1 &= (1 - \alpha)\pi_0 + (1 - \beta)\pi_1, \\ \pi_0 + \pi_1 &= 1\end{aligned}$$

which yields that

$$\pi_0 = \frac{\beta}{1 + \beta - \alpha}, \quad \pi_1 = \frac{1 - \alpha}{1 + \beta - \alpha}$$

For example if $\alpha = 0.7$ and $\beta = 0.4$, then the long-run proportion of rain is $\pi_0 = \frac{4}{7} = 0.571$. ■

Example 4.23. Consider Example 4.3 in which the mood of an individual is considered as a three-state Markov chain having a transition probability matrix

$$\mathbf{P} = \begin{bmatrix} 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix}$$

In the long run, what proportion of time is the process in each of the three states?

Solution: The long run proportions $\pi_i, i = 0, 1, 2$, are obtained by solving the set of equations in Eq. (4.7). In this case these equations are

$$\begin{aligned}\pi_0 &= 0.5\pi_0 + 0.3\pi_1 + 0.2\pi_2, \\ \pi_1 &= 0.4\pi_0 + 0.4\pi_1 + 0.3\pi_2, \\ \pi_2 &= 0.1\pi_0 + 0.3\pi_1 + 0.5\pi_2, \\ \pi_0 + \pi_1 + \pi_2 &= 1\end{aligned}$$

Solving yields

$$\pi_0 = \frac{21}{62}, \quad \pi_1 = \frac{23}{62}, \quad \pi_2 = \frac{18}{62} \quad \blacksquare$$

Example 4.24 (A Model of Class Mobility). A problem of interest to sociologists is to determine the proportion of society that has an upper- or lower-class occupation. One possible mathematical model would be to assume that transitions between social classes of the successive generations in a family can be regarded as transitions of a Markov chain. That is, we assume that the occupation of a child depends only on his or her parent's occupation. Let us suppose that such a model is appropriate and that the transition probability matrix is given by

$$\mathbf{P} = \begin{bmatrix} 0.45 & 0.48 & 0.07 \\ 0.05 & 0.70 & 0.25 \\ 0.01 & 0.50 & 0.49 \end{bmatrix} \quad (4.8)$$

That is, for instance, we suppose that the child of a middle-class worker will attain an upper-, middle-, or lower-class occupation with respective probabilities 0.05, 0.70, 0.25.

The long-run proportions π_i thus satisfy

$$\pi_0 = 0.45\pi_0 + 0.05\pi_1 + 0.01\pi_2,$$

$$\pi_1 = 0.48\pi_0 + 0.70\pi_1 + 0.50\pi_2,$$

$$\pi_2 = 0.07\pi_0 + 0.25\pi_1 + 0.49\pi_2,$$

$$\pi_0 + \pi_1 + \pi_2 = 1$$

Hence,

$$\pi_0 = 0.07, \quad \pi_1 = 0.62, \quad \pi_2 = 0.31$$

In other words, a society in which social mobility between classes can be described by a Markov chain with transition probability matrix given by Eq. (4.8) has, in the long run, 7 percent of its people in upper-class jobs, 62 percent of its people in middle-class jobs, and 31 percent in lower-class jobs.

Example 4.25 (The Hardy–Weinberg Law and a Markov Chain in Genetics). Consider a large population of individuals, each of whom possesses a particular pair of genes, of which each individual gene is classified as being of type A or type a . Assume that the proportions of individuals whose gene pairs are AA , aa , or Aa are, respectively, p_0 , q_0 , and r_0 ($p_0 + q_0 + r_0 = 1$). When two individuals mate, each contributes one of his or her genes, chosen at random, to the resultant offspring. Assuming that the mating occurs at random, in that each individual is equally likely to mate with any other individual, we are interested in determining the proportions of individuals in the next generation whose genes are AA , aa , or Aa . Calling these proportions p , q , and r , they are easily obtained by focusing attention on an individual of the next generation and then determining the probabilities for the gene pair of that individual.

To begin, note that randomly choosing a parent and then randomly choosing one of its genes is equivalent to just randomly choosing a gene from the total gene population. By conditioning on the gene pair of the parent, we see that a randomly chosen gene will be type A with probability

$$\begin{aligned} P\{A\} &= P\{A|AA\}p_0 + P\{A|aa\}q_0 + P\{A|Aa\}r_0 \\ &= p_0 + r_0/2 \end{aligned}$$

Similarly, it will be type a with probability

$$P\{a\} = q_0 + r_0/2$$

Thus, under random mating a randomly chosen member of the next generation will be type AA with probability p , where

$$p = P\{A\}P\{A\} = (p_0 + r_0/2)^2$$

Similarly, the randomly chosen member will be type aa with probability

$$q = P\{a\}P\{a\} = (q_0 + r_0/2)^2$$

and will be type Aa with probability

$$r = 2P\{A\}P\{a\} = 2(p_0 + r_0/2)(q_0 + r_0/2)$$

Since each member of the next generation will independently be of each of the three gene types with probabilities p, q, r , it follows that the percentages of the members of the next generation that are of type AA, aa , or Aa are respectively p, q , and r .

If we now consider the total gene pool of this next generation, then $p + r/2$, the fraction of its genes that are A , will be unchanged from the previous generation. This follows either by arguing that the total gene pool has not changed from generation to generation or by the following simple algebra:

$$\begin{aligned} p + r/2 &= (p_0 + r_0/2)^2 + (p_0 + r_0/2)(q_0 + r_0/2) \\ &= (p_0 + r_0/2)[p_0 + r_0/2 + q_0 + r_0/2] \\ &= p_0 + r_0/2 \quad \text{since } p_0 + r_0 + q_0 = 1 \\ &= P\{A\} \end{aligned} \tag{4.9}$$

Thus, the fractions of the gene pool that are A and a are the same as in the initial generation. From this it follows that, under random mating, in all successive generations after the initial one the percentages of the population having gene pairs AA, aa , and Aa will remain fixed at the values p, q , and r . This is known as the *Hardy–Weinberg law*.

Suppose now that the gene pair population has stabilized in the percentages p, q, r , and let us follow the genetic history of a single individual and her descendants. (For simplicity, assume that each individual has exactly one offspring.) So, for a given individual, let X_n denote the genetic state of her descendant in the n th generation. The transition probability matrix of this Markov chain, namely,

$$\begin{array}{c} \begin{array}{ccc} & AA & aa & Aa \\ \begin{array}{c} AA \\ aa \\ Aa \end{array} & \left\| \begin{array}{ccc} p + \frac{r}{2} & 0 & q + \frac{r}{2} \\ 0 & q + \frac{r}{2} & p + \frac{r}{2} \\ \frac{p}{2} + \frac{r}{4} & \frac{q}{2} + \frac{r}{4} & \frac{p}{2} + \frac{q}{2} + \frac{r}{2} \end{array} \right\| \end{array} \end{array}$$

is easily verified by conditioning on the state of the randomly chosen mate. It is quite intuitive (why?) that the limiting probabilities for this Markov chain (which also equal the fractions of the individual's descendants that are in each of the three genetic states) should just be p, q , and r . To verify this we must show that they satisfy Theorem 4.1.

Because one of the equations in Theorem 4.1 is redundant, it suffices to show that

$$\begin{aligned} p &= p \left(p + \frac{r}{2} \right) + r \left(\frac{p}{2} + \frac{r}{4} \right) = \left(p + \frac{r}{2} \right)^2, \\ q &= q \left(q + \frac{r}{2} \right) + r \left(\frac{q}{2} + \frac{r}{4} \right) = \left(q + \frac{r}{2} \right)^2, \\ p + q + r &= 1 \end{aligned}$$

But this follows from Eq. (4.9), and thus the result is established. $\blacksquare$

Example 4.26. Suppose that a production process changes states in accordance with an irreducible, positive recurrent Markov chain having transition probabilities P_{ij} , $i, j = 1, \dots, n$, and suppose that certain of the states are considered acceptable and the remaining unacceptable. Let A denote the acceptable states and A^c the unacceptable ones. If the production process is said to be “up” when in an acceptable state and “down” when in an unacceptable state, determine

1. the rate at which the production process goes from up to down (that is, the rate of breakdowns);
2. the average length of time the process remains down when it goes down; and
3. the average length of time the process remains up when it goes up.

Solution: Let π_k , $k = 1, \dots, n$, denote the long-run proportions. Now for $i \in A$ and $j \in A^c$ the rate at which the process enters state j from state i is

$$\text{rate enter } j \text{ from } i = \pi_i P_{ij}$$

and so the rate at which the production process enters state j from an acceptable state is

$$\text{rate enter } j \text{ from } A = \sum_{i \in A} \pi_i P_{ij}$$

Hence, the rate at which it enters an unacceptable state from an acceptable one (which is the rate at which breakdowns occur) is

$$\text{rate breakdowns occur} = \sum_{j \in A^c} \sum_{i \in A} \pi_i P_{ij} \quad (4.10)$$

Now let $\bar{U}$ and $\bar{D}$ denote the average time the process remains up when it goes up and down when it goes down. Because there is a single breakdown every $\bar{U} + \bar{D}$ time units on the average, it follows heuristically that

$$\text{rate at which breakdowns occur} = \frac{1}{\bar{U} + \bar{D}}$$

and so from Eq. (4.10),

$$\frac{1}{\bar{U} + \bar{D}} = \sum_{j \in A^c} \sum_{i \in A} \pi_i P_{ij} \quad (4.11)$$

To obtain a second equation relating $\bar{U}$ and $\bar{D}$, consider the percentage of time the process is up, which, of course, is equal to $\sum_{i \in A} \pi_i$. However, since the process is up on the average $\bar{U}$ out of every $\bar{U} + \bar{D}$ time units, it follows (again somewhat heuristically) that the

$$\text{proportion of up time} = \frac{\bar{U}}{\bar{U} + \bar{D}}$$

and so

$$\frac{\bar{U}}{\bar{U} + \bar{D}} = \sum_{i \in A} \pi_i \quad (4.12)$$

Hence, from Eqs. (4.11) and (4.12) we obtain

$$\begin{aligned} \bar{U} &= \frac{\sum_{i \in A} \pi_i}{\sum_{j \in A^c} \sum_{i \in A} \pi_i P_{ij}}, \\ \bar{D} &= \frac{1 - \sum_{i \in A} \pi_i}{\sum_{j \in A^c} \sum_{i \in A} \pi_i P_{ij}} \\ &= \frac{\sum_{i \in A^c} \pi_i}{\sum_{j \in A^c} \sum_{i \in A} \pi_i P_{ij}} \end{aligned}$$

For example, suppose the transition probability matrix is

$$\mathbf{P} = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} & 0 \\ 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{2} \end{pmatrix}$$

where the acceptable (up) states are 1, 2 and the unacceptable (down) ones are 3, 4. The long-run proportions satisfy

$$\begin{aligned} \pi_1 &= \pi_1 \frac{1}{4} + \pi_3 \frac{1}{4} + \pi_4 \frac{1}{4}, \\ \pi_2 &= \pi_1 \frac{1}{4} + \pi_2 \frac{1}{4} + \pi_3 \frac{1}{4} + \pi_4 \frac{1}{4}, \\ \pi_3 &= \pi_1 \frac{1}{2} + \pi_2 \frac{1}{2} + \pi_3 \frac{1}{4}, \\ \pi_1 + \pi_2 + \pi_3 + \pi_4 &= 1 \end{aligned}$$

These solve to yield

$$\pi_1 = \frac{3}{16}, \quad \pi_2 = \frac{1}{4}, \quad \pi_3 = \frac{14}{48}, \quad \pi_4 = \frac{13}{48}$$

and thus

$$\begin{aligned}
 \text{rate of breakdowns} &= \pi_1(P_{13} + P_{14}) + \pi_2(P_{23} + P_{24}) \\
 &= \frac{9}{32}, \\
 \bar{U} &= \frac{14}{9} \quad \text{and} \quad \bar{D} = 2
 \end{aligned}$$

Hence, on the average, breakdowns occur about $\frac{9}{32}$ (or 28 percent) of the time. They last, on the average, 2 time units, and then there follows a stretch of (on the average) $\frac{14}{9}$ time units when the system is up. ■

The long run proportions $\pi_j, j \geq 0$, are often called *stationary* probabilities. The reason being that if the initial state is chosen according to the probabilities $\pi_j, j \geq 0$, then the probability of being in state j at any time n is also equal to π_j . That is, if

$$P\{X_0 = j\} = \pi_j, \quad j \geq 0$$

then

$$P\{X_n = j\} = \pi_j \quad \text{for all } n, j \geq 0$$

The preceding is easily proven by induction, for it is true when $n = 0$ and if we suppose it true for $n - 1$, then writing

$$\begin{aligned}
 P\{X_n = j\} &= \sum_i P\{X_n = j | X_{n-1} = i\} P\{X_{n-1} = i\} \\
 &= \sum_i P_{ij} \pi_i \quad \text{by the induction hypothesis} \\
 &= \pi_j \quad \text{by Theorem 4.1}
 \end{aligned}$$

Example 4.27. Suppose the numbers of families that check into a hotel on successive days are independent Poisson random variables with mean λ . Also suppose that the number of days that a family stays in the hotel is a geometric random variable with parameter $p, 0 < p < 1$. (Thus, a family who spent the previous night in the hotel will, independently of how long they have already spent in the hotel, check out the next day with probability p .) Also suppose that all families act independently of each other. Under these conditions it is easy to see that if X_n denotes the number of families that are checked in the hotel at the beginning of day n then $\{X_n, n \geq 0\}$ is a Markov chain. Find

- (a) the transition probabilities of this Markov chain;
- (b) $E[X_n | X_0 = i]$;
- (c) the stationary probabilities of this Markov chain.

Solution: (a) To find $P_{i,j}$, suppose there are i families checked into the hotel at the beginning of a day. Because each of these i families will stay for another day with probability $q = 1 - p$ it follows that R_i , the number of these families that remain another day, is a binomial (i, q) random variable. So, letting N be the number of new families that check in that day, we see that

$$P_{i,j} = P(R_i + N = j)$$

Conditioning on R_i and using that N is Poisson with mean λ , we obtain

$$\begin{aligned}
 P_{i,j} &= \sum_{k=0}^i P(R_i + N = j | R_i = k) \binom{i}{k} q^k p^{i-k} \\
 &= \sum_{k=0}^i P(N = j - k | R_i = k) \binom{i}{k} q^k p^{i-k} \\
 &= \sum_{k=0}^{\min(i,j)} P(N = j - k) \binom{i}{k} q^k p^{i-k} \\
 &= \sum_{k=0}^{\min(i,j)} e^{-\lambda} \frac{\lambda^{j-k}}{(j-k)!} \binom{i}{k} q^k p^{i-k}
 \end{aligned}$$

(b) Using the preceding representation $R_i + N$ for the next state from state i , we see that

$$E[X_n | X_{n-1} = i] = E[R_i + N] = iq + \lambda$$

Consequently,

$$E[X_n | X_{n-1}] = X_{n-1}q + \lambda$$

Taking expectations of both sides yields

$$E[X_n] = \lambda + qE[X_{n-1}]$$

Iterating the preceding gives

$$\begin{aligned}
 E[X_n] &= \lambda + qE[X_{n-1}] \\
 &= \lambda + q(\lambda + qE[X_{n-2}]) \\
 &= \lambda + q\lambda + q^2E[X_{n-2}] \\
 &= \lambda + q\lambda + q^2(\lambda + qE[X_{n-3}]) \\
 &= \lambda + q\lambda + q^2\lambda + q^3E[X_{n-3}]
 \end{aligned}$$

showing that

$$E[X_n] = \lambda \left(1 + q + q^2 + \dots + q^{n-1} \right) + q^n E[X_0]$$

and yielding the result

$$E[X_n | X_0 = i] = \frac{\lambda(1 - q^n)}{p} + q^n i$$

(c) To find the stationary probabilities we will not directly use the complicated transition probabilities derived in part (a). Rather we will make use of the fact that

the stationary probability distribution is the only distribution on the initial state that results in the next state having the same distribution. Now, suppose that the initial state X_0 has a Poisson distribution with mean α . That is, assume that the number of families initially in the hotel is Poisson with mean α . Let R denote the number of these families that remain in the hotel at the beginning of the next day. Then, using the result of Example 3.24 that if each of a Poisson distributed (with mean α) number of events occurs with probability q , then the total number of these events that occur is Poisson distributed with mean αq , it follows that R is a Poisson random variable with mean αq . In addition, the number of new families that check in during the day, call it N , is Poisson with mean λ , and is independent of R . Hence, since the sum of independent Poisson random variables is also Poisson distributed, it follows that $R + N$, the number of guests at the beginning of the next day, is Poisson with mean $\lambda + \alpha q$. Consequently, if we choose α so that

$$\alpha = \lambda + \alpha q$$

then the distribution of X_1 would be the same as that of X_0 . But this means that when the initial distribution of X_0 is Poisson with mean $\alpha = \frac{\lambda}{p}$, then so is the distribution of X_1 , implying that this is the stationary distribution. That is, the stationary probabilities are

$$\pi_i = e^{-\lambda/p} (\lambda/p)^i / i!, \quad i \geq 0$$

The preceding model has an important generalization. Namely, consider an organization whose workers are of r distinct types. For instance, the organization could be a law firm and its lawyers could either be juniors, associates, or partners. Suppose that a worker who is currently type i will in the next period become type j with probability $q_{i,j}$ for $j = 1, \dots, r$ or will leave the organization with probability $1 - \sum_{j=1}^r q_{i,j}$. In addition, suppose that new workers are hired each period, and that the numbers of types $1, \dots, r$ workers hired are independent Poisson random variables with means $\lambda_1, \dots, \lambda_r$. If we let $\mathbf{X}_n = (X_n(1), \dots, X_n(r))$, where $X_n(i)$ is the number of type i workers in the organization at the beginning of period n , then $\mathbf{X}_n, n \geq 0$ is a Markov chain. To compute its stationary probability distribution, suppose that the initial state is chosen so that the number of workers of different types are independent Poisson random variables, with α_i being the mean number of type i workers. That is, suppose that $X_0(1), \dots, X_0(r)$ are independent Poisson random variables with respective means $\alpha_1, \dots, \alpha_r$. Also, let $N_j, j = 1, \dots, r$, be the number of new type j workers hired during the initial period. Now, fix i , and for $j = 1, \dots, r$, let $M_i(j)$ be the number of the $X_0(i)$ type i workers who become type j in the next period. Then because each of the Poisson number $X_0(i)$ of type i workers will independently become type j with probability $q_{i,j}, j = 1, \dots, r$, it follows from the remarks following Example 3.24 that $M_i(1), \dots, M_i(r)$ are independent Poisson random variables with $M_i(j)$ having mean $\alpha_i q_{i,j}$. Because $X_0(1), \dots, X_0(r)$ are, by assumption, independent, we can also conclude that the random variables $M_i(j), i, j = 1, \dots, r$ are all independent. Because the sum of independent Poisson random variables is also Poisson

distributed, the preceding yields that the random variables

$$X_1(j) = N_j + \sum_{i=1}^r M_i(j), \quad j = 1, \dots, r$$

are independent Poisson random variables with means

$$E[X_1(j)] = \lambda_j + \sum_{i=1}^r \alpha_i q_{i,j}$$

Hence, if $\alpha_1, \dots, \alpha_r$ satisfied

$$\alpha_j = \lambda_j + \sum_{i=1}^r \alpha_i q_{i,j}, \quad j = 1, \dots, r$$

then $\mathbf{X}_1$ would have the same distribution as $\mathbf{X}_0$. Consequently, if we let $\alpha_1^o, \dots, \alpha_r^o$ be such that

$$\alpha_j^o = \lambda_j + \sum_{i=1}^r \alpha_i^o q_{i,j}, \quad j = 1, \dots, r$$

then the stationary distribution of the Markov chain is the distribution that takes the number of workers in each type to be independent Poisson random variables with means $\alpha_1^o, \dots, \alpha_r^o$. That is, the long run proportions are

$$\pi_{k_1, \dots, k_r} = \prod_{i=1}^r e^{-\alpha_i^o} (\alpha_i^o)^{k_i} / k_i!$$

It can be shown that there will be such values $\alpha_j^o, j = 1, \dots, r$, provided that, with probability 1, each worker eventually leaves the organization. Also, because there is a unique stationary distribution, there can only be one such set of values. ■

The following example exploits the relationship $m_i = 1/\pi_i$, which states that the mean time between visits to a state is the inverse of the long run proportion of time the chain is in that state, to obtain a method for computing the mean time until a specified pattern appears when the data constitutes the successive states of a Markov chain.

Example 4.28 (Mean Pattern Times in Markov Chain Generated Data). Consider an irreducible Markov chain $\{X_n, n \geq 0\}$ with transition probabilities $P_{i,j}$ and stationary probabilities $\pi_j, j \geq 0$. Starting in state r , we are interested in determining the expected number of transitions until the pattern $i_1, i_2, \dots, i_k$ appears. That is, with

$$N(i_1, i_2, \dots, i_k) = \min\{n \geq k: X_{n-k+1} = i_1, \dots, X_n = i_k\}$$

we are interested in

$$E[N(i_1, i_2, \dots, i_k) | X_0 = r]$$

Note that even if $i_1 = r$, the initial state X_0 is not considered part of the pattern sequence.

Let $\mu(i, i_1)$ be the mean number of transitions for the chain to enter state i_1 , given that the initial state is i , $i \geq 0$. The quantities $\mu(i, i_1)$ can be determined as the solution of the following set of equations, obtained by conditioning on the first transition out of state i :

$$\mu(i, i_1) = 1 + \sum_{j \neq i_1} P_{i,j} \mu(j, i_1), \quad i \geq 0$$

For the Markov chain $\{X_n, n \geq 0\}$ associate a corresponding Markov chain, which we will refer to as the k -chain, whose state at any time is the sequence of the most recent k states of the original chain. (For instance, if $k = 3$ and $X_2 = 4, X_3 = 1, X_4 = 1$, then the state of the k -chain at time 4 is $(4, 1, 1)$.) Let $\pi(j_1, \dots, j_k)$ be the stationary probabilities for the k -chain. Because $\pi(j_1, \dots, j_k)$ is the proportion of time that the state of the original Markov chain k units ago was j_1 and the following $k - 1$ states, in sequence, were $j_2, \dots, j_k$, we can conclude that

$$\pi(j_1, \dots, j_k) = \pi_{j_1} P_{j_1, j_2} \cdots P_{j_{k-1}, j_k}$$

Moreover, because the mean number of transitions between successive visits of the k -chain to the state $i_1, i_2, \dots, i_k$ is equal to the inverse of the stationary probability of that state, we have that

$$\begin{aligned} E[\text{number of transitions between visits to } i_1, i_2, \dots, i_k] \\ = \frac{1}{\pi(i_1, \dots, i_k)} \end{aligned} \quad (4.13)$$

Let $A(i_1, \dots, i_m)$ be the additional number of transitions needed until the pattern appears, given that the first m transitions have taken the chain into states $X_1 = i_1, \dots, X_m = i_m$.

We will now consider whether the pattern has overlaps, where we say that the pattern $i_1, i_2, \dots, i_k$ has an overlap of size j , $j < k$, if the sequence of its final j elements is the same as that of its first j elements. That is, it has an overlap of size j if

$$(i_{k-j+1}, \dots, i_k) = (i_1, \dots, i_j), \quad j < k$$

Case 1. The pattern $i_1, i_2, \dots, i_k$ has no overlaps.

Because there is no overlap, Eq. (4.13) yields

$$E[N(i_1, i_2, \dots, i_k) | X_0 = i_k] = \frac{1}{\pi(i_1, \dots, i_k)}$$

Because the time until the pattern occurs is equal to the time until the chain enters state i_1 plus the additional time, we may write

$$E[N(i_1, i_2, \dots, i_k) | X_0 = i_k] = \mu(i_k, i_1) + E[A(i_1)]$$

The preceding two equations imply

$$E[A(i_1)] = \frac{1}{\pi(i_1, \dots, i_k)} - \mu(i_k, i_1)$$

Using that

$$E[N(i_1, i_2, \dots, i_k) | X_0 = r] = \mu(r, i_1) + E[A(i_1)]$$

gives the result

$$E[N(i_1, i_2, \dots, i_k) | X_0 = r] = \mu(r, i_1) + \frac{1}{\pi(i_1, \dots, i_k)} - \mu(i_k, i_1)$$

where

$$\pi(i_1, \dots, i_k) = \pi_{i_1} P_{i_1, i_2} \cdots P_{i_{k-1}, i_k}$$

Case 2. Now suppose that the pattern has overlaps and let its largest overlap be of size s . In this case the number of transitions between successive visits of the k -chain to the state $i_1, i_2, \dots, i_k$ is equal to the additional number of transitions of the original chain until the pattern appears given that it has already made s transitions with the results $X_1 = i_1, \dots, X_s = i_s$. Therefore, from Eq. (4.13)

$$E[A(i_1, \dots, i_s)] = \frac{1}{\pi(i_1, \dots, i_k)}$$

But because

$$N(i_1, i_2, \dots, i_k) = N(i_1, \dots, i_s) + A(i_1, \dots, i_s)$$

we have

$$E[N(i_1, i_2, \dots, i_k) | X_0 = r] = E[N(i_1, i_2, \dots, i_s) | X_0 = r] + \frac{1}{\pi(i_1, \dots, i_k)}$$

We can now repeat the same procedure on the pattern $i_1, \dots, i_s$, continuing to do so until we reach one that has no overlap, and then apply the result from Case 1.

For instance, suppose the desired pattern is 1, 2, 3, 1, 2, 3, 1, 2. Then

$$\begin{aligned} E[N(1, 2, 3, 1, 2, 3, 1, 2) | X_0 = r] &= E[N(1, 2, 3, 1, 2) | X_0 = r] \\ &\quad + \frac{1}{\pi(1, 2, 3, 1, 2, 3, 1, 2)} \end{aligned}$$

Because the largest overlap of the pattern (1, 2, 3, 1, 2) is of size 2, the same argument as in the preceding gives

$$E[N(1, 2, 3, 1, 2) | X_0 = r] = E[N(1, 2) | X_0 = r] + \frac{1}{\pi(1, 2, 3, 1, 2)}$$

Because the pattern (1, 2) has no overlap, we obtain from Case 1 that

$$E[N(1, 2)|X_0 = r] = \mu(r, 1) + \frac{1}{\pi(1, 2)} - \mu(2, 1)$$

Putting it together yields

$$\begin{aligned} E[N(1, 2, 3, 1, 2, 3, 1, 2)|X_0 = r] &= \mu(r, 1) + \frac{1}{\pi_1 P_{1,2}} - \mu(2, 1) \\ &\quad + \frac{1}{\pi_1 P_{1,2}^2 P_{2,3} P_{3,1}} + \frac{1}{\pi_1 P_{1,2}^3 P_{2,3}^2 P_{3,1}^2} \end{aligned}$$

If the generated data is a sequence of independent and identically distributed random variables, with each value equal to j with probability P_j , then the Markov chain has $P_{i,j} = P_j$. In this case, $\pi_j = P_j$. Also, because the time to go from state i to state j is a geometric random variable with parameter P_j , we have $\mu(i, j) = 1/P_j$. Thus, the expected number of data values that need be generated before the pattern 1, 2, 3, 1, 2, 3, 1, 2 appears would be

$$\begin{aligned} &\frac{1}{P_1} + \frac{1}{P_1 P_2} - \frac{1}{P_1} + \frac{1}{P_1^2 P_2^2 P_3} + \frac{1}{P_1^3 P_2^3 P_3^2} \\ &= \frac{1}{P_1 P_2} + \frac{1}{P_1^2 P_2^2 P_3} + \frac{1}{P_1^3 P_2^3 P_3^2} \end{aligned} \quad \blacksquare$$

The following result is quite useful.

Proposition 4.6. *Let $\{X_n, n \geq 1\}$ be an irreducible Markov chain with stationary probabilities $\pi_j, j \geq 0$, and let r be a bounded function on the state space. Then, with probability 1,*

$$\lim_{N \rightarrow \infty} \frac{\sum_{n=1}^N r(X_n)}{N} = \sum_{j=0}^{\infty} r(j) \pi_j$$

Proof. If we let $a_j(N)$ be the amount of time the Markov chain spends in state j during time periods $1, \dots, N$, then

$$\sum_{n=1}^N r(X_n) = \sum_{j=0}^{\infty} a_j(N) r(j)$$

Since $a_j(N)/N \rightarrow \pi_j$ the result follows from the preceding upon dividing by N and then letting $N \rightarrow \infty$. $\blacksquare$

If we suppose that we earn a reward $r(j)$ whenever the chain is in state j , then Proposition 4.6 states that our average reward per unit time is $\sum_j r(j) \pi_j$.

Example 4.29. For the four state Bonus Malus automobile insurance system specified in Example 4.7, find the average annual premium paid by a policyholder whose yearly number of claims is a Poisson random variable with mean $1/2$.

Solution: With $a_k = e^{-1/2} \frac{(1/2)^k}{k!}$, we have

$$a_0 = 0.6065, \quad a_1 = 0.3033, \quad a_2 = 0.0758$$

Therefore, the Markov chain of successive states has the following transition probability matrix:

$$\begin{pmatrix} 0.6065 & 0.3033 & 0.0758 & 0.0144 \\ 0.6065 & 0.0000 & 0.3033 & 0.0902 \\ 0.0000 & 0.6065 & 0.0000 & 0.3935 \\ 0.0000 & 0.0000 & 0.6065 & 0.3935 \end{pmatrix}$$

The stationary probabilities are given as the solution of

$$\pi_1 = 0.6065\pi_1 + 0.6065\pi_2,$$

$$\pi_2 = 0.3033\pi_1 + 0.6065\pi_3,$$

$$\pi_3 = 0.0758\pi_1 + 0.3033\pi_2 + 0.6065\pi_4,$$

$$\pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$$

Rewriting the first three of these equations gives

$$\pi_2 = \frac{1 - 0.6065}{0.6065}\pi_1,$$

$$\pi_3 = \frac{\pi_2 - 0.3033\pi_1}{0.6065},$$

$$\pi_4 = \frac{\pi_3 - 0.0758\pi_1 - 0.3033\pi_2}{0.6065}$$

or

$$\pi_2 = 0.6488\pi_1,$$

$$\pi_3 = 0.5697\pi_1,$$

$$\pi_4 = 0.4900\pi_1$$

Using that $\sum_{i=1}^4 \pi_i = 1$ gives the solution (rounded to four decimal places)

$$\pi_1 = 0.3692, \quad \pi_2 = 0.2395, \quad \pi_3 = 0.2103, \quad \pi_4 = 0.1809$$

Therefore, the average annual premium paid is

$$200\pi_1 + 250\pi_2 + 400\pi_3 + 600\pi_4 = 326.375$$



4.4.1 Limiting Probabilities

In Example 4.8 we considered a two-state Markov chain with transition probability matrix

$$\mathbf{P} = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$$

and showed that

$$\mathbf{P}^{(4)} = \begin{pmatrix} 0.5749 & 0.4251 \\ 0.5668 & 0.4332 \end{pmatrix}$$

From this it follows that $\mathbf{P}^{(8)} = \mathbf{P}^{(4)} \cdot \mathbf{P}^{(4)}$ is given (to three significant places) by

$$\mathbf{P}^{(8)} = \begin{pmatrix} 0.571 & 0.429 \\ 0.571 & 0.429 \end{pmatrix}$$

Note that the matrix $\mathbf{P}^{(8)}$ is almost identical to the matrix $\mathbf{P}^{(4)}$, and that each of the rows of $\mathbf{P}^{(8)}$ has almost identical values. Indeed, it seems that P_{ij}^n is converging to some value as $n \rightarrow \infty$, with this value not depending on i . Moreover, in Example 4.22 we showed that the long-run proportions for this chain are $\pi_0 = 4/7 \approx .571$, $\pi_1 = 3/7 \approx .429$, thus making it appear that these long-run proportions may also be limiting probabilities. Although this is indeed the case for the preceding chain, it is not always true that the long-run proportions are also limiting probabilities. To see why not, consider a two-state Markov chain having

$$P_{0,1} = P_{1,0} = 1$$

Because this Markov chain continually alternates between states 0 and 1, the long-run proportions of time it spends in these states are

$$\pi_0 = \pi_1 = 1/2$$

However,

$$P_{0,0}^n = \begin{cases} 1, & \text{if } n \text{ is even} \\ 0, & \text{if } n \text{ is odd} \end{cases}$$

and so $P_{0,0}^n$ does not have a limiting value as n goes to infinity. In general, a chain that can only return to a state in a multiple of $d > 1$ steps (where $d = 2$ in the preceding example) is said to be *periodic* and does not have limiting probabilities. However, for an irreducible chain that is not periodic, and such chains are called *aperiodic*, the limiting probabilities will always exist and will not depend on the initial state. Moreover, the limiting probability that the chain will be in state j will equal π_j , the long-run proportion of time the chain is in state j . That the limiting probabilities, when they exist, will equal the long-run proportions can be seen by letting

$$\alpha_j = \lim_{n \rightarrow \infty} P(X_n = j)$$

and using that

$$P(X_{n+1} = j) = \sum_{i=0}^{\infty} P(X_{n+1} = j | X_n = i) P(X_n = i) = \sum_{i=0}^{\infty} P_{ij} P(X_n = i)$$

and

$$1 = \sum_{i=0}^{\infty} P(X_n = i)$$

Letting $n \rightarrow \infty$ in the preceding two equations yields, upon assuming that we can bring the limit inside the summation, that

$$\begin{aligned} \alpha_j &= \sum_{i=0}^{\infty} \alpha_i P_{ij} \\ 1 &= \sum_{i=0}^{\infty} \alpha_i \end{aligned}$$

Hence, $\{\alpha_j, j \geq 0\}$ satisfies the equations for which $\{\pi_j, j \geq 0\}$ is the unique solution, showing that $\alpha_j = \pi_j, j \geq 0$.

An irreducible, positive recurrent, aperiodic Markov chain is said to be *ergodic*.

4.5 Some Applications

4.5.1 The Gambler's Ruin Problem

Consider a gambler who at each play of the game has probability p of winning one unit and probability $q = 1 - p$ of losing one unit. Assuming that successive plays of the game are independent, what is the probability that, starting with i units, the gambler's fortune will reach N before reaching 0?

If we let X_n denote the player's fortune at time n , then the process $\{X_n, n = 0, 1, 2, \dots\}$ is a Markov chain with transition probabilities

$$\begin{aligned} P_{00} &= P_{NN} = 1, \\ P_{i,i+1} &= p = 1 - P_{i,i-1}, \quad i = 1, 2, \dots, N-1 \end{aligned}$$

This Markov chain has three classes, namely, $\{0\}$, $\{1, 2, \dots, N-1\}$, and $\{N\}$; the first and third class being recurrent and the second transient. Since each transient state is visited only finitely often, it follows that, after some finite amount of time, the gambler will either attain his goal of N or go broke.

Let $P_i, i = 0, 1, \dots, N$, denote the probability that, starting with i , the gambler's fortune will eventually reach N . By conditioning on the outcome of the initial play of

the game we obtain

$$P_i = pP_{i+1} + qP_{i-1}, \quad i = 1, 2, \dots, N-1$$

or equivalently, since $p + q = 1$,

$$pP_i + qP_i = pP_{i+1} + qP_{i-1}$$

or

$$P_{i+1} - P_i = \frac{q}{p}(P_i - P_{i-1}), \quad i = 1, 2, \dots, N-1$$

Hence, since $P_0 = 0$, we obtain from the preceding line that

$$\begin{aligned} P_2 - P_1 &= \frac{q}{p}(P_1 - P_0) = \frac{q}{p}P_1, \\ P_3 - P_2 &= \frac{q}{p}(P_2 - P_1) = \left(\frac{q}{p}\right)^2 P_1, \\ &\vdots \\ P_i - P_{i-1} &= \frac{q}{p}(P_{i-1} - P_{i-2}) = \left(\frac{q}{p}\right)^{i-1} P_1, \\ &\vdots \\ P_N - P_{N-1} &= \left(\frac{q}{p}\right)(P_{N-1} - P_{N-2}) = \left(\frac{q}{p}\right)^{N-1} P_1 \end{aligned}$$

Adding the first $i-1$ of these equations yields

$$P_i - P_1 = P_1 \left[\left(\frac{q}{p}\right) + \left(\frac{q}{p}\right)^2 + \dots + \left(\frac{q}{p}\right)^{i-1} \right]$$

or

$$P_i = \begin{cases} \frac{1 - (q/p)^i}{1 - (q/p)} P_1, & \text{if } \frac{q}{p} \neq 1 \\ iP_1, & \text{if } \frac{q}{p} = 1 \end{cases}$$

Now, using the fact that $P_N = 1$, we obtain

$$P_1 = \begin{cases} \frac{1 - (q/p)}{1 - (q/p)^N}, & \text{if } p \neq \frac{1}{2} \\ \frac{1}{N}, & \text{if } p = \frac{1}{2} \end{cases}$$

and hence

$$P_i = \begin{cases} \frac{1 - (q/p)^i}{1 - (q/p)^N}, & \text{if } p \neq \frac{1}{2} \\ \frac{i}{N}, & \text{if } p = \frac{1}{2} \end{cases} \quad (4.14)$$

Note that, as $N \rightarrow \infty$,

$$P_i \rightarrow \begin{cases} 1 - \left(\frac{q}{p}\right)^i, & \text{if } p > \frac{1}{2} \\ 0, & \text{if } p \leq \frac{1}{2} \end{cases}$$

Thus, if $p > \frac{1}{2}$, there is a positive probability that the gambler's fortune will increase indefinitely; while if $p \leq \frac{1}{2}$, the gambler will, with probability 1, go broke against an infinitely rich adversary.

Example 4.30. Suppose Max and Patty decide to flip pennies; the one coming closest to the wall wins. Patty, being the better player, has a probability 0.6 of winning on each flip. (a) If Patty starts with five pennies and Max with ten, what is the probability that Patty will wipe Max out? (b) What if Patty starts with 10 and Max with 20?

Solution: (a) The desired probability is obtained from Eq. (4.14) by letting $i = 5$, $N = 15$, and $p = 0.6$. Hence, the desired probability is

$$\frac{1 - \left(\frac{2}{3}\right)^5}{1 - \left(\frac{2}{3}\right)^{15}} \approx 0.87$$

(b) The desired probability is

$$\frac{1 - \left(\frac{2}{3}\right)^{10}}{1 - \left(\frac{2}{3}\right)^{30}} \approx 0.98$$

■

For an application of the gambler's ruin problem to drug testing, suppose that two new drugs have been developed for treating a certain disease. Drug i has a cure rate P_i , $i = 1, 2$, in the sense that each patient treated with drug i will be cured with probability P_i . These cure rates, however, are not known, and suppose we are interested in a method for deciding whether $P_1 > P_2$ or $P_2 > P_1$. To decide upon one of these alternatives, consider the following test: Pairs of patients are treated sequentially with one member of the pair receiving drug 1 and the other drug 2. The results for each pair are determined, and the testing stops when the cumulative number of cures using one of the drugs exceeds the cumulative number of cures when using the other by some

fixed predetermined number. More formally, let

$$X_j = \begin{cases} 1, & \text{if the patient in the } j\text{th pair to receive drug number 1 is cured} \\ 0, & \text{otherwise} \end{cases}$$

$$Y_j = \begin{cases} 1, & \text{if the patient in the } j\text{th pair to receive drug number 2 is cured} \\ 0, & \text{otherwise} \end{cases}$$

For a predetermined positive integer M the test stops after pair N where N is the first value of n such that either

$$X_1 + \cdots + X_n - (Y_1 + \cdots + Y_n) = M$$

or

$$X_1 + \cdots + X_n - (Y_1 + \cdots + Y_n) = -M$$

In the former case we then assert that $P_1 > P_2$, and in the latter that $P_2 > P_1$.

In order to help ascertain whether the preceding is a good test, one thing we would like to know is the probability of it leading to an incorrect decision. That is, for given P_1 and P_2 where $P_1 > P_2$, what is the probability that the test will incorrectly assert that $P_2 > P_1$? To determine this probability, note that after each pair is checked the cumulative difference of cures using drug 1 versus drug 2 will either go up by 1 with probability $P_1(1 - P_2)$ —since this is the probability that drug 1 leads to a cure and drug 2 does not—or go down by 1 with probability $(1 - P_1)P_2$, or remain the same with probability $P_1P_2 + (1 - P_1)(1 - P_2)$. Hence, if we only consider those pairs in which the cumulative difference changes, then the difference will go up 1 with probability

$$p = P\{\text{up 1} | \text{up 1 or down 1}\}$$

$$= \frac{P_1(1 - P_2)}{P_1(1 - P_2) + (1 - P_1)P_2}$$

and down 1 with probability

$$q = 1 - p = \frac{P_2(1 - P_1)}{P_1(1 - P_2) + (1 - P_1)P_2}$$

Hence, the probability that the test will assert that $P_2 > P_1$ is equal to the probability that a gambler who wins each (one unit) bet with probability p will go down M before going up M . But Eq. (4.14) with $i = M$, $N = 2M$, shows that this probability is given by

$$P\{\text{test asserts that } P_2 > P_1\} = 1 - \frac{1 - (q/p)^M}{1 - (q/p)^{2M}}$$

$$= \frac{1}{1 + (p/q)^M}$$

Thus, for instance, if $P_1 = 0.6$ and $P_2 = 0.4$ then the probability of an incorrect decision is 0.017 when $M = 5$ and reduces to 0.0003 when $M = 10$.

4.5.2 A Model for Algorithmic Efficiency

The following optimization problem is called a linear program:

$$\begin{aligned} &\text{minimize } \mathbf{c}\mathbf{x}, \\ &\text{subject to } \mathbf{A}\mathbf{x} = \mathbf{b}, \\ &\quad \mathbf{x} \geq \mathbf{0} \end{aligned}$$

where $\mathbf{A}$ is an $m \times n$ matrix of fixed constants; $\mathbf{c} = (c_1, \dots, c_n)$ and $\mathbf{b} = (b_1, \dots, b_m)$ are vectors of fixed constants; and $\mathbf{x} = (x_1, \dots, x_n)$ is the n -vector of nonnegative values that is to be chosen to minimize $\mathbf{c}\mathbf{x} \equiv \sum_{i=1}^n c_i x_i$. Supposing that $n > m$, it can be shown that the optimal $\mathbf{x}$ can always be chosen to have at least $n - m$ components equal to 0—that is, it can always be taken to be one of the so-called extreme points of the feasibility region.

The simplex algorithm solves this linear program by moving from an extreme point of the feasibility region to a better (in terms of the objective function $\mathbf{c}\mathbf{x}$) extreme point (via the pivot operation) until the optimal is reached. Because there can be as many as $N \equiv \binom{n}{m}$ such extreme points, it would seem that this method might take many iterations, but, surprisingly to some, this does not appear to be the case in practice.

To obtain a feel for whether or not the preceding statement is surprising, let us consider a simple probabilistic (Markov chain) model as to how the algorithm moves along the extreme points. Specifically, we will suppose that if at any time the algorithm is at the j th best extreme point then after the next pivot the resulting extreme point is equally likely to be any of the $j - 1$ best. Under this assumption, we show that the time to get from the N th best to the best extreme point has approximately, for large N , a normal distribution with mean and variance equal to the logarithm (base e) of N .

Consider a Markov chain for which $P_{11} = 1$ and

$$P_{ij} = \frac{1}{i-1}, \quad j = 1, \dots, i-1, \quad i > 1$$

and let T_i denote the number of transitions needed to go from state i to state 1. A recursive formula for $E[T_i]$ can be obtained by conditioning on the initial transition:

$$E[T_i] = 1 + \frac{1}{i-1} \sum_{j=1}^{i-1} E[T_j]$$

Starting with $E[T_1] = 0$, we successively see that

$$\begin{aligned} E[T_2] &= 1, \\ E[T_3] &= 1 + \frac{1}{2}, \end{aligned}$$

$$E[T_4] = 1 + \frac{1}{3}(1 + 1 + \frac{1}{2}) = 1 + \frac{1}{2} + \frac{1}{3}$$

and it is not difficult to guess and then prove inductively that

$$E[T_i] = \sum_{j=1}^{i-1} 1/j$$

However, to obtain a more complete description of T_N , we will use the representation

$$T_N = \sum_{j=1}^{N-1} I_j$$

where

$$I_j = \begin{cases} 1, & \text{if the process ever enters } j \\ 0, & \text{otherwise} \end{cases}$$

The importance of the preceding representation stems from the following:

Proposition 4.7. $I_1, \dots, I_{N-1}$ are independent and

$$P\{I_j = 1\} = 1/j, \quad 1 \leq j \leq N-1$$

Proof. Given $I_{j+1}, \dots, I_N$, let $n = \min\{i: i > j, I_i = 1\}$ denote the lowest numbered state, greater than j , that is entered. Thus we know that the process enters state n and the next state entered is one of the states $1, 2, \dots, j$. Hence, as the next state from state n is equally likely to be any of the lower number states $1, 2, \dots, n-1$ we see that

$$P\{I_j = 1 | I_{j+1}, \dots, I_N\} = \frac{1/(n-1)}{j/(n-1)} = 1/j$$

Hence, $P\{I_j = 1\} = 1/j$, and independence follows since the preceding conditional probability does not depend on $I_{j+1}, \dots, I_N$. ■

Corollary 4.8. (i) $E[T_N] = \sum_{j=1}^{N-1} 1/j$.

(ii) $\text{Var}(T_N) = \sum_{j=1}^{N-1} (1/j)(1 - 1/j)$.

(iii) For N large, T_N has approximately a normal distribution with mean $\log N$ and variance $\log N$.

Proof. Parts (i) and (ii) follow from Proposition 4.7 and the representation $T_N = \sum_{j=1}^{N-1} I_j$. Part (iii) follows from the central limit theorem since

$$\int_1^N \frac{dx}{x} < \sum_{j=1}^{N-1} 1/j < 1 + \int_1^{N-1} \frac{dx}{x}$$

or

$$\log N < \sum_{j=1}^{N-1} 1/j < 1 + \log(N-1)$$

and so

$$\log N \approx \sum_{j=1}^{N-1} 1/j$$

■

Returning to the simplex algorithm, if we assume that n , m , and $n - m$ are all large, we have by Stirling's approximation that

$$N = \binom{n}{m} \sim \frac{n^{n+1/2}}{(n-m)^{n-m+1/2} m^{m+1/2} \sqrt{2\pi}}$$

and so, letting $c = n/m$,

$$\begin{aligned} \log N \sim (mc + \tfrac{1}{2}) \log(mc) - (m(c-1) + \tfrac{1}{2}) \log(m(c-1)) \\ - (m + \tfrac{1}{2}) \log m - \tfrac{1}{2} \log(2\pi) \end{aligned}$$

or

$$\log N \sim m \left[c \log \frac{c}{c-1} + \log(c-1) \right]$$

Now, as $\lim_{x \rightarrow \infty} x \log[x/(x-1)] = 1$, it follows that, when c is large,

$$\log N \sim m[1 + \log(c-1)]$$

Thus, for instance, if $n = 8000$, $m = 1000$, then the number of necessary transitions is approximately normally distributed with mean and variance equal to $1000(1 + \log 7) \approx 3000$. Hence, the number of necessary transitions would be roughly between

$$3000 \pm 2\sqrt{3000} \quad \text{or roughly} \quad 3000 \pm 110$$

95 percent of the time.

4.5.3 Using a Random Walk to Analyze a Probabilistic Algorithm for the Satisfiability Problem

Consider a Markov chain with states $0, 1, \dots, n$ having

$$P_{0,1} = 1, \quad P_{i,i+1} = p, \quad P_{i,i-1} = q = 1 - p, \quad 1 \leq i < n$$

and suppose that we are interested in studying the time that it takes for the chain to go from state 0 to state n . One approach to obtaining the mean time to reach state n

would be to let m_i denote the mean time to go from state i to state n , $i = 0, \dots, n-1$. If we then condition on the initial transition, we obtain the following set of equations:

$$\begin{aligned} m_0 &= 1 + m_1, \\ m_i &= E[\text{time to reach } n | \text{next state is } i+1]p \\ &\quad + E[\text{time to reach } n | \text{next state is } i-1]q \\ &= (1 + m_{i+1})p + (1 + m_{i-1})q \\ &= 1 + pm_{i+1} + qm_{i-1}, \quad i = 1, \dots, n-1 \end{aligned}$$

Whereas the preceding equations can be solved for m_i , $i = 0, \dots, n-1$, we do not pursue their solution; we instead make use of the special structure of the Markov chain to obtain a simpler set of equations. To start, let N_i denote the number of additional transitions that it takes the chain when it first enters state i until it enters state $i+1$. By the Markovian property, it follows that these random variables N_i , $i = 0, \dots, n-1$ are independent. Also, we can express $N_{0,n}$, the number of transitions that it takes the chain to go from state 0 to state n , as

$$N_{0,n} = \sum_{i=0}^{n-1} N_i \quad (4.15)$$

Letting $\mu_i = E[N_i]$ we obtain, upon conditioning on the next transition after the chain enters state i , that for $i = 1, \dots, n-1$

$$\mu_i = 1 + E[\text{number of additional transitions to reach } i+1 | \text{chain to } i-1]q$$

Now, if the chain next enters state $i-1$, then in order for it to reach $i+1$ it must first return to state i and must then go from state i to state $i+1$. Hence, we have from the preceding that

$$\mu_i = 1 + E[N_{i-1}^* + N_i^*]q$$

where N_{i-1}^* and N_i^* are, respectively, the additional number of transitions to return to state i from $i-1$ and the number to then go from i to $i+1$. Now, it follows from the Markovian property that these random variables have, respectively, the same distributions as N_{i-1} and N_i . In addition, they are independent (although we will only use this when we compute the variance of $N_{0,n}$). Hence, we see that

$$\mu_i = 1 + q(\mu_{i-1} + \mu_i)$$

or

$$\mu_i = \frac{1}{p} + \frac{q}{p}\mu_{i-1}, \quad i = 1, \dots, n-1$$

Starting with $\mu_0 = 1$, and letting $\alpha = q/p$, we obtain from the preceding recursion that

$$\mu_1 = 1/p + \alpha,$$

$$\begin{aligned}\mu_2 &= 1/p + \alpha(1/p + \alpha) = 1/p + \alpha/p + \alpha^2, \\ \mu_3 &= 1/p + \alpha(1/p + \alpha/p + \alpha^2) \\ &= 1/p + \alpha/p + \alpha^2/p + \alpha^3\end{aligned}$$

In general, we see that

$$\mu_i = \frac{1}{p} \sum_{j=0}^{i-1} \alpha^j + \alpha^i, \quad i = 1, \dots, n-1 \quad (4.16)$$

Using Eq. (4.15), we now get

$$E[N_{0,n}] = 1 + \frac{1}{p} \sum_{i=1}^{n-1} \sum_{j=0}^{i-1} \alpha^j + \sum_{i=1}^{n-1} \alpha^i$$

When $p = \frac{1}{2}$, and so $\alpha = 1$, we see from the preceding that

$$E[N_{0,n}] = 1 + (n-1)n + n-1 = n^2$$

When $p \neq \frac{1}{2}$, we obtain

$$\begin{aligned}E[N_{0,n}] &= 1 + \frac{1}{p(1-\alpha)} \sum_{i=1}^{n-1} (1-\alpha^i) + \frac{\alpha - \alpha^n}{1-\alpha} \\ &= 1 + \frac{1+\alpha}{1-\alpha} \left[n-1 - \frac{(\alpha - \alpha^n)}{1-\alpha} \right] + \frac{\alpha - \alpha^n}{1-\alpha} \\ &= 1 + \frac{2\alpha^{n+1} - (n+1)\alpha^2 + n-1}{(1-\alpha)^2}\end{aligned}$$

where the second equality used the fact that $p = 1/(1+\alpha)$. Therefore, we see that when $\alpha > 1$, or equivalently when $p < \frac{1}{2}$, the expected number of transitions to reach n is an exponentially increasing function of n . On the other hand, when $p = \frac{1}{2}$, $E[N_{0,n}] = n^2$, and when $p > \frac{1}{2}$, $E[N_{0,n}]$ is, for large n , essentially linear in n .

Let us now compute $\text{Var}(N_{0,n})$. To do so, we will again make use of the representation given by Eq. (4.15). Letting $v_i = \text{Var}(N_i)$, we start by determining the v_i recursively by using the conditional variance formula. Let $S_i = 1$ if the first transition out of state i is into state $i+1$, and let $S_i = -1$ if the transition is into state $i-1$, $i = 1, \dots, n-1$. Then,

$$\text{given that } S_i = 1: \quad N_i = 1$$

$$\text{given that } S_i = -1: \quad N_i = 1 + N_{i-1}^* + N_i^*$$

Hence,

$$E[N_i | S_i = 1] = 1,$$

$$E[N_i | S_i = -1] = 1 + \mu_{i-1} + \mu_i$$

implying that

$$\begin{aligned} \text{Var}(E[N_i | S_i]) &= \text{Var}(E[N_i | S_i] - 1) \\ &= (\mu_{i-1} + \mu_i)^2 q - (\mu_{i-1} + \mu_i)^2 q^2 \\ &= qp(\mu_{i-1} + \mu_i)^2 \end{aligned}$$

Also, since N_{i-1}^* and N_i^* , the numbers of transitions to return from state $i - 1$ to i and to then go from state i to state $i + 1$ are, by the Markovian property, independent random variables having the same distributions as N_{i-1} and N_i , respectively, we see that

$$\begin{aligned} \text{Var}(N_i | S_i = 1) &= 0, \\ \text{Var}(N_i | S_i = -1) &= v_{i-1} + v_i \end{aligned}$$

Hence,

$$E[\text{Var}(N_i | S_i)] = q(v_{i-1} + v_i)$$

From the conditional variance formula, we thus obtain

$$v_i = pq(\mu_{i-1} + \mu_i)^2 + q(v_{i-1} + v_i)$$

or, equivalently,

$$v_i = q(\mu_{i-1} + \mu_i)^2 + \alpha v_{i-1}, \quad i = 1, \dots, n - 1$$

Starting with $v_0 = 0$, we obtain from the preceding recursion that

$$\begin{aligned} v_1 &= q(\mu_0 + \mu_1)^2, \\ v_2 &= q(\mu_1 + \mu_2)^2 + \alpha q(\mu_0 + \mu_1)^2, \\ v_3 &= q(\mu_2 + \mu_3)^2 + \alpha q(\mu_1 + \mu_2)^2 + \alpha^2 q(\mu_0 + \mu_1)^2 \end{aligned}$$

In general, we have for $i > 0$,

$$v_i = q \sum_{j=1}^i \alpha^{i-j} (\mu_{j-1} + \mu_j)^2 \quad (4.17)$$

Therefore, we see that

$$\text{Var}(N_{0,n}) = \sum_{i=0}^{n-1} v_i = q \sum_{i=1}^{n-1} \sum_{j=1}^i \alpha^{i-j} (\mu_{j-1} + \mu_j)^2$$

where μ_j is given by Eq. (4.16).

We see from Eqs. (4.16) and (4.17) that when $p \geq \frac{1}{2}$, and so $\alpha \leq 1$, that μ_i and v_i , the mean and variance of the number of transitions to go from state i to $i + 1$, do not increase too rapidly in i . For instance, when $p = \frac{1}{2}$ it follows from Eqs. (4.16) and (4.17) that

$$\mu_i = 2i + 1$$

and

$$v_i = \frac{1}{2} \sum_{j=1}^i (4j)^2 = 8 \sum_{j=1}^i j^2$$

Hence, since $N_{0,n}$ is the sum of independent random variables, which are of roughly similar magnitudes when $p \geq \frac{1}{2}$, it follows in this case from the central limit theorem that $N_{0,n}$ is, for large n , approximately normally distributed. In particular, when $p = \frac{1}{2}$, $N_{0,n}$ is approximately normal with mean n^2 and variance

$$\begin{aligned} \text{Var}(N_{0,n}) &= 8 \sum_{i=1}^{n-1} \sum_{j=1}^i j^2 \\ &= 8 \sum_{j=1}^{n-1} \sum_{i=j}^{n-1} j^2 \\ &= 8 \sum_{j=1}^{n-1} (n-j) j^2 \\ &\approx 8 \int_1^{n-1} (n-x) x^2 dx \\ &\approx \frac{2}{3} n^4 \end{aligned}$$

Example 4.31 (The Satisfiability Problem). A Boolean variable x is one that takes on either of two values: TRUE or FALSE. If $x_i, i \geq 1$ are Boolean variables, then a Boolean clause of the form

$$x_1 + \bar{x}_2 + x_3$$

is TRUE if x_1 is TRUE, or if x_2 is FALSE, or if x_3 is TRUE. That is, the symbol “+” means “or” and $\bar{x}$ is TRUE if x is FALSE and vice versa. A Boolean formula is a combination of clauses such as

$$(x_1 + \bar{x}_2) * (x_1 + x_3) * (x_2 + \bar{x}_3) * (\bar{x}_1 + \bar{x}_2) * (x_1 + x_2)$$

In the preceding, the terms between the parentheses represent clauses, and the formula is TRUE if all the clauses are TRUE, and is FALSE otherwise. For a given Boolean formula, the *satisfiability problem* is either to determine values for the variables that

result in the formula being TRUE, or to determine that the formula is never true. For instance, one set of values that makes the preceding formula TRUE is to set $x_1 = \text{TRUE}$, $x_2 = \text{FALSE}$, and $x_3 = \text{FALSE}$.

Consider a formula of the n Boolean variables $x_1, \dots, x_n$ and suppose that each clause in this formula refers to exactly two variables. We will now present a *probabilistic algorithm* that will either find values that satisfy the formula or determine to a high probability that it is not possible to satisfy it. To begin, start with an arbitrary setting of values. Then, at each stage choose a clause whose value is FALSE, and randomly choose one of the Boolean variables in that clause and change its value. That is, if the variable has value TRUE then change its value to FALSE, and vice versa. If this new setting makes the formula TRUE then stop, otherwise continue in the same fashion. If you have not stopped after $n^2(1 + 4\sqrt{\frac{2}{3}})$ repetitions, then declare that the formula cannot be satisfied. We will now argue that if there is a satisfiable assignment then this algorithm will find such an assignment with a probability very close to 1.

Let us start by assuming that there is a satisfiable assignment of truth values and let $\mathcal{A}$ be such an assignment. At each stage of the algorithm there is a certain assignment of values. Let Y_j denote the number of the n variables whose values at the j th stage of the algorithm agree with their values in $\mathcal{A}$. For instance, suppose that $n = 3$ and $\mathcal{A}$ consists of the settings $x_1 = x_2 = x_3 = \text{TRUE}$. If the assignment of values at the j th step of the algorithm is $x_1 = \text{TRUE}$, $x_2 = x_3 = \text{FALSE}$, then $Y_j = 1$. Now, at each stage, the algorithm considers a clause that is not satisfied, thus implying that at least one of the values of the two variables in this clause does not agree with its value in $\mathcal{A}$. As a result, when we randomly choose one of the variables in this clause then there is a probability of at least $\frac{1}{2}$ that $Y_{j+1} = Y_j + 1$ and at most $\frac{1}{2}$ that $Y_{j+1} = Y_j - 1$. That is, independent of what has previously transpired in the algorithm, at each stage the number of settings in agreement with those in $\mathcal{A}$ will either increase or decrease by 1 and the probability of an increase is at least $\frac{1}{2}$ (it is 1 if both variables have values different from their values in $\mathcal{A}$). Thus, even though the process Y_j , $j \geq 0$ is not itself a Markov chain (why not?) it is intuitively clear that both the expectation and the variance of the number of stages of the algorithm needed to obtain the values of $\mathcal{A}$ will be less than or equal to the expectation and variance of the number of transitions to go from state 0 to state n in the Markov chain of Section 4.5.2. Hence, if the algorithm has not yet terminated because it found a set of satisfiable values different from that of $\mathcal{A}$, it will do so within an expected time of at most n^2 and with a standard deviation of at most $n^2\sqrt{\frac{2}{3}}$. In addition, since the time for the Markov chain to go from 0 to n is approximately normal when n is large we can be quite certain that a satisfiable assignment will be reached by $n^2 + 4(n^2\sqrt{\frac{2}{3}})$ stages, and thus if one has not been found by this number of stages of the algorithm we can be quite certain that there is no satisfiable assignment.

Our analysis also makes it clear why we assumed that there are only two variables in each clause. For if there were k , $k > 2$, variables in a clause then as any clause that is not presently satisfied may have only one incorrect setting, a randomly chosen variable whose value is changed might only increase the number of values in agreement with $\mathcal{A}$ with probability $1/k$ and so we could only conclude from our prior Markov chain

results that the mean time to obtain the values in $\mathcal{A}$ is an exponential function of n , which is not an efficient algorithm when n is large. ■

4.6 Mean Time Spent in Transient States

Consider now a finite state Markov chain and suppose that the states are numbered so that $T = \{1, 2, \dots, t\}$ denotes the set of transient states. Let

$$\mathbf{P}_T = \begin{bmatrix} P_{11} & P_{12} & \cdots & P_{1t} \\ \vdots & \vdots & \vdots & \vdots \\ P_{t1} & P_{t2} & \cdots & P_{tt} \end{bmatrix}$$

and note that since $\mathbf{P}_T$ specifies only the transition probabilities from transient states into transient states, some of its row sums are less than 1 (otherwise, T would be a closed class of states).

For transient states i and j , let s_{ij} denote the expected number of time periods that the Markov chain is in state j , given that it starts in state i . Let $\delta_{i,j} = 1$ when $i = j$ and let it be 0 otherwise. Condition on the initial transition to obtain

$$\begin{aligned} s_{ij} &= \delta_{i,j} + \sum_k P_{ik} s_{kj} \\ &= \delta_{i,j} + \sum_{k=1}^t P_{ik} s_{kj} \end{aligned} \quad (4.18)$$

where the final equality follows since it is impossible to go from a recurrent to a transient state, implying that $s_{kj} = 0$ when k is a recurrent state.

Let $\mathbf{S}$ denote the matrix of values s_{ij} , $i, j = 1, \dots, t$. That is,

$$\mathbf{S} = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1t} \\ \vdots & \vdots & \vdots & \vdots \\ s_{t1} & s_{t2} & \cdots & s_{tt} \end{bmatrix}$$

In matrix notation, Eq. (4.18) can be written as

$$\mathbf{S} = \mathbf{I} + \mathbf{P}_T \mathbf{S}$$

where $\mathbf{I}$ is the identity matrix of size t . Because the preceding equation is equivalent to

$$(\mathbf{I} - \mathbf{P}_T) \mathbf{S} = \mathbf{I}$$

we obtain, upon multiplying both sides by $(\mathbf{I} - \mathbf{P}_T)^{-1}$,

$$\mathbf{S} = (\mathbf{I} - \mathbf{P}_T)^{-1}$$

That is, the quantities s_{ij} , $i \in T$, $j \in T$, can be obtained by inverting the matrix $\mathbf{I} - \mathbf{P}_T$. (The existence of the inverse is easily established.)

Example 4.32. Consider the gambler's ruin problem with $p = 0.4$ and $N = 7$. Starting with 3 units, determine

- (a) the expected amount of time the gambler has 5 units,
- (b) the expected amount of time the gambler has 2 units.

Solution: The matrix $\mathbf{P}_T$, which specifies P_{ij} , $i, j \in \{1, 2, 3, 4, 5, 6\}$, is as follows:

$$\mathbf{P}_T = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & 0 & 0.4 & 0 & 0 & 0 & 0 \\ 2 & 0.6 & 0 & 0.4 & 0 & 0 & 0 \\ 3 & 0 & 0.6 & 0 & 0.4 & 0 & 0 \\ 4 & 0 & 0 & 0.6 & 0 & 0.4 & 0 \\ 5 & 0 & 0 & 0 & 0.6 & 0 & 0.4 \\ 6 & 0 & 0 & 0 & 0 & 0.6 & 0 \end{array}$$

Inverting $\mathbf{I} - \mathbf{P}_T$ gives

$$\mathbf{S} = (\mathbf{I} - \mathbf{P}_T)^{-1} = \begin{bmatrix} 1.6149 & 1.0248 & 0.6314 & 0.3691 & 0.1943 & 0.0777 \\ 1.5372 & 2.5619 & 1.5784 & 0.9228 & 0.4857 & 0.1943 \\ 1.4206 & 2.3677 & 2.9990 & 1.7533 & 0.9228 & 0.3691 \\ 1.2458 & 2.0763 & 2.6299 & 2.9990 & 1.5784 & 0.6314 \\ 0.9835 & 1.6391 & 2.0763 & 2.3677 & 2.5619 & 1.0248 \\ 0.5901 & 0.9835 & 1.2458 & 1.4206 & 1.5372 & 1.6149 \end{bmatrix}$$

Hence,

$$s_{3,5} = 0.9228, \quad s_{3,2} = 2.3677 \quad \blacksquare$$

For $i \in T$, $j \in T$, the quantity f_{ij} , equal to the probability that the Markov chain ever makes a transition into state j given that it starts in state i , is easily determined from $\mathbf{P}_T$. To determine the relationship, let us start by deriving an expression for s_{ij} by conditioning on whether state j is ever entered. This yields

$$\begin{aligned} s_{ij} &= E[\text{time in } j | \text{start in } i, \text{ ever transit to } j] f_{ij} \\ &\quad + E[\text{time in } j | \text{start in } i, \text{ never transit to } j] (1 - f_{ij}) \\ &= (\delta_{i,j} + s_{jj}) f_{ij} + \delta_{i,j} (1 - f_{ij}) \\ &= \delta_{i,j} + f_{ij} s_{jj} \end{aligned}$$

since s_{jj} is the expected number of additional time periods spent in state j given that it is eventually entered from state i . Solving the preceding equation yields

$$f_{ij} = \frac{s_{ij} - \delta_{i,j}}{s_{jj}}$$

Example 4.33. In Example 4.32, what is the probability that the gambler ever has a fortune of 1?

Solution: Since $s_{3,1} = 1.4206$ and $s_{1,1} = 1.6149$, then

$$f_{3,1} = \frac{s_{3,1}}{s_{1,1}} = 0.8797$$

As a check, note that $f_{3,1}$ is just the probability that a gambler starting with 3 reaches 1 before 7. That is, it is the probability that the gambler's fortune will go down 2 before going up 4; which is the probability that a gambler starting with 2 will go broke before reaching 6. Therefore,

$$f_{3,1} = 1 - \frac{1 - (0.6/0.4)^2}{1 - (0.6/0.4)^6} = 0.8797$$

which checks with our earlier answer. ■

Suppose we are interested in the expected time until the Markov chain enters some sets of states A , which need not be the set of recurrent states. We can reduce this back to the previous situation by making all states in A absorbing states. That is, reset the transition probabilities of states in A to satisfy

$$P_{i,i} = 1, \quad i \in A$$

This transforms the states of A into recurrent states, and transforms any state outside of A from which an eventual transition into A is possible into a transient state. Thus, our previous approach can be used.

4.7 Branching Processes

In this section we consider a class of Markov chains, known as *branching processes*, which have a wide variety of applications in the biological, sociological, and engineering sciences.

Consider a population consisting of individuals able to produce offspring of the same kind. Suppose that each individual will, by the end of its lifetime, have produced j new offspring with probability P_j , $j \geq 0$, independently of the numbers produced by other individuals. We suppose that $P_j < 1$ for all $j \geq 0$. The number of individuals initially present, denoted by X_0 , is called the size of the zeroth generation. All offspring of the zeroth generation constitute the first generation and their number is denoted by X_1 . In general, let X_n denote the size of the n th generation. It follows that $\{X_n, n = 0, 1, \dots\}$ is a Markov chain having as its state space the set of nonnegative integers.

Note that state 0 is a recurrent state, since clearly $P_{00} = 1$. Also, if $P_0 > 0$, all other states are transient. This follows since $P_{i0} = P_0^i$, which implies that starting with i

individuals there is a positive probability of at least P_0^i that no later generation will ever consist of i individuals. Moreover, since any finite set of transient states $\{1, 2, \dots, n\}$ will be visited only finitely often, this leads to the important conclusion that, if $P_0 > 0$, *then the population will either die out or its size will converge to infinity.*

Let

$$\mu = \sum_{j=0}^{\infty} j P_j$$

denote the mean number of offspring of a single individual, and let

$$\sigma^2 = \sum_{j=0}^{\infty} (j - \mu)^2 P_j$$

be the variance of the number of offspring produced by a single individual.

Let us suppose that $X_0 = 1$, that is, initially there is a single individual present. We calculate $E[X_n]$ and $\text{Var}(X_n)$ by first noting that we may write

$$X_n = \sum_{i=1}^{X_{n-1}} Z_i$$

where Z_i represents the number of offspring of the i th individual of the $(n - 1)$ st generation. By conditioning on X_{n-1} , we obtain

$$\begin{aligned} E[X_n] &= E[E[X_n | X_{n-1}]] \\ &= E \left[E \left[\sum_{i=1}^{X_{n-1}} Z_i | X_{n-1} \right] \right] \\ &= E[X_{n-1} \mu] \\ &= \mu E[X_{n-1}] \end{aligned}$$

where we have used the fact that $E[Z_i] = \mu$. Since $E[X_0] = 1$, the preceding yields

$$\begin{aligned} E[X_1] &= \mu, \\ E[X_2] &= \mu E[X_1] = \mu^2, \\ &\vdots \\ E[X_n] &= \mu E[X_{n-1}] = \mu^n \end{aligned}$$

Similarly, $\text{Var}(X_n)$ may be obtained by using the conditional variance formula

$$\text{Var}(X_n) = E[\text{Var}(X_n | X_{n-1})] + \text{Var}(E[X_n | X_{n-1}])$$

Now, given X_{n-1} , X_n is just the sum of X_{n-1} independent random variables each having the distribution $\{P_j, j \geq 0\}$. Hence,

$$E[X_n|X_{n-1}] = X_{n-1}\mu, \quad \text{Var}(X_n|X_{n-1}) = X_{n-1}\sigma^2$$

The conditional variance formula now yields

$$\begin{aligned} \text{Var}(X_n) &= E[X_{n-1}\sigma^2] + \text{Var}(X_{n-1}\mu) \\ &= \sigma^2\mu^{n-1} + \mu^2\text{Var}(X_{n-1}) \\ &= \sigma^2\mu^{n-1} + \mu^2(\sigma^2\mu^{n-2} + \mu^2\text{Var}(X_{n-2})) \\ &= \sigma^2(\mu^{n-1} + \mu^n) + \mu^4\text{Var}(X_{n-2}) \\ &= \sigma^2(\mu^{n-1} + \mu^n) + \mu^4(\sigma^2\mu^{n-3} + \mu^2\text{Var}(X_{n-3})) \\ &= \sigma^2(\mu^{n-1} + \mu^n + \mu^{n+1}) + \mu^6\text{Var}(X_{n-3}) \\ &= \dots \\ &= \sigma^2(\mu^{n-1} + \mu^n + \dots + \mu^{2n-2}) + \mu^{2n}\text{Var}(X_0) \\ &= \sigma^2(\mu^{n-1} + \mu^n + \dots + \mu^{2n-2}) \end{aligned}$$

Therefore,

$$\text{Var}(X_n) = \begin{cases} \sigma^2\mu^{n-1} \left(\frac{1-\mu^n}{1-\mu} \right), & \text{if } \mu \neq 1 \\ n\sigma^2, & \text{if } \mu = 1 \end{cases} \quad (4.19)$$

Let π_0 denote the probability that the population will eventually die out (under the assumption that $X_0 = 1$). More formally,

$$\pi_0 = \lim_{n \rightarrow \infty} P\{X_n = 0 | X_0 = 1\}$$

The problem of determining the value of π_0 was first raised in connection with the extinction of family surnames by Galton in 1889.

We first note that $\pi_0 = 1$ if $\mu < 1$. This follows since

$$\begin{aligned} \mu^n &= E[X_n] = \sum_{j=1}^{\infty} j P\{X_n = j\} \\ &\geq \sum_{j=1}^{\infty} 1 \cdot P\{X_n = j\} \\ &= P\{X_n \geq 1\} \end{aligned}$$

Since $\mu^n \rightarrow 0$ when $\mu < 1$, it follows that $P\{X_n \geq 1\} \rightarrow 0$, and hence $P\{X_n = 0\} \rightarrow 1$.

In fact, it can be shown that $\pi_0 = 1$ even when $\mu = 1$. When $\mu > 1$, it turns out that $\pi_0 < 1$, and an equation determining π_0 may be derived by conditioning on the

number of offspring of the initial individual, as follows:

$$\begin{aligned}\pi_0 &= P\{\text{population dies out}\} \\ &= \sum_{j=0}^{\infty} P\{\text{population dies out} | X_1 = j\} P_j\end{aligned}$$

Now, given that $X_1 = j$, the population will eventually die out if and only if each of the j families started by the members of the first generation eventually dies out. Since each family is assumed to act independently, and since the probability that any particular family dies out is just π_0 , this yields

$$P\{\text{population dies out} | X_1 = j\} = \pi_0^j$$

and thus π_0 satisfies

$$\pi_0 = \sum_{j=0}^{\infty} \pi_0^j P_j \quad (4.20)$$

In fact when $\mu > 1$, it can be shown that π_0 is the smallest positive number satisfying Eq. (4.20).

Example 4.34. If $P_0 = \frac{1}{2}$, $P_1 = \frac{1}{4}$, $P_2 = \frac{1}{4}$, then determine π_0 .

Solution: Since $\mu = \frac{3}{4} \leq 1$, it follows that $\pi_0 = 1$. ■

Example 4.35. If $P_0 = \frac{1}{4}$, $P_1 = \frac{1}{4}$, $P_2 = \frac{1}{2}$, then determine π_0 .

Solution: π_0 satisfies

$$\pi_0 = \frac{1}{4} + \frac{1}{4}\pi_0 + \frac{1}{2}\pi_0^2$$

or

$$2\pi_0^2 - 3\pi_0 + 1 = 0$$

The smallest positive solution of this quadratic equation is $\pi_0 = \frac{1}{2}$. ■

Example 4.36. In Examples 4.34 and 4.35, what is the probability that the population will die out if it initially consists of n individuals?

Solution: Since the population will die out if and only if the families of each of the members of the initial generation die out, the desired probability is π_0^n . For Example 4.34 this yields $\pi_0^n = 1$, and for Example 4.35, $\pi_0^n = \left(\frac{1}{2}\right)^n$. ■

4.8 Time Reversible Markov Chains

Consider a stationary ergodic Markov chain (that is, an ergodic Markov chain that has been in operation for a long time) having transition probabilities P_{ij} and stationary probabilities π_i , and suppose that starting at some time we trace the sequence of states going backward in time. That is, starting at time n , consider the sequence of states $X_n, X_{n-1}, X_{n-2}, \dots$. It turns out that this sequence of states is itself a Markov chain with transition probabilities Q_{ij} defined by

$$\begin{aligned} Q_{ij} &= P\{X_m = j | X_{m+1} = i\} \\ &= \frac{P\{X_m = j, X_{m+1} = i\}}{P\{X_{m+1} = i\}} \\ &= \frac{P\{X_m = j\}P\{X_{m+1} = i | X_m = j\}}{P\{X_{m+1} = i\}} \\ &= \frac{\pi_j P_{ji}}{\pi_i} \end{aligned}$$

To prove that the reversed process is indeed a Markov chain, we must verify that

$$P\{X_m = j | X_{m+1} = i, X_{m+2}, X_{m+3}, \dots\} = P\{X_m = j | X_{m+1} = i\}$$

To see that this is so, suppose that the present time is $m+1$. Now, since $X_0, X_1, X_2, \dots$ is a Markov chain, it follows that the conditional distribution of the future $X_{m+2}, X_{m+3}, \dots$ given the present state X_{m+1} is independent of the past state X_m . However, independence is a symmetric relationship (that is, if A is independent of B , then B is independent of A), and so this means that given X_{m+1} , X_m is independent of $X_{m+2}, X_{m+3}, \dots$. But this is exactly what we had to verify.

Thus, the reversed process is also a Markov chain with transition probabilities given by

$$Q_{ij} = \frac{\pi_j P_{ji}}{\pi_i}$$

If $Q_{ij} = P_{ij}$ for all i, j , then the Markov chain is said to be *time reversible*. The condition for time reversibility, namely, $Q_{ij} = P_{ij}$, can also be expressed as

$$\pi_i P_{ij} = \pi_j P_{ji} \quad \text{for all } i, j \quad (4.21)$$

The condition in Eq. (4.21) can be stated that, for all states i and j , the rate at which the process goes from i to j (namely, $\pi_i P_{ij}$) is equal to the rate at which it goes from j to i (namely, $\pi_j P_{ji}$). It is worth noting that this is an obvious necessary condition for time reversibility since a transition from i to j going backward in time is equivalent to a transition from j to i going forward in time; that is, if $X_m = i$ and $X_{m-1} = j$, then a transition from i to j is observed if we are looking backward, and one from j to i if we are looking forward in time. Thus, the rate at which the forward process makes a transition from j to i is always equal to the rate at which the reverse process makes a

transition from i to j ; if time reversible, this must equal the rate at which the forward process makes a transition from i to j .

If we can find nonnegative numbers, summing to one, that satisfy Eq. (4.21), then it follows that the Markov chain is time reversible and the numbers represent the limiting probabilities. This is so since if

$$x_i P_{ij} = x_j P_{ji} \quad \text{for all } i, j, \quad \sum_i x_i = 1 \quad (4.22)$$

then summing over i yields

$$\sum_i x_i P_{ij} = x_j \sum_i P_{ji} = x_j, \quad \sum_i x_i = 1$$

and, because the limiting probabilities π_i are the unique solution of the preceding, it follows that $x_i = \pi_i$ for all i .

Example 4.37. Consider a random walk with states $0, 1, \dots, M$ and transition probabilities

$$\begin{aligned} P_{i,i+1} &= \alpha_i = 1 - P_{i,i-1}, \quad i = 1, \dots, M-1, \\ P_{0,1} &= \alpha_0 = 1 - P_{0,0}, \\ P_{M,M} &= \alpha_M = 1 - P_{M,M-1} \end{aligned}$$

Without the need for any computations, it is possible to argue that this Markov chain, which can only make transitions from a state to one of its two nearest neighbors, is time reversible. This follows by noting that the number of transitions from i to $i+1$ must at all times be within 1 of the number from $i+1$ to i . This is so because between any two transitions from i to $i+1$ there must be one from $i+1$ to i (and conversely) since the only way to reenter i from a higher state is via state $i+1$. Hence, it follows that the rate of transitions from i to $i+1$ equals the rate from $i+1$ to i , and so the process is time reversible.

We can easily obtain the limiting probabilities by equating for each state $i = 0, 1, \dots, M-1$ the rate at which the process goes from i to $i+1$ with the rate at which it goes from $i+1$ to i . This yields

$$\begin{aligned} \pi_0 \alpha_0 &= \pi_1 (1 - \alpha_1), \\ \pi_1 \alpha_1 &= \pi_2 (1 - \alpha_2), \\ &\vdots \\ \pi_i \alpha_i &= \pi_{i+1} (1 - \alpha_{i+1}), \quad i = 0, 1, \dots, M-1 \end{aligned}$$

Solving in terms of π_0 yields

$$\pi_1 = \frac{\alpha_0}{1 - \alpha_1} \pi_0,$$

$$\pi_2 = \frac{\alpha_1}{1 - \alpha_2} \pi_1 = \frac{\alpha_1 \alpha_0}{(1 - \alpha_2)(1 - \alpha_1)} \pi_0$$

and, in general,

$$\pi_i = \frac{\alpha_{i-1} \cdots \alpha_0}{(1 - \alpha_i) \cdots (1 - \alpha_1)} \pi_0, \quad i = 1, 2, \dots, M$$

Since $\sum_0^M \pi_i = 1$, we obtain

$$\pi_0 \left[1 + \sum_{j=1}^M \frac{\alpha_{j-1} \cdots \alpha_0}{(1 - \alpha_j) \cdots (1 - \alpha_1)} \right] = 1$$

or

$$\pi_0 = \left[1 + \sum_{j=1}^M \frac{\alpha_{j-1} \cdots \alpha_0}{(1 - \alpha_j) \cdots (1 - \alpha_1)} \right]^{-1} \quad (4.23)$$

and

$$\pi_i = \frac{\alpha_{i-1} \cdots \alpha_0}{(1 - \alpha_i) \cdots (1 - \alpha_1)} \pi_0, \quad i = 1, \dots, M \quad (4.24)$$

For instance, if $\alpha_i \equiv \alpha$, then

$$\begin{aligned} \pi_0 &= \left[1 + \sum_{j=1}^M \left(\frac{\alpha}{1 - \alpha} \right)^j \right]^{-1} \\ &= \frac{1 - \beta}{1 - \beta^{M+1}} \end{aligned}$$

and, in general,

$$\pi_i = \frac{\beta^i (1 - \beta)}{1 - \beta^{M+1}}, \quad i = 0, 1, \dots, M$$

where

$$\beta = \frac{\alpha}{1 - \alpha} \quad \blacksquare$$

Another special case of Example 4.37 is the following urn model, proposed by the physicists P. and T. Ehrenfest to describe the movements of molecules. Suppose that M molecules are distributed among two urns; and at each time point one of the molecules is chosen at random, removed from its urn, and placed in the other one. The number of molecules in urn 1 is a special case of the Markov chain of Example 4.37 having

$$\alpha_i = \frac{M - i}{M}, \quad i = 0, 1, \dots, M$$

Hence, using Eqs. (4.23) and (4.24) the limiting probabilities in this case are

$$\begin{aligned}\pi_0 &= \left[1 + \sum_{j=1}^M \frac{(M-j+1) \cdots (M-1)M}{j(j-1) \cdots 1} \right]^{-1} \\ &= \left[\sum_{j=0}^M \binom{M}{j} \right]^{-1} \\ &= \left(\frac{1}{2} \right)^M\end{aligned}$$

where we have used the identity

$$\begin{aligned}1 &= \left(\frac{1}{2} + \frac{1}{2} \right)^M \\ &= \sum_{j=0}^M \binom{M}{j} \left(\frac{1}{2} \right)^M\end{aligned}$$

Hence, from Eq. (4.24)

$$\pi_i = \binom{M}{i} \left(\frac{1}{2} \right)^M, \quad i = 0, 1, \dots, M$$

Because the preceding are just the binomial probabilities, it follows that in the long run, the positions of each of the M balls are independent and each one is equally likely to be in either urn. This, however, is quite intuitive, for if we focus on any one ball, it becomes quite clear that its position will be independent of the positions of the other balls (since no matter where the other $M-1$ balls are, the ball under consideration at each stage will be moved with probability $1/M$) and by symmetry, it is equally likely to be in either urn.

Example 4.38. Consider an arbitrary connected graph (see Section 3.6 for definitions) having a number w_{ij} associated with arc (i, j) for each arc. One instance of such a graph is given by Fig. 4.1. Now consider a particle moving from node to node in this

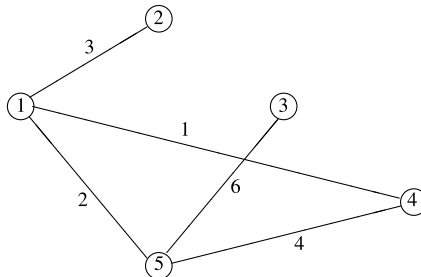


Figure 4.1 A connected graph with arc weights.

manner: If at any time the particle resides at node i , then it will next move to node j with probability P_{ij} where

$$P_{ij} = \frac{w_{ij}}{\sum_j w_{ij}}$$

and where w_{ij} is 0 if (i, j) is not an arc. For instance, for the graph of Fig. 4.1, $P_{12} = 3/(3 + 1 + 2) = \frac{1}{2}$.

The time reversibility equations

$$\pi_i P_{ij} = \pi_j P_{ji}$$

reduce to

$$\pi_i \frac{w_{ij}}{\sum_j w_{ij}} = \pi_j \frac{w_{ji}}{\sum_i w_{ji}}$$

or, equivalently, since $w_{ij} = w_{ji}$

$$\frac{\pi_i}{\sum_j w_{ij}} = \frac{\pi_j}{\sum_i w_{ji}}$$

which is equivalent to

$$\frac{\pi_i}{\sum_j w_{ij}} = c$$

or

$$\pi_i = c \sum_j w_{ij}$$

or, since $1 = \sum_i \pi_i$

$$\pi_i = \frac{\sum_j w_{ij}}{\sum_i \sum_j w_{ij}}$$

Because the π_i s given by this equation satisfy the time reversibility equations, it follows that the process is time reversible with these limiting probabilities.

For the graph of Fig. 4.1 we have that

$$\pi_1 = \frac{6}{32}, \quad \pi_2 = \frac{3}{32}, \quad \pi_3 = \frac{6}{32}, \quad \pi_4 = \frac{5}{32}, \quad \pi_5 = \frac{12}{32} \quad \blacksquare$$

If we try to solve Eq. (4.22) for an arbitrary Markov chain with states $0, 1, \dots, M$, it will usually turn out that no solution exists. For example, from Eq. (4.22),

$$x_i P_{ij} = x_j P_{ji},$$

$$x_k P_{kj} = x_j P_{jk}$$

implying (if $P_{ij}P_{jk} > 0$) that

$$\frac{x_i}{x_k} = \frac{P_{ji}P_{kj}}{P_{ij}P_{jk}}$$

which in general need not equal P_{ki}/P_{ik} . Thus, we see that a necessary condition for time reversibility is that

$$P_{ik}P_{kj}P_{ji} = P_{ij}P_{jk}P_{ki} \quad \text{for all } i, j, k \quad (4.25)$$

which is equivalent to the statement that, starting in state i , the path $i \rightarrow k \rightarrow j \rightarrow i$ has the same probability as the reversed path $i \rightarrow j \rightarrow k \rightarrow i$. To understand the necessity of this, note that time reversibility implies that the rate at which a sequence of transitions from i to k to j to i occurs must equal the rate of ones from i to j to k to i (why?), and so we must have

$$\pi_i P_{ik} P_{kj} P_{ji} = \pi_i P_{ij} P_{jk} P_{ki}$$

implying Eq. (4.25) when $\pi_i > 0$.

In fact, we can show the following.

Theorem 4.2. *A stationary Markov chain for which $P_{ij} = 0$ whenever $P_{ji} = 0$ is time reversible if and only if starting in state i , any path back to i has the same probability as the reversed path. That is, if*

$$P_{i,i_1} P_{i_1,i_2} \cdots P_{i_k,i} = P_{i,i_k} P_{i_k,i_{k-1}} \cdots P_{i_1,i} \quad (4.26)$$

for all states $i, i_1, \dots, i_k$.

Proof. We have already proven necessity. To prove sufficiency, fix states i and j and rewrite (4.26) as

$$P_{i,i_1} P_{i_1,i_2} \cdots P_{i_k,j} P_{ji} = P_{ij} P_{j,i_k} \cdots P_{i_1,i}$$

Summing the preceding over all states $i_1, \dots, i_k$ yields

$$P_{ij}^{k+1} P_{ji} = P_{ij} P_{ji}^{k+1}$$

Consequently,

$$\frac{P_{ji} \sum_{k=1}^m P_{ij}^{k+1}}{m} = \frac{P_{ij} \sum_{k=1}^m P_{ji}^{k+1}}{m}$$

Letting $m \rightarrow \infty$ yields

$$P_{ji} \pi_j = P_{ij} \pi_i$$

which proves the theorem. ■

Example 4.39. Suppose we are given a set of n elements, numbered 1 through n , which are to be arranged in some ordered list. At each unit of time a request is made to retrieve one of these elements, element i being requested (independently of the past) with probability P_i . After being requested, the element then is put back but not necessarily in the same position. In fact, let us suppose that the element requested is moved one closer to the front of the list; for instance, if the present list ordering is 1, 3, 4, 2, 5 and element 2 is requested, then the new ordering becomes 1, 3, 2, 4, 5. We are interested in the long-run average position of the element requested.

For any given probability vector $P = (P_1, \dots, P_n)$, the preceding can be modeled as a Markov chain with $n!$ states, with the state at any time being the list order at that time. We shall show that this Markov chain is time reversible and then use this to show that the average position of the element requested when this one-closer rule is in effect is less than when the rule of always moving the requested element to the front of the line is used. The time reversibility of the resulting Markov chain when the one-closer reordering rule is in effect easily follows from Theorem 4.2. For instance, suppose $n = 3$ and consider the following path from state (1, 2, 3) to itself:

$$\begin{aligned} (1, 2, 3) &\rightarrow (2, 1, 3) \rightarrow (2, 3, 1) \rightarrow (3, 2, 1) \\ &\rightarrow (3, 1, 2) \rightarrow (1, 3, 2) \rightarrow (1, 2, 3) \end{aligned}$$

The product of the transition probabilities in the forward direction is

$$P_2 P_3 P_3 P_1 P_1 P_2 = P_1^2 P_2^2 P_3^2$$

whereas in the reverse direction, it is

$$P_3 P_3 P_2 P_2 P_1 P_1 = P_1^2 P_2^2 P_3^2$$

Because the general result follows in much the same manner, the Markov chain is indeed time reversible. (For a formal argument note that if f_i denotes the number of times element i moves forward in the path, then as the path goes from a fixed state back to itself, it follows that element i will also move backward f_i times. Therefore, since the backward moves of element i are precisely the times that it moves forward in the reverse path, it follows that the product of the transition probabilities for both the path and its reversal will equal

$$\prod_i P_i^{f_i + r_i}$$

where r_i is equal to the number of times that element i is in the first position and the path (or the reverse path) does not change states.)

For any permutation $i_1, i_2, \dots, i_n$ of $1, 2, \dots, n$, let $\pi(i_1, i_2, \dots, i_n)$ denote the limiting probability under the one-closer rule. By time reversibility we have

$$P_{i_{j+1}} \pi(i_1, \dots, i_j, i_{j+1}, \dots, i_n) = P_{i_j} \pi(i_1, \dots, i_{j+1}, i_j, \dots, i_n) \quad (4.27)$$

for all permutations.

Now the average position of the element requested can be expressed (as in Section 3.6.1) as

$$\begin{aligned}
 \text{Average position} &= \sum_i P_i E[\text{Position of element } i] \\
 &= \sum_i P_i \left[1 + \sum_{j \neq i} P\{\text{element } j \text{ precedes element } i\} \right] \\
 &= 1 + \sum_i \sum_{j \neq i} P_i P\{e_j \text{ precedes } e_i\} \\
 &= 1 + \sum_{i < j} \sum [P_i P\{e_j \text{ precedes } e_i\} + P_j P\{e_i \text{ precedes } e_j\}] \\
 &= 1 + \sum_{i < j} \sum [P_i P\{e_j \text{ precedes } e_i\} \\
 &\quad + P_j (1 - P\{e_j \text{ precedes } e_i\})] \\
 &= 1 + \sum_{i < j} \sum (P_i - P_j) P\{e_j \text{ precedes } e_i\} + \sum_{i < j} \sum P_j
 \end{aligned}$$

Hence, to minimize the average position of the element requested, we would want to make $P\{e_j \text{ precedes } e_i\}$ as large as possible when $P_j > P_i$ and as small as possible when $P_i > P_j$. Under the front-of-the-line rule we showed in Section 3.6.1,

$$P\{e_j \text{ precedes } e_i\} = \frac{P_j}{P_j + P_i}$$

(since under the front-of-the-line rule element j will precede element i if and only if the last request for either i or j was for j).

Therefore, to show that the one-closer rule is better than the front-of-the-line rule, it suffices to show that under the one-closer rule

$$P\{e_j \text{ precedes } e_i\} > \frac{P_j}{P_j + P_i} \quad \text{when } P_j > P_i$$

Now consider any state where element i precedes element j , say, $(\dots, i, i_1, \dots, i_k, j, \dots)$. By successive transpositions using Eq. (4.27), we have

$$\pi(\dots, i, i_1, \dots, i_k, j, \dots) = \left(\frac{P_i}{P_j} \right)^{k+1} \pi(\dots, j, i_1, \dots, i_k, i, \dots) \quad (4.28)$$

For instance,

$$\begin{aligned}
 \pi(1, 2, 3) &= \frac{P_2}{P_3} \pi(1, 3, 2) = \frac{P_2}{P_3} \frac{P_1}{P_3} \pi(3, 1, 2) \\
 &= \frac{P_2}{P_3} \frac{P_1}{P_3} \frac{P_1}{P_2} \pi(3, 2, 1) = \left(\frac{P_1}{P_3} \right)^2 \pi(3, 2, 1)
 \end{aligned}$$

Now when $P_j > P_i$, Eq. (4.28) implies that

$$\pi(\dots, i, i_1, \dots, i_k, j, \dots) < \frac{P_i}{P_j} \pi(\dots, j, i_1, \dots, i_k, i, \dots)$$

Letting $\alpha(i, j) = P\{e_i \text{ precedes } e_j\}$, we see by summing over all states for which i precedes j and by using the preceding that

$$\alpha(i, j) < \frac{P_i}{P_j} \alpha(j, i)$$

which, since $\alpha(i, j) = 1 - \alpha(j, i)$, yields

$$\alpha(j, i) > \frac{P_j}{P_j + P_i}$$

Hence, the average position of the element requested is indeed smaller under the one-closer rule than under the front-of-the-line rule. ■

The concept of the reversed chain is useful even when the process is not time reversible. To illustrate this, we start with the following proposition whose proof is left as an exercise.

Proposition 4.9. *Consider an irreducible Markov chain with transition probabilities P_{ij} . If we can find positive numbers $\pi_i, i \geq 0$, summing to one, and a transition probability matrix $\mathbf{Q} = [Q_{ij}]$ such that*

$$\pi_i P_{ij} = \pi_j Q_{ji} \quad (4.29)$$

then the Q_{ij} are the transition probabilities of the reversed chain and the π_i are the stationary probabilities both for the original and reversed chain.

The importance of the preceding proposition is that, by thinking backward, we can sometimes guess at the nature of the reversed chain and then use the set of Eqs. (4.29) to obtain both the stationary probabilities and the Q_{ij} .

Example 4.40. A single bulb is necessary to light a given room. When the bulb in use fails, it is replaced by a new one at the beginning of the next day. Let X_n equal i if the bulb in use at the beginning of day n is in its i th day of use (that is, if its present age is i). For instance, if a bulb fails on day $n - 1$, then a new bulb will be put in use at the beginning of day n and so $X_n = 1$. If we suppose that each bulb, independently, fails on its i th day of use with probability $p_i, i \geq 1$, then it is easy to see that $\{X_n, n \geq 1\}$ is a Markov chain whose transition probabilities are as follows:

$$\begin{aligned} P_{i,1} &= P\{\text{bulb, on its } i\text{th day of use, fails}\} \\ &= P\{\text{life of bulb} = i | \text{life of bulb} \geq i\} \\ &= \frac{P\{L = i\}}{P\{L \geq i\}} \end{aligned}$$

where L , a random variable representing the lifetime of a bulb, is such that $P\{L = i\} = p_i$. Also,

$$P_{i,i+1} = 1 - P_{i,1}$$

Suppose now that this chain has been in operation for a long (in theory, an infinite) time and consider the sequence of states going backward in time. Since, in the forward direction, the state is always increasing by 1 until it reaches the age at which the item fails, it is easy to see that the reverse chain will always decrease by 1 until it reaches 1 and then it will jump to a random value representing the lifetime of the (in real time) previous bulb. Thus, it seems that the reverse chain should have transition probabilities given by

$$\begin{aligned} Q_{i,i-1} &= 1, & i > 1 \\ Q_{1,i} &= p_i, & i \geq 1 \end{aligned}$$

To check this, and at the same time determine the stationary probabilities, we must see if we can find, with the $Q_{i,j}$ as previously given, positive numbers $\{\pi_i\}$ such that

$$\pi_i P_{i,j} = \pi_j Q_{j,i}$$

To begin, let $j = 1$ and consider the resulting equations:

$$\pi_i P_{i,1} = \pi_1 Q_{1,i}$$

This is equivalent to

$$\pi_i \frac{P\{L = i\}}{P\{L \geq i\}} = \pi_1 P\{L = i\}$$

or

$$\pi_i = \pi_1 P\{L \geq i\}$$

Summing over all i yields

$$1 = \sum_{i=1}^{\infty} \pi_i = \pi_1 \sum_{i=1}^{\infty} P\{L \geq i\} = \pi_1 E[L]$$

and so, for the preceding $Q_{i,j}$ to represent the reverse transition probabilities, it is necessary for the stationary probabilities to be

$$\pi_i = \frac{P\{L \geq i\}}{E[L]}, \quad i \geq 1$$

To finish the proof that the reverse transition probabilities and stationary probabilities are as given, all that remains is to show that they satisfy

$$\pi_i P_{i,i+1} = \pi_{i+1} Q_{i+1,i}$$

which is equivalent to

$$\frac{P\{L \geq i\}}{E[L]} \left(1 - \frac{P\{L = i\}}{P\{L \geq i\}}\right) = \frac{P\{L \geq i + 1\}}{E[L]}$$

and which is true since $P\{L \geq i\} - P\{L = i\} = P\{L \geq i + 1\}$. ■

4.9 Markov Chain Monte Carlo Methods

Let $\mathbf{X}$ be a discrete random vector whose set of possible values is $\mathbf{x}_j, j \geq 1$. Let the probability mass function of $\mathbf{X}$ be given by $P\{\mathbf{X} = \mathbf{x}_j\}, j \geq 1$, and suppose that we are interested in calculating

$$\theta = E[h(\mathbf{X})] = \sum_{j=1}^{\infty} h(\mathbf{x}_j) P\{\mathbf{X} = \mathbf{x}_j\}$$

for some specified function h . In situations where it is computationally difficult to evaluate the function $h(\mathbf{x}_j), j \geq 1$, we often turn to simulation to approximate θ . The usual approach, called *Monte Carlo simulation*, is to use random numbers to generate a partial sequence of independent and identically distributed random vectors $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ having the mass function $P\{\mathbf{X} = \mathbf{x}_j\}, j \geq 1$ (see Chapter 11 for a discussion as to how this can be accomplished). Since the strong law of large numbers yields

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{h(\mathbf{X}_i)}{n} = \theta \quad (4.30)$$

it follows that we can estimate θ by letting n be large and using the average of the values of $h(\mathbf{X}_i), i = 1, \dots, n$ as the estimator.

It often, however, turns out that it is difficult to generate a random vector having the specified probability mass function, particularly if $\mathbf{X}$ is a vector of dependent random variables. In addition, its probability mass function is sometimes given in the form $P\{\mathbf{X} = \mathbf{x}_j\} = C b_j, j \geq 1$, where the b_j are specified, but C must be computed, and in many applications it is not computationally feasible to sum the b_j so as to determine C . Fortunately, however, there is another way of using simulation to estimate θ in these situations. It works by generating a sequence, not of independent random vectors, but of the successive states of a vector-valued Markov chain $\mathbf{X}_1, \mathbf{X}_2, \dots$ whose stationary probabilities are $P\{\mathbf{X} = \mathbf{x}_j\}, j \geq 1$. If this can be accomplished, then it would follow from Proposition 4.6 that Eq. (4.30) remains valid, implying that we can then use $\sum_{i=1}^n h(\mathbf{X}_i)/n$ as an estimator of θ .

We now show how to generate a Markov chain with arbitrary stationary probabilities that may only be specified up to a multiplicative constant. Let $b(j), j = 1, 2, \dots$ be positive numbers whose sum $B = \sum_{j=1}^{\infty} b(j)$ is finite. The following, known as

the *Hastings–Metropolis algorithm*, can be used to generate a time reversible Markov chain whose stationary probabilities are

$$\pi(j) = b(j)/B, \quad j = 1, 2, \dots$$

To begin, let $\mathbf{Q}$ be any specified irreducible Markov transition probability matrix on the integers, with $q(i, j)$ representing the row i column j element of $\mathbf{Q}$. Now define a Markov chain $\{X_n, n \geq 0\}$ as follows. When $X_n = i$, generate a random variable Y such that $P\{Y = j\} = q(i, j)$, $j = 1, 2, \dots$. If $Y = j$, then set X_{n+1} equal to j with probability $\alpha(i, j)$, and set it equal to i with probability $1 - \alpha(i, j)$. Under these conditions, it is easy to see that the sequence of states constitutes a Markov chain with transition probabilities $P_{i,j}$ given by

$$\begin{aligned} P_{i,j} &= q(i, j)\alpha(i, j), \quad \text{if } j \neq i \\ P_{i,i} &= q(i, i) + \sum_{k \neq i} q(i, k)(1 - \alpha(i, k)) \end{aligned}$$

This Markov chain will be time reversible and have stationary probabilities $\pi(j)$ if

$$\pi(i)P_{i,j} = \pi(j)P_{j,i} \quad \text{for } j \neq i$$

which is equivalent to

$$\pi(i)q(i, j)\alpha(i, j) = \pi(j)q(j, i)\alpha(j, i) \quad (4.31)$$

But if we take $\pi(j) = b(j)/B$ and set

$$\alpha(i, j) = \min\left(\frac{\pi(j)q(j, i)}{\pi(i)q(i, j)}, 1\right) \quad (4.32)$$

then Eq. (4.31) is easily seen to be satisfied. For if

$$\alpha(i, j) = \frac{\pi(j)q(j, i)}{\pi(i)q(i, j)}$$

then $\alpha(j, i) = 1$ and Eq. (4.31) follows, and if $\alpha(i, j) = 1$ then

$$\alpha(j, i) = \frac{\pi(i)q(i, j)}{\pi(j)q(j, i)}$$

and again Eq. (4.31) holds, thus showing that the Markov chain is time reversible with stationary probabilities $\pi(j)$. Also, since $\pi(j) = b(j)/B$, we see from (4.32) that

$$\alpha(i, j) = \min\left(\frac{b(j)q(j, i)}{b(i)q(i, j)}, 1\right)$$

which shows that the value of B is not needed to define the Markov chain, because the values $b(j)$ suffice. Also, it is almost always the case that $\pi(j)$, $j \geq 1$ will not only be stationary probabilities but will also be limiting probabilities. (Indeed, a sufficient condition is that $P_{i,i} > 0$ for some i .)

Example 4.41. Suppose that we want to generate a uniformly distributed element in $\mathcal{S}$, the set of all permutations $(x_1, \dots, x_n)$ of the numbers $(1, \dots, n)$ for which $\sum_{j=1}^n jx_j > a$ for a given constant a . To utilize the Hastings–Metropolis algorithm we need to define an irreducible Markov transition probability matrix on the state space $\mathcal{S}$. To accomplish this, we first define a concept of “neighboring” elements of $\mathcal{S}$, and then construct a graph whose vertex set is $\mathcal{S}$. We start by putting an arc between each pair of neighboring elements in $\mathcal{S}$, where any two permutations in $\mathcal{S}$ are said to be neighbors if one results from an interchange of two of the positions of the other. That is, $(1, 2, 3, 4)$ and $(1, 2, 4, 3)$ are neighbors whereas $(1, 2, 3, 4)$ and $(1, 3, 4, 2)$ are not. Now, define the q transition probability function as follows. With $N(s)$ defined as the set of neighbors of s , and $|N(s)|$ equal to the number of elements in the set $N(s)$, let

$$q(s, t) = \frac{1}{|N(s)|} \quad \text{if } t \in N(s)$$

That is, the candidate next state from s is equally likely to be any of its neighbors. Since the desired limiting probabilities of the Markov chain are $\pi(s) = C$, it follows that $\pi(s) = \pi(t)$, and so

$$\alpha(s, t) = \min(|N(s)|/|N(t)|, 1)$$

That is, if the present state of the Markov chain is s then one of its neighbors is randomly chosen, say, t . If t is a state with fewer neighbors than s (in graph theory language, if the degree of vertex t is less than that of vertex s), then the next state is t . If not, a uniform $(0, 1)$ random number U is generated and the next state is t if $U < |N(s)|/|N(t)|$ and is s otherwise. The limiting probabilities of this Markov chain are $\pi(s) = 1/|\mathcal{S}|$, where $|\mathcal{S}|$ is the (unknown) number of permutations in $\mathcal{S}$. ■

The most widely used version of the Hastings–Metropolis algorithm is the *Gibbs sampler*. Let $\mathbf{X} = (X_1, \dots, X_n)$ be a discrete random vector with probability mass function $p(\mathbf{x})$ that is only specified up to a multiplicative constant, and suppose that we want to generate a random vector whose distribution is that of $\mathbf{X}$. That is, we want to generate a random vector having mass function

$$p(\mathbf{x}) = Cg(\mathbf{x})$$

where $g(\mathbf{x})$ is known, but C is not. Utilization of the Gibbs sampler assumes that for any i and values x_j , $j \neq i$, we can generate a random variable X having the probability mass function

$$P\{X = x\} = P\{X_i = x | X_j = x_j, j \neq i\}$$

It operates by using the Hasting–Metropolis algorithm on a Markov chain with states $\mathbf{x} = (x_1, \dots, x_n)$, and with transition probabilities defined as follows. Whenever the present state is $\mathbf{x}$, a coordinate that is equally likely to be any of $1, \dots, n$ is chosen. If coordinate i is chosen, then a random variable X with probability mass function $P\{X = x\} = P\{X_i = x | X_j = x_j, j \neq i\}$ is generated. If $X = x$, then the state $\mathbf{y} =$

$(x_1, \dots, x_{i-1}, x, x_{i+1}, \dots, x_n)$ is considered as the candidate next state. In other words, with $\mathbf{x}$ and $\mathbf{y}$ as given, the Gibbs sampler uses the Hastings–Metropolis algorithm with

$$q(\mathbf{x}, \mathbf{y}) = \frac{1}{n} P\{X_i = x | X_j = x_j, j \neq i\} = \frac{p(\mathbf{y})}{n P\{X_j = x_j, j \neq i\}}$$

Because we want the limiting mass function to be p , we see from Eq. (4.32) that the vector $\mathbf{y}$ is then accepted as the new state with probability

$$\begin{aligned} \alpha(\mathbf{x}, \mathbf{y}) &= \min\left(\frac{p(\mathbf{y})q(\mathbf{y}, \mathbf{x})}{p(\mathbf{x})q(\mathbf{x}, \mathbf{y})}, 1\right) \\ &= \min\left(\frac{p(\mathbf{y})p(\mathbf{x})}{p(\mathbf{x})p(\mathbf{y})}, 1\right) \\ &= 1 \end{aligned}$$

Hence, when utilizing the Gibbs sampler, the candidate state is always accepted as the next state of the chain.

Example 4.42. Suppose that we want to generate n uniformly distributed points in the circle of radius 1 centered at the origin, conditional on the event that no two points are within a distance d of each other, when the probability of this conditioning event is small. This can be accomplished by using the Gibbs sampler as follows. Start with any n points $\mathbf{x}_1, \dots, \mathbf{x}_n$ in the circle that have the property that no two of them are within d of the other; then generate the value of I , equally likely to be any of the values $1, \dots, n$. Then continually generate a random point in the circle until you obtain one that is not within d of any of the other $n - 1$ points excluding $\mathbf{x}_I$. At this point, replace $\mathbf{x}_I$ by the generated point and then repeat the operation. After a large number of iterations of this algorithm, the set of n points will approximately have the desired distribution. ■

Example 4.43. Let $X_i, i = 1, \dots, n$, be independent exponential random variables with respective rates $\lambda_i, i = 1, \dots, n$. Let $S = \sum_{i=1}^n X_i$, and suppose that we want to generate the random vector $\mathbf{X} = (X_1, \dots, X_n)$, conditional on the event that $S > c$ for some large positive constant c . That is, we want to generate the value of a random vector whose density function is

$$f(x_1, \dots, x_n) = \frac{1}{P\{S > c\}} \prod_{i=1}^n \lambda_i e^{-\lambda_i x_i}, \quad x_i \geq 0, \sum_{i=1}^n x_i > c$$

This is easily accomplished by starting with an initial vector $\mathbf{x} = (x_1, \dots, x_n)$ satisfying $x_i > 0, i = 1, \dots, n, \sum_{i=1}^n x_i > c$. Then generate a random variable I that is equally likely to be any of $1, \dots, n$. Next, generate an exponential random variable X with rate λ_I conditional on the event that $X + \sum_{j \neq I} x_j > c$. This latter step, which calls for generating the value of an exponential random variable given that it exceeds $c - \sum_{j \neq I} x_j$, is easily accomplished by using the fact that an exponential conditioned to be greater than a positive constant is distributed as the constant plus the exponential.

Consequently, to obtain X , first generate an exponential random variable Y with rate λ_I , and then set

$$X = Y + \left(c - \sum_{j \neq I} x_j \right)^+$$

where $a^+ = \max(a, 0)$.

The value of x_I should then be reset as X and a new iteration of the algorithm begun. ■

Remark. As can be seen by Examples 4.42 and 4.43, although the theory for the Gibbs sampler was represented under the assumption that the distribution to be generated was discrete, it also holds when this distribution is continuous.

4.10 Markov Decision Processes

Consider a process that is observed at discrete time points to be in any one of M possible states, which we number by $1, 2, \dots, M$. After observing the state of the process, an action must be chosen, and we let A , assumed finite, denote the set of all possible actions.

If the process is in state i at time n and action a is chosen, then the next state of the system is determined according to the transition probabilities $P_{ij}(a)$. If we let X_n denote the state of the process at time n and a_n the action chosen at time n , then the preceding is equivalent to stating that

$$P\{X_{n+1} = j | X_0, a_0, X_1, a_1, \dots, X_n = i, a_n = a\} = P_{ij}(a)$$

Thus, the transition probabilities are functions only of the present state and the subsequent action.

By a policy, we mean a rule for choosing actions. We shall restrict ourselves to policies that are of the form that the action they prescribe at any time depends only on the state of the process at that time (and not on any information concerning prior states and actions). However, we shall allow the policy to be “randomized” in that its instructions may be to choose actions according to a probability distribution. In other words, a policy β is a set of numbers $\beta = \{\beta_i(a), a \in A, i = 1, \dots, M\}$ with the interpretation that if the process is in state i , then action a is to be chosen with probability $\beta_i(a)$. Of course, we need have

$$\begin{aligned} 0 &\leq \beta_i(a) \leq 1, & \text{for all } i, a \\ \sum_a \beta_i(a) &= 1, & \text{for all } i \end{aligned}$$

Under any given policy β , the sequence of states $\{X_n, n = 0, 1, \dots\}$ constitutes a Markov chain with transition probabilities $P_{ij}(\beta)$ given by²

$$\begin{aligned} P_{ij}(\beta) &= P_{\beta}\{X_{n+1} = j | X_n = i\} \\ &= \sum_a P_{ij}(a) \beta_i(a) \end{aligned}$$

where the last equality follows by conditioning on the action chosen when in state i . Let us suppose that for every choice of a policy β , the resultant Markov chain $\{X_n, n = 0, 1, \dots\}$ is ergodic.

For any policy β , let π_{ia} denote the limiting (or steady-state) probability that the process will be in state i and action a will be chosen if policy β is employed. That is,

$$\pi_{ia} = \lim_{n \rightarrow \infty} P_{\beta}\{X_n = i, a_n = a\}$$

The vector $\pi = (\pi_{ia})$ must satisfy

$$\begin{aligned} \text{(i)} \quad & \pi_{ia} \geq 0 \quad \text{for all } i, a, \\ \text{(ii)} \quad & \sum_i \sum_a \pi_{ia} = 1, \\ \text{(iii)} \quad & \sum_a \pi_{ja} = \sum_i \sum_a \pi_{ia} P_{ij}(a) \quad \text{for all } j \end{aligned} \tag{4.33}$$

Eqs. (i) and (ii) are obvious, and Eq. (iii), which is an analogue of Theorem 4.1, follows as the left-hand side equals the steady-state probability of being in state j and the right-hand side is the same probability computed by conditioning on the state and action chosen one stage earlier.

Thus for any policy β , there is a vector $\pi = (\pi_{ia})$ that satisfies (i)–(iii) and with the interpretation that π_{ia} is equal to the steady-state probability of being in state i and choosing action a when policy β is employed. Moreover, it turns out that the reverse is also true. Namely, for any vector $\pi = (\pi_{ia})$ that satisfies (i)–(iii), there exists a policy β such that if β is used, then the steady-state probability of being in i and choosing action a equals π_{ia} . To verify this last statement, suppose that $\pi = (\pi_{ia})$ is a vector that satisfies (i)–(iii). Then, let the policy $\beta = (\beta_i(a))$ be

$$\begin{aligned} \beta_i(a) &= P\{\beta \text{ chooses } a | \text{state is } i\} \\ &= \frac{\pi_{ia}}{\sum_a \pi_{ia}} \end{aligned}$$

Now let P_{ia} denote the limiting probability of being in i and choosing a when policy β is employed. We need to show that $P_{ia} = \pi_{ia}$. To do so, first note that $\{P_{ia}, i = 1, \dots, M, a \in A\}$ are the limiting probabilities of the two-dimensional Markov chain $\{(X_n, a_n), n \geq 0\}$. Hence, by the fundamental Theorem 4.1, they are the unique solu-

² We use the notation P_{β} to signify that the probability is conditional on the fact that policy β is used.

tion of

$$\begin{aligned} \text{(i')} \quad & P_{ia} \geq 0, \\ \text{(ii')} \quad & \sum_i \sum_a P_{ia} = 1, \\ \text{(iii')} \quad & P_{ja} = \sum_i \sum_{a'} P_{ia'} P_{ij}(a') \beta_j(a) \end{aligned}$$

where (iii') follows since

$$P\{X_{n+1} = j, a_{n+1} = a | X_n = i, a_n = a'\} = P_{ij}(a') \beta_j(a)$$

Because

$$\beta_j(a) = \frac{\pi_{ja}}{\sum_a \pi_{ja}}$$

we see that (P_{ia}) is the unique solution of

$$\begin{aligned} P_{ia} &\geq 0, \\ \sum_i \sum_a P_{ia} &= 1, \\ P_{ja} &= \sum_i \sum_{a'} P_{ia'} P_{ij}(a') \frac{\pi_{ja}}{\sum_a \pi_{ja}} \end{aligned}$$

Hence, to show that $P_{ia} = \pi_{ia}$, we need show that

$$\begin{aligned} \pi_{ia} &\geq 0, \\ \sum_i \sum_a \pi_{ia} &= 1, \\ \pi_{ja} &= \sum_i \sum_{a'} \pi_{ia'} P_{ij}(a') \frac{\pi_{ja}}{\sum_a \pi_{ja}} \end{aligned}$$

The top two equations follow from (i) and (ii) of Eq. (4.33), and the third, which is equivalent to

$$\sum_a \pi_{ja} = \sum_i \sum_{a'} \pi_{ia'} P_{ij}(a')$$

follows from condition (iii) of Eq. (4.33).

Thus we have shown that a vector $\pi = (\pi_{ia})$ will satisfy (i), (ii), and (iii) of Eq. (4.33) if and only if there exists a policy β such that π_{ia} is equal to the steady-state probability of being in state i and choosing action a when β is used. In fact, the policy β is defined by $\beta_i(a) = \pi_{ia} / \sum_a \pi_{ia}$.

The preceding is quite important in the determination of “optimal” policies. For instance, suppose that a reward $R(i, a)$ is earned whenever action a is chosen in state i . Since $R(X_i, a_i)$ would then represent the reward earned at time i , the expected average

reward per unit time under policy β can be expressed as

$$\text{expected average reward under } \beta = \lim_{n \rightarrow \infty} E_{\beta} \left[\frac{\sum_{i=1}^n R(X_i, a_i)}{n} \right]$$

Now, if π_{ia} denotes the steady-state probability of being in state i and choosing action a , it follows that the limiting expected reward at time n equals

$$\lim_{n \rightarrow \infty} E[R(X_n, a_n)] = \sum_i \sum_a \pi_{ia} R(i, a)$$

which implies that

$$\text{expected average reward under } \beta = \sum_i \sum_a \pi_{ia} R(i, a)$$

Hence, the problem of determining the policy that maximizes the expected average reward is

$$\begin{aligned} & \text{maximize } \sum_i \sum_a \pi_{ia} R(i, a) \\ & \text{subject to } \pi_{ia} \geq 0, \quad \text{for all } i, a, \\ & \sum_i \sum_a \pi_{ia} = 1, \\ & \sum_a \pi_{ja} = \sum_i \sum_a \pi_{ia} P_{ij}(a), \quad \text{for all } j \end{aligned} \tag{4.34}$$

However, the preceding maximization problem is a special case of what is known as a *linear program* and can be solved by a standard linear programming algorithm known as the *simplex algorithm*.³ If $\pi^* = (\pi_{ia}^*)$ maximizes the preceding, then the optimal policy will be given by β^* where

$$\beta_i^*(a) = \frac{\pi_{ia}^*}{\sum_a \pi_{ia}^*}$$

- Remarks.** (i) It can be shown that there is a π^* maximizing Eq. (4.34) that has the property that for each i , π_{ia}^* is zero for all but one value of a , which implies that the optimal policy is nonrandomized. That is, the action it prescribes when in state i is a deterministic function of i .
- (ii) The linear programming formulation also often works when there are restrictions placed on the class of allowable policies. For instance, suppose there is a restriction on the fraction of time the process spends in some state, say, state 1. Specifically, suppose that we are allowed to consider only policies having the

³ It is called a linear program since the objective function $\sum_i \sum_a R(i, a) \pi_{ia}$ and the constraints are all linear functions of the π_{ia} . For a heuristic analysis of the simplex algorithm, see Section 4.5.2.

property that their use results in the process being in state 1 less than 100α percent of time. To determine the optimal policy subject to this requirement, we add to the linear programming problem the additional constraint

$$\sum_a \pi_{1a} \leq \alpha$$

since $\sum_a \pi_{1a}$ represents the proportion of time that the process is in state 1.

4.11 Hidden Markov Chains

Let $\{X_n, n = 1, 2, \dots\}$ be a Markov chain with transition probabilities $P_{i,j}$ and initial state probabilities $p_i = P\{X_1 = i\}$, $i \geq 0$. Suppose that there is a finite set $\mathcal{S}$ of signals, and that a signal from $\mathcal{S}$ is emitted each time the Markov chain enters a state. Further, suppose that when the Markov chain enters state j then, independently of previous Markov chain states and signals, the signal emitted is s with probability $p(s|j)$, $\sum_{s \in \mathcal{S}} p(s|j) = 1$. That is, if S_n represents the n th signal emitted, then

$$\begin{aligned} P\{S_1 = s | X_1 = j\} &= p(s|j), \\ P\{S_n = s | X_1, S_1, \dots, X_{n-1}, S_{n-1}, X_n = j\} &= p(s|j) \end{aligned}$$

A model of the preceding type in which the sequence of signals $S_1, S_2, \dots$ is observed, while the sequence of underlying Markov chain states $X_1, X_2, \dots$ is unobserved, is called a *hidden Markov chain* model.

Example 4.44. Consider a production process that in each period is either in a good state (state 1) or in a poor state (state 2). If the process is in state 1 during a period then, independent of the past, with probability 0.9 it will be in state 1 during the next period and with probability 0.1 it will be in state 2. Once in state 2, it remains in that state forever. Suppose that a single item is produced each period and that each item produced when the process is in state 1 is of acceptable quality with probability 0.99, while each item produced when the process is in state 2 is of acceptable quality with probability 0.96.

If the status, either acceptable or unacceptable, of each successive item is observed, while the process states are unobservable, then the preceding is a hidden Markov chain model. The signal is the status of the item produced, and has value either a or u , depending on whether the item is acceptable or unacceptable. The signal probabilities are

$$\begin{aligned} p(u|1) &= 0.01, & p(a|1) &= 0.99, \\ p(u|2) &= 0.04, & p(a|2) &= 0.96 \end{aligned}$$

while the transition probabilities of the underlying Markov chain are

$$P_{1,1} = 0.9 = 1 - P_{1,2}, \quad P_{2,2} = 1$$



Although $\{S_n, n \geq 1\}$ is not a Markov chain, it should be noted that, conditional on the current state X_n , the sequence $S_n, X_{n+1}, S_{n+1}, \dots$ of future signals and states is independent of the sequence $X_1, S_1, \dots, X_{n-1}, S_{n-1}$ of past states and signals.

Let $\mathbf{S}^n = (S_1, \dots, S_n)$ be the random vector of the first n signals. For a fixed sequence of signals $s_1, \dots, s_n$, let $\mathbf{s}_k = (s_1, \dots, s_k), k \leq n$. To begin, let us determine the conditional probability of the Markov chain state at time n given that $\mathbf{S}^n = \mathbf{s}_n$. To obtain this probability, let

$$F_n(j) = P\{\mathbf{S}^n = \mathbf{s}_n, X_n = j\}$$

and note that

$$\begin{aligned} P\{X_n = j | \mathbf{S}^n = \mathbf{s}_n\} &= \frac{P\{\mathbf{S}^n = \mathbf{s}_n, X_n = j\}}{P\{\mathbf{S}^n = \mathbf{s}_n\}} \\ &= \frac{F_n(j)}{\sum_i F_n(i)} \end{aligned}$$

Now,

$$\begin{aligned} F_n(j) &= P\{\mathbf{S}^{n-1} = \mathbf{s}_{n-1}, S_n = s_n, X_n = j\} \\ &= \sum_i P\{\mathbf{S}^{n-1} = \mathbf{s}_{n-1}, X_{n-1} = i, X_n = j, S_n = s_n\} \\ &= \sum_i F_{n-1}(i) P\{X_n = j, S_n = s_n | \mathbf{S}^{n-1} = \mathbf{s}_{n-1}, X_{n-1} = i\} \\ &= \sum_i F_{n-1}(i) P\{X_n = j, S_n = s_n | X_{n-1} = i\} \\ &= \sum_i F_{n-1}(i) P_{i,j} p(s_n | j) \\ &= p(s_n | j) \sum_i F_{n-1}(i) P_{i,j} \end{aligned} \tag{4.35}$$

where the preceding used that

$$\begin{aligned} P\{X_n = j, S_n = s_n | X_{n-1} = i\} &= P\{X_n = j | X_{n-1} = i\} \times P\{S_n = s_n | X_n = j, X_{n-1} = i\} \\ &= P_{i,j} P\{S_n = s_n | X_n = j\} \\ &= P_{i,j} p(s_n | j) \end{aligned}$$

Starting with

$$F_1(i) = P\{X_1 = i, S_1 = s_1\} = p_i p(s_1 | i)$$

we can use Eq. (4.35) to recursively determine the functions $F_2(i), F_3(i), \dots$, up to $F_n(i)$.

Example 4.45. Suppose in Example 4.44 that $P\{X_1 = 1\} = 0.8$. It is given that the successive conditions of the first three items produced are a, u, a .

- (a) What is the probability that the process was in its good state when the third item was produced?
- (b) What is the probability that X_4 is 1?
- (c) What is the probability that the next item produced is acceptable?

Solution: With $\mathbf{s}_3 = (a, u, a)$, we have

$$F_1(1) = (0.8)(0.99) = 0.792,$$

$$F_1(2) = (0.2)(0.96) = 0.192$$

$$F_2(1) = 0.01[0.792(0.9) + 0.192(0)] = 0.007128,$$

$$F_2(2) = 0.04[0.792(0.1) + 0.192(1)] = 0.010848$$

$$F_3(1) = 0.99[(0.007128)(0.9)] \approx 0.006351,$$

$$F_3(2) = 0.96[(0.007128)(0.1) + 0.010848] \approx 0.011098$$

Therefore, the answer to part (a) is

$$P\{X_3 = 1|\mathbf{s}_3\} \approx \frac{0.006351}{0.006351 + 0.011098} \approx 0.364$$

To compute $P\{X_4 = 1|\mathbf{s}_3\}$, condition on X_3 to obtain

$$\begin{aligned} P\{X_4 = 1|\mathbf{s}_3\} &= P\{X_4 = 1|X_3 = 1, \mathbf{s}_3\}P\{X_3 = 1|\mathbf{s}_3\} \\ &\quad + P\{X_4 = 1|X_3 = 2, \mathbf{s}_3\}P\{X_3 = 2|\mathbf{s}_3\} \\ &= P\{X_4 = 1|X_3 = 1, \mathbf{s}_3\}(0.364) \\ &\quad + P\{X_4 = 1|X_3 = 2, \mathbf{s}_3\}(0.636) \\ &= 0.364P_{1,1} + 0.636P_{2,1} \\ &= 0.3276 \end{aligned}$$

To compute $P\{S_4 = a|\mathbf{s}_3\}$, condition on X_4 to obtain

$$\begin{aligned} P\{S_4 = a|\mathbf{s}_3\} &= P\{S_4 = a|X_4 = 1, \mathbf{s}_3\}P\{X_4 = 1|\mathbf{s}_3\} \\ &\quad + P\{S_4 = a|X_4 = 2, \mathbf{s}_3\}P\{X_4 = 2|\mathbf{s}_3\} \\ &= P\{S_4 = a|X_4 = 1\}(0.3276) \\ &\quad + P\{S_4 = a|X_4 = 2\}(1 - 0.3276) \\ &= (0.99)(0.3276) + (0.96)(0.6724) = 0.9698 \end{aligned} \quad \blacksquare$$

To compute $P\{\mathbf{S}^n = \mathbf{s}_n\}$, use the identity $P\{\mathbf{S}^n = \mathbf{s}_n\} = \sum_i F_n(i)$ along with Eq. (4.35). If there are N states of the Markov chain, this requires computing nN quantities $F_n(i)$, with each computation requiring a summation over N terms. This

can be compared with a computation of $P\{\mathbf{S}^n = \mathbf{s}_n\}$ based on conditioning on the first n states of the Markov chain to obtain

$$\begin{aligned} P\{\mathbf{S}^n = \mathbf{s}_n\} &= \sum_{i_1, \dots, i_n} P\{\mathbf{S}^n = \mathbf{s}_n | X_1 = i_1, \dots, X_n = i_n\} P\{X_1 = i_1, \dots, X_n = i_n\} \\ &= \sum_{i_1, \dots, i_n} p(s_1 | i_1) \cdots p(s_n | i_n) p_{i_1} P_{i_1, i_2} P_{i_2, i_3} \cdots P_{i_{n-1}, i_n} \end{aligned}$$

The use of the preceding identity to compute $P\{\mathbf{S}^n = \mathbf{s}_n\}$ would thus require a summation over N^n terms, with each term being a product of $2n$ values, indicating that it is not competitive with the previous approach.

The computation of $P\{\mathbf{S}^n = \mathbf{s}_n\}$ by recursively determining the functions $F_k(i)$ is known as the *forward approach*. There also is a *backward approach*, which is based on the quantities $B_k(i)$, defined by

$$B_k(i) = P\{S_{k+1} = s_{k+1}, \dots, S_n = s_n | X_k = i\}$$

A recursive formula for $B_k(i)$ can be obtained by conditioning on X_{k+1} .

$$\begin{aligned} B_k(i) &= \sum_j P\{S_{k+1} = s_{k+1}, \dots, S_n = s_n | X_k = i, X_{k+1} = j\} \\ &\quad \times P\{X_{k+1} = j | X_k = i\} \\ &= \sum_j P\{S_{k+1} = s_{k+1}, \dots, S_n = s_n | X_{k+1} = j\} P_{i,j} \\ &= \sum_j P\{S_{k+1} = s_{k+1} | X_{k+1} = j\} \\ &\quad \times P\{S_{k+2} = s_{k+2}, \dots, S_n = s_n | S_{k+1} = s_{k+1}, X_{k+1} = j\} P_{i,j} \\ &= \sum_j p(s_{k+1} | j) P\{S_{k+2} = s_{k+2}, \dots, S_n = s_n | X_{k+1} = j\} P_{i,j} \\ &= \sum_j p(s_{k+1} | j) B_{k+1}(j) P_{i,j} \end{aligned} \tag{4.36}$$

Starting with

$$\begin{aligned} B_{n-1}(i) &= P\{S_n = s_n | X_{n-1} = i\} \\ &= \sum_j P_{i,j} p(s_n | j) \end{aligned}$$

we would then use Eq. (4.36) to determine the function $B_{n-2}(i)$, then $B_{n-3}(i)$, and so on, down to $B_1(i)$. This would then yield $P\{\mathbf{S}^n = \mathbf{s}_n\}$ via

$$P\{\mathbf{S}^n = \mathbf{s}_n\} = \sum_i P\{S_1 = s_1, \dots, S_n = s_n | X_1 = i\} p_i$$

$$\begin{aligned}
&= \sum_i P\{S_1 = s_1 | X_1 = i\} \\
&\quad \times P\{S_2 = s_2, \dots, S_n = s_n | S_1 = s_1, X_1 = i\} p_i \\
&= \sum_i p(s_1 | i) P\{S_2 = s_2, \dots, S_n = s_n | X_1 = i\} p_i \\
&= \sum_i p(s_1 | i) B_1(i) p_i
\end{aligned}$$

Another approach to obtaining $P\{\mathbf{S}^n = \mathbf{s}_n\}$ is to combine both the forward and backward approaches. Suppose that for some k we have computed both functions $F_k(j)$ and $B_k(j)$. Because

$$\begin{aligned}
P\{\mathbf{S}^n = \mathbf{s}_n, X_k = j\} &= P\{\mathbf{S}^k = \mathbf{s}_k, X_k = j\} \\
&\quad \times P\{S_{k+1} = s_{k+1}, \dots, S_n = s_n | \mathbf{S}^k = \mathbf{s}_k, X_k = j\} \\
&= P\{\mathbf{S}^k = \mathbf{s}_k, X_k = j\} P\{S_{k+1} = s_{k+1}, \dots, S_n = s_n | X_k = j\} \\
&= F_k(j) B_k(j)
\end{aligned}$$

we see that

$$P\{\mathbf{S}^n = \mathbf{s}_n\} = \sum_j F_k(j) B_k(j)$$

The beauty of using the preceding identity to determine $P\{\mathbf{S}^n = \mathbf{s}_n\}$ is that we may simultaneously compute the sequence of forward functions, starting with F_1 , as well as the sequence of backward functions, starting at B_{n-1} . The parallel computations can then be stopped once we have computed both F_k and B_k for some k .

4.11.1 Predicting the States

Suppose the first n observed signals are $\mathbf{s}_n = (s_1, \dots, s_n)$, and that given this data we want to predict the first n states of the Markov chain. The best predictor depends on what we are trying to accomplish. If our objective is to maximize the expected number of states that are correctly predicted, then for each $k = 1, \dots, n$ we need to compute $P\{X_k = j | \mathbf{S}^n = \mathbf{s}_n\}$ and then let the value of j that maximizes this quantity be the predictor of X_k . (That is, we take the mode of the conditional probability mass function of X_k , given the sequence of signals, as the predictor of X_k .) To do so, we must first compute this conditional probability mass function, which is accomplished as follows. For $k \leq n$,

$$\begin{aligned}
P\{X_k = j | \mathbf{S}^n = \mathbf{s}_n\} &= \frac{P\{\mathbf{S}^n = \mathbf{s}_n, X_k = j\}}{P\{\mathbf{S}^n = \mathbf{s}_n\}} \\
&= \frac{F_k(j) B_k(j)}{\sum_j F_k(j) B_k(j)}
\end{aligned}$$

Thus, given that $\mathbf{S}^n = \mathbf{s}_n$, the optimal predictor of X_k is the value of j that maximizes $F_k(j)B_k(j)$.

A different variant of the prediction problem arises when we regard the sequence of states as a single entity. In this situation, our objective is to choose that sequence of states whose conditional probability, given the sequence of signals, is maximal. For instance, in signal processing, while $X_1, \dots, X_n$ might be the actual message sent, $S_1, \dots, S_n$ would be what is received, and so the objective would be to predict the actual message in its entirety.

Letting $\mathbf{X}_k = (X_1, \dots, X_k)$ be the vector of the first k states, the problem of interest is to find the sequence of states $i_1, \dots, i_n$ that maximizes $P\{\mathbf{X}_n = (i_1, \dots, i_n) | \mathbf{S}^n = \mathbf{s}_n\}$. Because

$$P\{\mathbf{X}_n = (i_1, \dots, i_n) | \mathbf{S}^n = \mathbf{s}_n\} = \frac{P\{\mathbf{X}_n = (i_1, \dots, i_n), \mathbf{S}^n = \mathbf{s}_n\}}{P\{\mathbf{S}^n = \mathbf{s}_n\}}$$

this is equivalent to finding the sequence of states $i_1, \dots, i_n$ that maximizes $P\{\mathbf{X}_n = (i_1, \dots, i_n), \mathbf{S}^n = \mathbf{s}_n\}$.

To solve the preceding problem let, for $k \leq n$,

$$V_k(j) = \max_{i_1, \dots, i_{k-1}} P\{\mathbf{X}_{k-1} = (i_1, \dots, i_{k-1}), X_k = j, \mathbf{S}^k = \mathbf{s}_k\}$$

To recursively solve for $V_k(j)$, use that

$$\begin{aligned} V_k(j) &= \max_i \max_{i_1, \dots, i_{k-2}} P\{\mathbf{X}_{k-2} = (i_1, \dots, i_{k-2}), X_{k-1} = i, X_k = j, \mathbf{S}^k = \mathbf{s}_k\} \\ &= \max_i \max_{i_1, \dots, i_{k-2}} P\{\mathbf{X}_{k-2} = (i_1, \dots, i_{k-2}), X_{k-1} = i, \mathbf{S}^{k-1} = \mathbf{s}_{k-1}, \\ &\quad X_k = j, S_k = s_k\} \\ &= \max_i \max_{i_1, \dots, i_{k-2}} P\{\mathbf{X}_{k-2} = (i_1, \dots, i_{k-2}), X_{k-1} = i, \mathbf{S}^{k-1} = \mathbf{s}_{k-1}\} \\ &\quad \times P\{X_k = j, S_k = s_k | \mathbf{X}_{k-2} = (i_1, \dots, i_{k-2}), X_{k-1} = i, \mathbf{S}^{k-1} = \mathbf{s}_{k-1}\} \\ &= \max_i \max_{i_1, \dots, i_{k-2}} P\{\mathbf{X}_{k-2} = (i_1, \dots, i_{k-2}), X_{k-1} = i, \mathbf{S}^{k-1} = \mathbf{s}_{k-1}\} \\ &\quad \times P\{X_k = j, S_k = s_k | X_{k-1} = i\} \\ &= \max_i P\{X_k = j, S_k = s_k | X_{k-1} = i\} \\ &\quad \times \max_{i_1, \dots, i_{k-2}} P\{\mathbf{X}_{k-2} = (i_1, \dots, i_{k-2}), X_{k-1} = i, \mathbf{S}^{k-1} = \mathbf{s}_{k-1}\} \\ &= \max_i P_{i,j} p(s_k | j) V_{k-1}(i) \\ &= p(s_k | j) \max_i P_{i,j} V_{k-1}(i) \end{aligned} \tag{4.37}$$

Starting with

$$V_1(j) = P\{X_1 = j, S_1 = s_1\} = p_j p(s_1 | j)$$

we now use the recursive identity (4.37) to determine $V_2(j)$ for each j ; then $V_3(j)$ for each j ; and so on, up to $V_n(j)$ for each j .

To obtain the maximizing sequence of states, we work in the reverse direction. Let j_n be the value (or any of the values if there are more than one) of j that maximizes $V_n(j)$. Thus j_n is the final state of a maximizing state sequence. Also, for $k < n$, let $i_k(j)$ be a value of i that maximizes $P_{i,j} V_k(i)$. Then

$$\begin{aligned}
 & \max_{i_1, \dots, i_n} P\{\mathbf{X}_n = (i_1, \dots, i_n), \mathbf{S}^n = \mathbf{s}_n\} \\
 &= \max_j V_n(j) \\
 &= V_n(j_n) \\
 &= \max_{i_1, \dots, i_{n-1}} P\{\mathbf{X}_n = (i_1, \dots, i_{n-1}, j_n), \mathbf{S}^n = \mathbf{s}_n\} \\
 &= p(s_n | j_n) \max_i P_{i, j_n} V_{n-1}(i) \\
 &= p(s_n | j_n) P_{i_{n-1}(j_n), j_n} V_{n-1}(i_{n-1}(j_n))
 \end{aligned}$$

Thus, $i_{n-1}(j_n)$ is the next to last state of the maximizing sequence. Continuing in this manner, the second from the last state of the maximizing sequence is $i_{n-2}(i_{n-1}(j_n))$, and so on.

The preceding approach to finding the most likely sequence of states given a prescribed sequence of signals is known as the *Viterbi Algorithm*.

Exercises

- *1. Three white and three black balls are distributed in two urns in such a way that each contains three balls. We say that the system is in state $i, i = 0, 1, 2, 3$, if the first urn contains i white balls. At each step, we draw one ball from each urn and place the ball drawn from the first urn into the second, and conversely with the ball from the second urn. Let X_n denote the state of the system after the n th step. Explain why $\{X_n, n = 0, 1, 2, \dots\}$ is a Markov chain and calculate its transition probability matrix.
2. Each individual in a population independently has a random number of offspring that is Poisson distributed with mean λ . Those initially present constitute the zeroth generation. Offspring of zeroth generation people constitute the first generation; their offspring constitute the second generation, and so on. If X_n denotes the size of generation n , is $\{X_n, n \geq 0\}$ a Markov chain. If it is, give its transition probabilities $P_{i,j}$; if it is not, explain why it is not.
3. There are k players, with player i having value $v_i > 0, i = 1, \dots, k$. In every period, two of the players play a game, while the other $k - 2$ wait in an ordered line. The loser of a game joins the end of the line, and the winner then plays a new game against the player who is first in line. Whenever i and j play, i wins with probability $\frac{v_i}{v_i + v_j}$.

- (a) Define a Markov chain that is useful in analyzing this model.
 - (b) How many states does the Markov chain have?
 - (c) Give the transition probabilities of the chain.
4. Let $\mathbf{P}$ and $\mathbf{Q}$ be transition probability matrices on states $1, \dots, m$, with respective transition probabilities $P_{i,j}$ and $Q_{i,j}$. Consider processes $\{X_n, n \geq 0\}$ and $\{Y_n, n \geq 0\}$ defined as follows:
- (a) $X_0 = 1$. A coin that comes up heads with probability p is then flipped. If the coin lands heads, the subsequent states $X_1, X_2, \dots$, are obtained by using the transition probability matrix $\mathbf{P}$; if it lands tails, the subsequent states $X_1, X_2, \dots$, are obtained by using the transition probability matrix $\mathbf{Q}$. (In other words, if the coin lands heads (tails) then the sequence of states is a Markov chain with transition probability matrix $\mathbf{P}$ ($\mathbf{Q}$).) Is $\{X_n, n \geq 0\}$ a Markov chain. If it is, give its transition probabilities. If it is not, tell why not.
 - (b) $Y_0 = 1$. If the current state is i , then the next state is determined by first flipping a coin that comes up heads with probability p . If the coin lands heads then the next state is j with probability $P_{i,j}$; if it lands tails then the next state is j with probability $Q_{i,j}$. Is $\{Y_n, n \geq 0\}$ a Markov chain. If it is, give its transition probabilities. If it is not, tell why not.
5. A Markov chain $\{X_n, n \geq 0\}$ with states $0, 1, 2$, has the transition probability matrix

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

If $P\{X_0 = 0\} = P\{X_0 = 1\} = \frac{1}{4}$, find $E[X_3]$.

6. Let the transition probability matrix of a two-state Markov chain be given, as in Example 4.2, by

$$\mathbf{P} = \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}$$

Show by mathematical induction that

$$\mathbf{P}^{(n)} = \begin{bmatrix} \frac{1}{2} + \frac{1}{2}(2p-1)^n & \frac{1}{2} - \frac{1}{2}(2p-1)^n \\ \frac{1}{2} - \frac{1}{2}(2p-1)^n & \frac{1}{2} + \frac{1}{2}(2p-1)^n \end{bmatrix}$$

7. In Example 4.4 suppose that it has rained neither yesterday nor the day before yesterday. What is the probability that it will rain tomorrow?
8. An urn initially contains 2 balls, one of which is red and the other blue. At each stage a ball is randomly selected. If the selected ball is red, then it is replaced with a red ball with probability .7 or with a blue ball with probability .3; if the selected ball is blue, then it is equally likely to be replaced by either a red or blue ball.

- (a) Let X_n equal 1 if the n th ball selected is red, and let it equal 0 otherwise. Is $\{X_n, n \geq 1\}$ a Markov chain? If so, give its transition probability matrix.
- (b) Let Y_n denote the number of red balls in the urn immediately before the n th ball is selected. Is $\{Y_n, n \geq 1\}$ a Markov chain? If so, give its transition probability matrix.
- (c) Find the probability that the second ball selected is red.
- (d) Find the probability that the fourth ball selected is red.
- *9. In a sequence of independent flips of a coin that comes up heads with probability .6, what is the probability that there is a run of three consecutive heads within the first 10 flips?
10. In Example 4.3, Gary is currently in a cheerful mood. What is the probability that he is not in a glum mood on any of the following three days?
11. In Example 4.13, give the transition probabilities of the Y_n Markov chain in terms of the transition probabilities $P_{i,j}$ of the X_n chain.
12. For a Markov chain $\{X_n, n \geq 0\}$ with transition probabilities $P_{i,j}$, consider the conditional probability that $X_n = m$ given that the chain started at time 0 in state i and has not yet entered state r by time n , where r is a specified state not equal to either i or m . We are interested in whether this conditional probability is equal to the n stage transition probability of a Markov chain whose state space does not include state r and whose transition probabilities are

$$Q_{i,j} = \frac{P_{i,j}}{1 - P_{i,r}}, \quad i, j \neq r$$

Either prove the equality

$$P\{X_n = m | X_0 = i, X_k \neq r, k = 1, \dots, n\} = Q_{i,m}^n$$

or construct a counterexample.

13. Let $\mathbf{P}$ be the transition probability matrix of a Markov chain. Argue that if for some positive integer r , $\mathbf{P}^r$ has all positive entries, then so does $\mathbf{P}^n$, for all integers $n \geq r$.
14. Specify the classes of the following Markov chains, and determine whether they are transient or recurrent:

$$\mathbf{P}_1 = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}, \quad \mathbf{P}_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix},$$

$$\mathbf{P}_3 = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}, \quad \mathbf{P}_4 = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & \frac{2}{3} & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

15. Prove that if the number of states in a Markov chain is M , and if state j can be reached from state i , then it can be reached in M steps or less.
- *16. Show that if state i is recurrent and state i does not communicate with state j , then $P_{ij} = 0$. This implies that once a process enters a recurrent class of states it can never leave that class. For this reason, a recurrent class is often referred to as a *closed* class.
17. For the random walk of Example 4.19 use the strong law of large numbers to give another proof that the Markov chain is transient when $p \neq \frac{1}{2}$.
- Hint:** Note that the state at time n can be written as $\sum_{i=1}^n Y_i$ where the Y_i s are independent and $P\{Y_i = 1\} = p = 1 - P\{Y_i = -1\}$. Argue that if $p > \frac{1}{2}$, then, by the strong law of large numbers, $\sum_{i=1}^n Y_i \rightarrow \infty$ as $n \rightarrow \infty$ and hence the initial state 0 can be visited only finitely often, and hence must be transient. A similar argument holds when $p < \frac{1}{2}$.
18. Coin 1 comes up heads with probability 0.6 and coin 2 with probability 0.5. A coin is continually flipped until it comes up tails, at which time that coin is put aside and we start flipping the other one.
- (a) What proportion of flips use coin 1?
- (b) If we start the process with coin 1 what is the probability that coin 2 is used on the fifth flip?
- (c) What proportion of flips land heads?
19. For Example 4.4, calculate the proportion of days that it rains.
20. A transition probability matrix $\mathbf{P}$ is said to be doubly stochastic if the sum over each column equals one; that is,

$$\sum_i P_{ij} = 1, \quad \text{for all } j$$

If such a chain is irreducible and consists of $M + 1$ states $0, 1, \dots, M$, show that the long-run proportions are given by

$$\pi_j = \frac{1}{M+1}, \quad j = 0, 1, \dots, M$$

- *21. A DNA nucleotide has any of four values. A standard model for a mutational change of the nucleotide at a specific location is a Markov chain model that supposes that in going from period to period the nucleotide does not change with probability $1 - 3\alpha$, and if it does change then it is equally likely to change to any of the other three values, for some $0 < \alpha < \frac{1}{3}$.

(a) Show that $P_{1,1}^n = \frac{1}{4} + \frac{3}{4}(1 - 4\alpha)^n$.

(b) What is the long-run proportion of time the chain is in each state?

22. Let Y_n be the sum of n independent rolls of a fair die. Find

$$\lim_{n \rightarrow \infty} P\{Y_n \text{ is a multiple of } 13\}$$

Hint: Define an appropriate Markov chain and apply the results of Exercise 20.

23. In a good weather year the number of storms is Poisson distributed with mean 1; in a bad year it is Poisson distributed with mean 3. Suppose that any year's weather conditions depends on past years only through the previous year's condition. Suppose that a good year is equally likely to be followed by either a good or a bad year, and that a bad year is twice as likely to be followed by a bad year as by a good year. Suppose that last year—call it year 0—was a good year.

(a) Find the expected total number of storms in the next two years (that is, in years 1 and 2).

(b) Find the probability there are no storms in year 3.

(c) Find the long-run average number of storms per year.

(d) Find the proportion of years that have no storms.

24. Consider three urns, one colored red, one white, and one blue. The red urn contains 1 red and 4 blue balls; the white urn contains 3 white balls, 2 red balls, and 2 blue balls; the blue urn contains 4 white balls, 3 red balls, and 2 blue balls. At the initial stage, a ball is randomly selected from the red urn and then returned to that urn. At every subsequent stage, a ball is randomly selected from the urn whose color is the same as that of the ball previously selected and is then returned to that urn. In the long run, what proportion of the selected balls are red? What proportion are white? What proportion are blue?

25. Each morning an individual leaves his house and goes for a run. He is equally likely to leave either from his front or back door. Upon leaving the house, he chooses a pair of running shoes (or goes running barefoot if there are no shoes at the door from which he departed). On his return he is equally likely to enter, and leave his running shoes, either by the front or back door. If he owns a total of k pairs of running shoes, what proportion of the time does he run barefooted?

26. Consider the following approach to shuffling a deck of n cards. Starting with any initial ordering of the cards, one of the numbers $1, 2, \dots, n$ is randomly chosen in such a manner that each one is equally likely to be selected. If number i is chosen, then we take the card that is in position i and put it on top of the deck—that is, we put that card in position 1. We then repeatedly perform the same operation. Show that, in the limit, the deck is perfectly shuffled in the sense that the resultant ordering is equally likely to be any of the $n!$ possible orderings.

*27. Each individual in a population of size N is, in each period, either active or inactive. If an individual is active in a period then, independent of all else, that individual will be active in the next period with probability α . Similarly, if an

individual is inactive in a period then, independent of all else, that individual will be inactive in the next period with probability β . Let X_n denote the number of individuals that are active in period n .

- (a) Argue that $X_n, n \geq 0$ is a Markov chain.
- (b) Find $E[X_n | X_0 = i]$.
- (c) Derive an expression for its transition probabilities.
- (d) Find the long-run proportion of time that exactly j people are active.

Hint for (d): Consider first the case where $N = 1$.

- 28. Every time that the team wins a game, it wins its next game with probability 0.8; every time it loses a game, it wins its next game with probability 0.3. If the team wins a game, then it has dinner together with probability 0.7, whereas if the team loses then it has dinner together with probability 0.2. What proportion of games result in a team dinner?
- 29. An organization has N employees where N is a large number. Each employee has one of three possible job classifications and changes classifications (independently) according to a Markov chain with transition probabilities

$$\begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

What percentage of employees are in each classification?

- 30. Three out of every four trucks on the road are followed by a car, while only one out of every five cars is followed by a truck. What fraction of vehicles on the road are trucks?
- 31. A certain town never has two sunny days in a row. Each day is classified as being either sunny, cloudy (but dry), or rainy. If it is sunny one day, then it is equally likely to be either cloudy or rainy the next day. If it is rainy or cloudy one day, then there is one chance in two that it will be the same the next day, and if it changes then it is equally likely to be either of the other two possibilities. In the long run, what proportion of days are sunny? What proportion are cloudy?
- *32. Each of two switches is either on or off during a day. On day n , each switch will independently be on with probability

$$[1 + \text{number of on switches during day } n - 1]/4$$

For instance, if both switches are on during day $n - 1$, then each will independently be on during day n with probability $3/4$. What fraction of days are both switches on? What fraction are both off?

- 33. Two players are playing a sequence of points, which begin when one of the players serves. Suppose that player 1 wins each point she serves with probability p , and wins each point her opponent serves with probability q . Suppose the winner of a point becomes the server of the next point.
 - (a) Find the proportion of points that are won by player 1.
 - (b) Find the proportion of time that player 1 is the server.

34. A flea moves around the vertices of a triangle in the following manner: Whenever it is at vertex i it moves to its clockwise neighbor vertex with probability p_i and to the counterclockwise neighbor with probability $q_i = 1 - p_i$, $i = 1, 2, 3$.
- Find the proportion of time that the flea is at each of the vertices.
 - How often does the flea make a counterclockwise move that is then followed by five consecutive clockwise moves?
35. Consider a Markov chain with states $0, 1, 2, 3, 4$. Suppose $P_{0,4} = 1$; and suppose that when the chain is in state i , $i > 0$, the next state is equally likely to be any of the states $0, 1, \dots, i - 1$. Find the limiting probabilities of this Markov chain.
36. The state of a process changes daily according to a two-state Markov chain. If the process is in state i during one day, then it is in state j the following day with probability $P_{i,j}$, where

$$P_{0,0} = 0.4, \quad P_{0,1} = 0.6, \quad P_{1,0} = 0.2, \quad P_{1,1} = 0.8$$

Every day a message is sent. If the state of the Markov chain that day is i then the message sent is “good” with probability p_i and is “bad” with probability $q_i = 1 - p_i$, $i = 0, 1$

- If the process is in state 0 on Monday, what is the probability that a good message is sent on Tuesday?
 - If the process is in state 0 on Monday, what is the probability that a good message is sent on Friday?
 - In the long run, what proportion of messages are good?
 - Let Y_n equal 1 if a good message is sent on day n and let it equal 2 otherwise. Is $\{Y_n, n \geq 1\}$ a Markov chain? If so, give its transition probability matrix. If not, briefly explain why not.
37. Show that the stationary probabilities for the Markov chain having transition probabilities $P_{i,j}$ are also the stationary probabilities for the Markov chain whose transition probabilities $Q_{i,j}$ are given by

$$Q_{i,j} = P_{i,j}^k$$

for any specified positive integer k .

38. Capa plays either one or two chess games every day, with the number of games that she plays on successive days being a Markov chain with transition probabilities

$$P_{1,1} = .2, \quad P_{1,2} = .8 \quad P_{2,1} = .4, \quad P_{2,2} = .6$$

Capa wins each game with probability p . Suppose she plays two games on Monday.

- What is the probability that she wins all the games she plays on Tuesday?
- What is the expected number of games that she plays on Wednesday?
- In the long run, on what proportion of days does Capa win all her games.

39. Consider the one-dimensional symmetric random walk of Example 4.19, which was shown in that example to be recurrent. Let π_i denote the long-run proportion of time that the chain is in state i .
- (a) Argue that $\pi_i = \pi_0$ for all i .
 - (b) Show that $\sum_i \pi_i \neq 1$.
 - (c) Conclude that this Markov chain is null recurrent, and thus all $\pi_i = 0$.
40. A particle moves on 12 points situated on a circle. At each step it is equally likely to move one step in the clockwise or in the counterclockwise direction. Find the mean number of steps for the particle to return to its starting position.
- *41. Consider a Markov chain with states equal to the nonnegative integers, and suppose its transition probabilities satisfy $P_{i,j} = 0$ if $j \leq i$. Assume $X_0 = 0$, and let e_j be the probability that the Markov chain is ever in state j . (Note that $e_0 = 1$ because $X_0 = 0$.) Argue that for $j > 0$

$$e_j = \sum_{i=0}^{j-1} e_i P_{i,j}$$

If $P_{i,i+k} = 1/3$, $k = 1, 2, 3$, find e_i for $i = 1, \dots, 10$.

42. Let A be a set of states, and let A^c be the remaining states.
- (a) What is the interpretation of

$$\sum_{i \in A} \sum_{j \in A^c} \pi_i P_{ij}?$$

- (b) What is the interpretation of

$$\sum_{i \in A^c} \sum_{j \in A} \pi_i P_{ij}?$$

- (c) Explain the identity

$$\sum_{i \in A} \sum_{j \in A^c} \pi_i P_{ij} = \sum_{i \in A^c} \sum_{j \in A} \pi_i P_{ij}$$

43. Each day, one of n possible elements is requested, the i th one with probability P_i , $i \geq 1$, $\sum_1^n P_i = 1$. These elements are at all times arranged in an ordered list that is revised as follows: The element selected is moved to the front of the list with the relative positions of all the other elements remaining unchanged. Define the state at any time to be the list ordering at that time and note that there are $n!$ possible states.
- (a) Argue that the preceding is a Markov chain.
 - (b) For any state $i_1, \dots, i_n$ (which is a permutation of $1, 2, \dots, n$), let $\pi(i_1, \dots, i_n)$ denote the limiting probability. In order for the state to be $i_1, \dots, i_n$, it is necessary for the last request to be for i_1 , the last non- i_1 request for i_2 , the last non- i_1 or i_2 request for i_3 , and so on. Hence, it

appears intuitive that

$$\pi(i_1, \dots, i_n) = P_{i_1} \frac{P_{i_2}}{1 - P_{i_1}} \frac{P_{i_3}}{1 - P_{i_1} - P_{i_2}} \cdots \frac{P_{i_{n-1}}}{1 - P_{i_1} - \cdots - P_{i_{n-2}}}$$

Verify when $n = 3$ that the preceding are indeed the limiting probabilities.

44. Suppose that a population consists of a fixed number, say, m , of genes in any generation. Each gene is one of two possible genetic types. If exactly i (of the m) genes of any generation are of type 1, then the next generation will have j type 1 (and $m - j$ type 2) genes with probability

$$\binom{m}{j} \left(\frac{i}{m}\right)^j \left(\frac{m-i}{m}\right)^{m-j}, \quad j = 0, 1, \dots, m$$

Let X_n denote the number of type 1 genes in the n th generation, and assume that $X_0 = i$.

- (a) Find $E[X_n]$.
 (b) What is the probability that eventually all the genes will be type 1?
45. Consider an irreducible finite Markov chain with states $0, 1, \dots, N$.
 (a) Starting in state i , what is the probability the process will ever visit state j ? Explain!
 (b) Let $x_i = P\{\text{visit state } N \text{ before state } 0 | \text{start in } i\}$. Compute a set of linear equations that the x_i satisfy, $i = 0, 1, \dots, N$.
 (c) If $\sum_j j P_{ij} = i$ for $i = 1, \dots, N - 1$, show that $x_i = i/N$ is a solution to the equations in part (b).
46. An individual possesses r umbrellas that he employs in going from his home to office, and vice versa. If he is at home (the office) at the beginning (end) of a day and it is raining, then he will take an umbrella with him to the office (home), provided there is one to be taken. If it is not raining, then he never takes an umbrella. Assume that, independent of the past, it rains at the beginning (end) of a day with probability p .
 (a) Define a Markov chain with $r + 1$ states, which will help us to determine the proportion of time that our man gets wet. (Note: He gets wet if it is raining, and all umbrellas are at his other location.)
 (b) Show that the limiting probabilities are given by

$$\pi_i = \begin{cases} \frac{q}{r+q}, & \text{if } i = 0 \\ \frac{1}{r+q}, & \text{if } i = 1, \dots, r \end{cases} \quad \text{where } q = 1 - p$$

- (c) What fraction of time does our man get wet?
 (d) When $r = 3$, what value of p maximizes the fraction of time he gets wet?
- *47. Let $\{X_n, n \geq 0\}$ denote an ergodic Markov chain with limiting probabilities π_i . Define the process $\{Y_n, n \geq 1\}$ by $Y_n = (X_{n-1}, X_n)$. That is, Y_n keeps track of

the last two states of the original chain. Is $\{Y_n, n \geq 1\}$ a Markov chain? If so, determine its transition probabilities and find

$$\lim_{n \rightarrow \infty} P\{Y_n = (i, j)\}$$

48. Consider a Markov chain in steady state. Say that a k length run of zeroes ends at time m if

$$X_{m-k-1} \neq 0, \quad X_{m-k} = X_{m-k+1} = \dots = X_{m-1} = 0, \quad X_m \neq 0$$

Show that the probability of this event is $\pi_0(P_{0,0})^{k-1}(1 - P_{0,0})^2$, where π_0 is the limiting probability of state 0.

49. Consider a Markov chain with states 1, 2, 3 having transition probability matrix

$$\begin{pmatrix} .5 & .3 & .2 \\ 0 & .4 & .6 \\ .8 & 0 & .2 \end{pmatrix}$$

- (a) If the chain is currently in state 1, find the probability that after two transitions it will be in state 2.
 - (b) Suppose you receive a reward $r(i) = i^2$ whenever the Markov chain is in state i , $i = 1, 2, 3$. Find your long run average reward per unit time. Let N_i denote the number of transitions, starting in state i , until the Markov chain enters state 3.
 - (c) Find $E[N_1]$.
 - (d) Find $P(N_1 \leq 4)$.
 - (e) Find $P(N_1 = 4)$.
50. A Markov chain with states 1, ..., 6 has transition probability matrix

$$\begin{pmatrix} .2 & .4 & 0 & .3 & 0 & .1 \\ .1 & .3 & 0 & .4 & 0 & .2 \\ 0 & 0 & .3 & .7 & 0 & 0 \\ 0 & 0 & .6 & .4 & 0 & 0 \\ 0 & 0 & 0 & 0 & .5 & .5 \\ 0 & 0 & 0 & 0 & .2 & .8 \end{pmatrix}$$

- (a) Give the classes and tell which are recurrent and which are transient.
 - (b) Find $\lim_{n \rightarrow \infty} P_{1,2}^n$.
 - (c) Find $\lim_{n \rightarrow \infty} P_{5,6}^n$.
 - (d) Find $\lim_{n \rightarrow \infty} P_{1,3}^n$.
51. In Example 4.3, Gary is in a cheerful mood today. Find the expected number of days until he has been glum for three consecutive days.
52. A taxi driver provides service in two zones of a city. Fares picked up in zone A will have destinations in zone A with probability 0.6 or in zone B with probability 0.4. Fares picked up in zone B will have destinations in zone A with probability 0.3 or in zone B with probability 0.7. The driver's expected

profit for a trip entirely in zone A is 6; for a trip entirely in zone B is 8; and for a trip that involves both zones is 12. Find the taxi driver's average profit per trip.

53. Find the average premium received per policyholder of the insurance company of Example 4.29 if $\lambda = 1/4$ for one-third of its clients, and $\lambda = 1/2$ for two-thirds of its clients.
54. Consider the Ehrenfest urn model in which M molecules are distributed between two urns, and at each time point one of the molecules is chosen at random and is then removed from its urn and placed in the other one. Let X_n denote the number of molecules in urn 1 after the n th switch and let $\mu_n = E[X_n]$. Show that
- (a) $\mu_{n+1} = 1 + (1 - 2/M)\mu_n$.
 - (b) Use (a) to prove that

$$\mu_n = \frac{M}{2} + \left(\frac{M-2}{M}\right)^n \left(E[X_0] - \frac{M}{2}\right)$$

55. Consider a population of individuals each of whom possesses two genes that can be either type A or type a . Suppose that in outward appearance type A is dominant and type a is recessive. (That is, an individual will have only the outward characteristics of the recessive gene if its pair is aa .) Suppose that the population has stabilized, and the percentages of individuals having respective gene pairs AA , aa , and Aa are p , q , and r . Call an individual dominant or recessive depending on the outward characteristics it exhibits. Let S_{11} denote the probability that an offspring of two dominant parents will be recessive; and let S_{10} denote the probability that the offspring of one dominant and one recessive parent will be recessive. Compute S_{11} and S_{10} to show that $S_{11} = S_{10}^2$. (The quantities S_{10} and S_{11} are known in the genetics literature as *Snyder's ratios*.)
56. Suppose that on each play of the game a gambler either wins 1 with probability p or loses 1 with probability $1 - p$. The gambler continues betting until she or he is either up n or down m . What is the probability that the gambler quits a winner?
57. A particle moves among $n + 1$ vertices that are situated on a circle in the following manner. At each step it moves one step either in the clockwise direction with probability p or the counterclockwise direction with probability $q = 1 - p$. Starting at a specified state, call it state 0, let T be the time of the first return to state 0. Find the probability that all states have been visited by time T .

Hint: Condition on the initial transition and then use results from the gambler's ruin problem.

58. In the gambler's ruin problem of Section 4.5.1, suppose the gambler's fortune is presently i , and suppose that we know that the gambler's fortune will eventually reach N (before it goes to 0). Given this information, show that the

probability he wins the next gamble is

$$\frac{p[1 - (q/p)^{i+1}]}{1 - (q/p)^i}, \quad \text{if } p \neq \frac{1}{2}$$

$$\frac{i+1}{2i}, \quad \text{if } p = \frac{1}{2}$$

Hint: The probability we want is

$$P\{X_{n+1} = i+1 | X_n = i, \lim_{m \rightarrow \infty} X_m = N\}$$

$$= \frac{P\{X_{n+1} = i+1, \lim_m X_m = N | X_n = i\}}{P\{\lim_m X_m = N | X_n = i\}}$$

59. For the gambler's ruin model of Section 4.5.1, let M_i denote the mean number of games that must be played until the gambler either goes broke or reaches a fortune of N , given that he starts with i , $i = 0, 1, \dots, N$. Show that M_i satisfies

$$M_0 = M_N = 0; \quad M_i = 1 + pM_{i+1} + qM_{i-1}, \quad i = 1, \dots, N-1$$

Solve these equations to obtain

$$M_i = i(N-i), \quad \text{if } p = \frac{1}{2}$$

$$= \frac{i}{q-p} - \frac{N}{q-p} \frac{1 - (q/p)^i}{1 - (q/p)^N}, \quad \text{if } p \neq \frac{1}{2}$$

60. The following is the transition probability matrix of a Markov chain with states 1, 2, 3, 4

$$\mathbf{P} = \begin{pmatrix} .4 & .3 & .2 & .1 \\ .2 & .2 & .2 & .4 \\ .25 & .25 & .5 & 0 \\ .2 & .1 & .4 & .3 \end{pmatrix}$$

If $X_0 = 1$

- find the probability that state 3 is entered before state 4;
 - find the mean number of transitions until either state 3 or state 4 is entered.
61. Suppose in the gambler's ruin problem that the probability of winning a bet depends on the gambler's present fortune. Specifically, suppose that α_i is the probability that the gambler wins a bet when his or her fortune is i . Given that the gambler's initial fortune is i , let $P(i)$ denote the probability that the gambler's fortune reaches N before 0.
- Derive a formula that relates $P(i)$ to $P(i-1)$ and $P(i+1)$.
 - Using the same approach as in the gambler's ruin problem, solve the equation of part (a) for $P(i)$.

- (c) Suppose that i balls are initially in urn 1 and $N - i$ are in urn 2, and suppose that at each stage one of the N balls is randomly chosen, taken from whichever urn it is in, and placed in the other urn. Find the probability that the first urn becomes empty before the second.
- *62. Consider the particle from Exercise 57. What is the expected number of steps the particle takes to return to the starting position? What is the probability that all other positions are visited before the particle returns to its starting state?
63. For the Markov chain with states 1, 2, 3, 4 whose transition probability matrix $\mathbf{P}$ is as specified below find f_{i3} and s_{i3} for $i = 1, 2, 3$.

$$\mathbf{P} = \begin{bmatrix} 0.4 & 0.2 & 0.1 & 0.3 \\ 0.1 & 0.5 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.2 & 0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

64. Consider a branching process having $\mu < 1$. Show that if $X_0 = 1$, then the expected number of individuals that ever exist in this population is given by $1/(1 - \mu)$. What if $X_0 = n$?
65. In a branching process having $X_0 = 1$ and $\mu > 1$, prove that π_0 is the *smallest* positive number satisfying Eq. (4.20).

Hint: Let π be any solution of $\pi = \sum_{j=0}^{\infty} \pi^j P_j$. Show by mathematical induction that $\pi \geq P\{X_n = 0\}$ for all n , and let $n \rightarrow \infty$. In using the induction argue that

$$P\{X_n = 0\} = \sum_{j=0}^{\infty} (P\{X_{n-1} = 0\})^j P_j$$

66. For a branching process, calculate π_0 when
- $P_0 = \frac{1}{4}, P_2 = \frac{3}{4}$.
 - $P_0 = \frac{1}{4}, P_1 = \frac{1}{2}, P_2 = \frac{1}{4}$.
 - $P_0 = \frac{1}{6}, P_1 = \frac{1}{2}, P_3 = \frac{1}{3}$.
67. At all times, an urn contains N balls—some white balls and some black balls. At each stage, a coin having probability $p, 0 < p < 1$, of landing heads is flipped. If heads appears, then a ball is chosen at random from the urn and is replaced by a white ball; if tails appears, then a ball is chosen from the urn and is replaced by a black ball. Let X_n denote the number of white balls in the urn after the n th stage.
- Is $\{X_n, n \geq 0\}$ a Markov chain? If so, explain why.
 - What are its classes? What are their periods? Are they transient or recurrent?
 - Compute the transition probabilities P_{ij} .
 - Let $N = 2$. Find the proportion of time in each state.
 - Based on your answer in part (d) and your intuition, guess the answer for the limiting probability in the general case.

- (f) Prove your guess in part (e) either by showing that Theorem (4.1) is satisfied or by using the results of Example 4.37.
- (g) If $p = 1$, what is the expected time until there are only white balls in the urn if initially there are i white and $N - i$ black?
- *68. (a) Show that the limiting probabilities of the reversed Markov chain are the same as for the forward chain by showing that they satisfy the equations

$$\pi_j = \sum_i \pi_i Q_{ij}$$

- (b) Give an intuitive explanation for the result of part (a).
69. M balls are initially distributed among m urns. At each stage one of the balls is selected at random, taken from whichever urn it is in, and then placed, at random, in one of the other $m - 1$ urns. Consider the Markov chain whose state at any time is the vector $(n_1, \dots, n_m)$ where n_i denotes the number of balls in urn i . Guess at the limiting probabilities for this Markov chain and then verify your guess and show at the same time that the Markov chain is time reversible.
70. A total of m white and m black balls are distributed among two urns, with each urn containing m balls. At each stage, a ball is randomly selected from each urn and the two selected balls are interchanged. Let X_n denote the number of black balls in urn 1 after the n th interchange.
- (a) Give the transition probabilities of the Markov chain $X_n, n \geq 0$.
- (b) Without any computations, what do you think are the limiting probabilities of this chain?
- (c) Find the limiting probabilities and show that the stationary chain is time reversible.
71. It follows from Theorem 4.2 that for a time reversible Markov chain

$$P_{ij} P_{jk} P_{ki} = P_{ik} P_{kj} P_{ji}, \quad \text{for all } i, j, k$$

It turns out that if the state space is finite and $P_{ij} > 0$ for all i, j , then the preceding is also a sufficient condition for time reversibility. (That is, in this case, we need only check Eq. (4.26) for paths from i to i that have only two intermediate states.) Prove this.

Hint: Fix i and show that the equations

$$\pi_j P_{jk} = \pi_k P_{kj}$$

are satisfied by $\pi_j = c P_{ij} / P_{ji}$, where c is chosen so that $\sum_j \pi_j = 1$.

72. For a time reversible Markov chain, argue that the rate at which transitions from i to j to k occur must equal the rate at which transitions from k to j to i occur.
73. There are k players, with player i having value $v_i > 0, i = 1, \dots, k$. In every period, two of the players play a game. Whoever wins then plays the next game against a randomly chosen one of the other $k - 1$ players (including the one

who has just lost). Suppose that whenever i and j play, i wins with probability $\frac{v_i}{v_i + v_j}$. Let X_n denote the winner of game n .

- (a) Give the transition probabilities of the Markov chain $\{X_n, n \geq 1\}$.
- (b) Give the stationarity equations that are uniquely satisfied by the π_j .
- (c) Give the time reversibility equations.
- (d) Find the proportion of all games that are won by j , $j = 1, \dots, k$.
- (e) Find the proportion of games that involve player j as one of the contestants.

74. A group of n processors is arranged in an ordered list. When a job arrives, the first processor in line attempts it; if it is unsuccessful, then the next in line tries it; if it too is unsuccessful, then the next in line tries it, and so on. When the job is successfully processed or after all processors have been unsuccessful, the job leaves the system. At this point we are allowed to reorder the processors, and a new job appears. Suppose that we use the one-closer reordering rule, which moves the processor that was successful one closer to the front of the line by interchanging its position with the one in front of it. If all processors were unsuccessful (or if the processor in the first position was successful), then the ordering remains the same. Suppose that each time processor i attempts a job then, independently of anything else, it is successful with probability p_i .

- (a) Define an appropriate Markov chain to analyze this model.
- (b) Show that this Markov chain is time reversible.
- (c) Find the long-run probabilities.

75. A Markov chain is said to be a tree process if

- (i) $P_{ij} > 0$ whenever $P_{ji} > 0$,
- (ii) for every pair of states i and j , $i \neq j$, there is a unique sequence of distinct states $i = i_0, i_1, \dots, i_{n-1}, i_n = j$ such that

$$P_{i_k, i_{k+1}} > 0, \quad k = 0, 1, \dots, n-1$$

In other words, a Markov chain is a tree process if for every pair of distinct states i and j there is a unique way for the process to go from i to j without reentering a state (and this path is the reverse of the unique path from j to i). Argue that an ergodic tree process is time reversible.

76. On a chessboard compute the expected number of plays it takes a knight, starting in one of the four corners of the chessboard, to return to its initial position if we assume that at each play it is equally likely to choose any of its legal moves. (No other pieces are on the board.)

Hint: Make use of Example 4.38.

77. In a Markov decision problem, another criterion often used, different than the expected average return per unit time, is that of the expected discounted return. In this criterion we choose a number α , $0 < \alpha < 1$, and try to choose a policy so as to maximize $E[\sum_{i=0}^{\infty} \alpha^i R(X_i, a_i)]$ (that is, rewards at time n are discounted at rate α^n). Suppose that the initial state is chosen according to the probabilities b_i . That is,

$$P\{X_0 = i\} = b_i, \quad i = 1, \dots, n$$

For a given policy β let y_{ja} denote the expected discounted time that the process is in state j and action a is chosen. That is,

$$y_{ja} = E_{\beta} \left[\sum_{n=0}^{\infty} \alpha^n I_{\{X_n=j, a_n=a\}} \right]$$

where for any event A the indicator variable I_A is defined by

$$I_A = \begin{cases} 1, & \text{if } A \text{ occurs} \\ 0, & \text{otherwise} \end{cases}$$

(a) Show that

$$\sum_a y_{ja} = E \left[\sum_{n=0}^{\infty} \alpha^n I_{\{X_n=j\}} \right]$$

or, in other words, $\sum_a y_{ja}$ is the expected discounted time in state j under β .

(b) Show that

$$\begin{aligned} \sum_j \sum_a y_{ja} &= \frac{1}{1-\alpha}, \\ \sum_a y_{ja} &= b_j + \alpha \sum_i \sum_a y_{ia} P_{ij}(a) \end{aligned}$$

Hint: For the second equation, use the identity

$$I_{\{X_{n+1}=j\}} = \sum_i \sum_a I_{\{X_n=i, a_n=a\}} I_{\{X_{n+1}=j\}}$$

Take expectations of the preceding to obtain

$$E[I_{\{X_{n+1}=j\}}] = \sum_i \sum_a E[I_{\{X_n=i, a_n=a\}}] P_{ij}(a)$$

(c) Let $\{y_{ja}\}$ be a set of numbers satisfying

$$\begin{aligned} \sum_j \sum_a y_{ja} &= \frac{1}{1-\alpha}, \\ \sum_a y_{ja} &= b_j + \alpha \sum_i \sum_a y_{ia} P_{ij}(a) \end{aligned} \tag{4.38}$$

Argue that y_{ja} can be interpreted as the expected discounted time that the process is in state j and action a is chosen when the initial state is chosen

according to the probabilities b_j and the policy β , given by

$$\beta_i(a) = \frac{y_{ia}}{\sum_a y_{ia}}$$

is employed.

Hint: Derive a set of equations for the expected discounted times when policy β is used and show that they are equivalent to Eq. (4.38).

(d) Argue that an optimal policy with respect to the expected discounted return criterion can be obtained by first solving the linear program

$$\begin{aligned} &\text{maximize} \quad \sum_j \sum_a y_{ja} R(j, a), \\ &\text{such that} \quad \sum_j \sum_a y_{ja} = \frac{1}{1 - \alpha}, \\ &\quad \quad \quad \sum_a y_{ja} = b_j + \alpha \sum_i \sum_a y_{ia} P_{ij}(a), \\ &\quad \quad \quad y_{ja} \geq 0, \quad \text{all } j, a; \end{aligned}$$

and then defining the policy β^* by

$$\beta_i^*(a) = \frac{y_{ia}^*}{\sum_a y_{ia}^*}$$

where the y_{ja}^* are the solutions of the linear program.

78. For the Markov chain of Exercise 5, suppose that $p(s|j)$ is the probability that signal s is emitted when the underlying Markov chain state is j , $j = 0, 1, 2$.
- (a) What proportion of emissions are signal s ?
 - (b) What proportion of those times in which signal s is emitted is 0 the underlying state?
79. In Example 4.45, what is the probability that the first 4 items produced are all acceptable?

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